ABM ENGINE — BUSINESS LOGIC BIBLE

DOCUMENT 2 OF 3: PRODUCTION ENGINE

Functional Specification — Full Rule Compendium with Resolved Conflicts

Decimal Technologies | KSA Banking & Financial Institutions | June 2026

DOCUMENT 2 SCOPE — PRODUCTION ENGINE (INCLUDES ALL MVP RULES)

This document is the complete production specification. It contains **every MVP rule plus all production-grade rules** — they are additive, never contradictory. Each rule is presented in full FSD style with its Rationale.

Production adds over MVP: Full Opportunity-anchored dual-scoring (Account Attractiveness feeding Effective Opportunity), multi-channel outreach with 48-hour channel-gap coordination, Autonomy Ladder A0-A2, EPIS Reality Confidence Model on all integrations, the Resource Allocation Operating System (§5), Trust Construction and Preservation (§7-8), Dynamic Relationship Management (§9), Reality Calibration (§10), Decision Partnership (§11), Distributed Reality Coordination (§12), the Opportunity State Machine (§14), and Market Intelligence (§17).

This is the engine that runs at scale.

DOCUMENT CONTENTS

Total rules in this document: 866

Section 1: Foundational Principles: Event-Driven + Enterprise Decision (200 rules)

ACCT Rules (16)

ACCT-COORD-001
When outreaching to multiple contacts at the same account, stagger initial touches by minimum 48 hours between contacts. Contact #1 Day 1, Contact #2 Day 3, Contact #3 Day 5.
Rationale: *Prevents the "everyone got emailed on the same day" problem that signals mass outreach rather than strategic engagement.*

ACCT-COORD-002
Maximum 3 contacts per account in active outreach simultaneously during MVP.
Rationale: *More than 3 creates coordination complexity and increases conflicting message risk.*

ACCT-COORD-003
Contact outreach priority order: (1) Role most directly mapping to Decimal's value (CTO, CDO, Head of Digital), (2) Most senior reachable contact, (3) Warmest existing relationship, (4) Verified email over LinkedIn-only.
Rationale: *Role relevance predicts response rate better than seniority alone.*

ACCT-COORD-004
Conflicting signals at the same account are flagged as "SIGNAL_CONFLICT." The system does NOT auto-resolve. Both signals are presented to the human reviewer. AI can suggest resolution but cannot execute autonomously.
Rationale: *Conflicting intelligence requires strategic judgment humans are better at.*

ACCT-COORD-005
Accounts can be linked as "related_accounts" with relationship type: PARENT_SUBSIDIARY, PARTNER, SHARED_BOARD, SAME_GROUP. When one account in a group reaches OUTREACH_ACTIVE, alert account_owner to manually coordinate.
Rationale: *Related accounts often have overlapping stakeholders — uncoordinated outreach across group entities looks uncoordinated.*

ACCT-COORD-006
— A contact appearing at TWO related accounts gets a "primary_account" designation. Outreach ONLY through their primary account.
Rationale: *Receiving outreach from two "different" Decimal campaigns simultaneously destroys credibility. Human-AI Review Logic*

ACCT-CORE-001
10-behavior Account-Centric Matrix: (1) Scoring at account level, (2) Tier at account level, (3) Outreach triggered by account evaluation then contact selection, (4) Reply from any contact pauses entire account, (5) Engagement rolls up to account, (6) One CRM opportunity per account, (7) Enrich account first then contacts, (8) AI reasons about organization + role, (9) Cooldown per account, (10) Analytics roll up to account.
Rationale: *Every integration defaults to contact-centric. These 10 behaviors must be explicitly enforced.*

ACCT-IDENTITY-001
Account = single legal entity or operationally independent business unit with its OWN CTO/IT leadership AND its OWN technology budget. If uncertain, default to SEPARATE accounts linked as related.
Rationale: *Merging loses granularity. Separating and linking preserves both independence and coordination.*

ACCT-IDENTITY-002
Account Lifecycle Events: MERGER: new merged account, migrate all data, archive old records, pause active sequences for review. ACQUISITION: acquired stays separate, relationship changes to PARENT_SUBSIDIARY, check if still independent procurement. SPIN-OFF: new account, copy relevant contacts, original keeps history. REBRAND: update name, add old name as alias.
Rationale: *Each lifecycle event requires different data handling.*

ACCT-IDENTITY-004
Geographic Scoping: MVP = KSA only. GCC subsidiary signals attributed to KSA parent at 70% relevance weight. UAE-HQ fintech opening KSA office = new account scoped to KSA, HQ signals weighted at 70%.
Rationale: *HQ decisions often flow to regional branches but aren't 100% applicable.*

ACCT-IDENTITY-006
Account Promotion: Watch->Active when P1/P2 signal + Fit >= 50. Active->Watch when COLD for 90+ days. Active->Archive only by manual human decision. Archive->Watch by manual decision.
Rationale: *Promotion is signal-driven. Demotion requires sustained inactivity. Archival requires human judgment. — 2F. Buying Committee Model (6 Rules)*

ACCT-COMMITTEE-004
Cross-Role Message Coherence: All messages to same account reference SAME signal, highlight SAME Decimal capability, propose SAME outcome. Only the ANGLE and PROOF POINTS differ by role. The NARRATIVE is consistent.
Rationale: *If CTO and Head of Lending compare notes, messages should tell same story from different angles.*

ACCT-COMMITTEE-006
Informal Influence Mapping: Each contact gets influence_modifier: STANDARD (matches title), ELEVATED (disproportionate influence — royal connection, CEO advisor, 15+ year tenure), REDUCED (less than title suggests — <6 months, interim, known to be leaving). Source: connector intel, BD knowledge, manual research only. ELEVATED contacts prioritized one seniority level higher.
Rationale: *Title does not equal power in KSA banking. — 2G. Failure Mode A: Account Misattribution (4 Rules)*

ACCT-PAUSE-001
Pause depends on reply type: POSITIVE (let's talk) = full account pause. REFERRAL (talk to [Name]) = pause this contact, redirect to referred person, continue others. NEGATIVE-PERSONAL (I'm not interested) = pause this contact, continue others unless reply says "our COMPANY isn't interested." NEGATIVE-ORGANIZATIONAL (we chose a vendor/no budget) = full pause + human review. AUTO-REPLY = no pause, resume after return. NEUTRAL (let me check) = pause this contact temporarily, slow others 50%.
Rationale: *Blanket pause for every auto-reply freezes the pipeline unnecessarily.*

ACCT-PAUSE-002
When system decides partial pause (not full): decision logged, account_owner notified, 4-hour override window. No override within 4h = decision stands.
Rationale: *Nuanced pause decisions need human oversight without requiring human initiation.*

ACCT-PAUSE-003
Pause Duration Limits: Day 7: reminder to account_owner. Day 14: escalate to manager. Day 21: forced decision — (a) "still in conversation" + reset timer, (b) "stalled" + revert to HOT + re-enable others, (c) "dead" + DORMANT.
Rationale: *Prevents "engaged zombie" accounts sitting paused forever because a human dropped the ball.*

ADVERSARIAL Rules (2)

ADVERSARIAL-001
"Input Manipulation Detection" — 5 attack vectors: (1) Fake signal injection: fake news about bank triggers outreach → mitigated by Source Reliability + Observability Score + two-stage classification. (2) Engagement inflation: competitor/troll opens all emails → detect bot patterns (every email opened within 30s), flag SUSPICIOUS_ENGAGEMENT, don't count for scoring. (3) LinkedIn honeypot: fake profile mimicking executive → cross-reference with Apollo/Clearbit, UNVERIFIED_IDENTITY blocks outreach until verified. (4) Competitor intelligence: competitor responds to learn our pitch → never reveal pricing/strategy in automated messages, only through human conversations. (5) System probing: testing how system responds → rate-limit signal ingestion per source, flag unusual patterns.
Rationale: *Any system trusting external data can be manipulated through that data.*

ADVERSARIAL-002
"Internal Gaming Detection": (a) Phantom meetings: owner marks low-quality as "booked" → cross-reference CRM calendar + follow-up activity. (b) Speed-approval: reviewer approves without reading → FAIL-RUBBER-001 mechanisms. (c) False manual signals: BD enters fake signals → random 20% monthly audit, signals never leading to activity are suspicious. (d) Suppressive overrides: owner suppresses accounts → override frequency tracking, persistent overrides without reason = review. Response: findings for PROCESS IMPROVEMENT not employee punishment. If incentives encourage gaming, fix incentives.
Rationale: *Internal gaming produces beautiful dashboards and zero pipeline.*

AIBOUNDARY Rules (3)

AIBOUNDARY-001
"AI Permission Matrix" — Exhaustive CAN/CANNOT list: CAN: gather public info, calculate scores, generate drafts, tailor messaging, recommend strategy, classify signals/sentiment/outcomes, analyze performance, validate outputs, propose send times, prepare CRM data. CANNOT: send messages without approval (MVP), change scoring formulas/weights/thresholds, reclassify human-verified items, override its own QC rejections, override business hours/holidays, create/modify/delete CRM deals, access private systems, execute strategy changes without confirmation. DEFAULT for unlisted actions: AI CANNOT.
Rationale: *Without explicit boundaries, AI scope creeps.*

AIBOUNDARY-002
"Approved Claims Library" — AI can ONLY make claims about Decimal from a pre-approved library: verified client references (REFERENCEABLE only), specific product capabilities (reviewed by product team), performance metrics (verified by data), compliance certifications (verified by legal), integration capabilities (verified by engineering). Claims not in library = "CLAIM_NOT_VERIFIED" flag + placeholder. AI must NEVER: guarantee outcomes ("WILL increase lending 40%"), make comparative claims ("better than Temenos"), claim non-existent capabilities, reference NDA-protected info.
Rationale: *One false capability claim to a bank CTO creates compliance and trust crisis.*

AIBOUNDARY-003
"Cultural Sensitivity Rules" — KSA-specific norms hardcoded: FORMALITY: formal titles always in initial outreach, never initiate informality. RELIGION: never reference practices/beliefs unless contextually appropriate (Ramadan greetings OK during Ramadan). GENDER: no assumptions about capabilities based on gender, same professional tone regardless. HIERARCHY: never suggest lower-ranking should "convince" superiors — frame as "sharing information." NATIONAL PRIDE: frame Decimal as supporting Vision 2030, never as "teaching Saudi banks." COMPETITOR RESPECT: respectful tone even when positioning against competitors.
Rationale: *Cultural missteps in KSA banking are relationship-ending. DEEP DRILL: Failure Modes 1-6 (11 Rules) FM1 — Rubber-Stamping (2 Rules)*

AIHUMAN Rules (5)

AIHUMAN-FOUNDATION-001
"The AI-Human Boundary Principle": AI operates in the domain of DATA (signals, scores, research, patterns, drafts). Humans operate in the domain of RELATIONSHIPS (trust, judgment, negotiation, reputation). The boundary between these domains shifts as AI proves reliability, but the RELATIONSHIP domain is permanently human. The system must be designed so AI never takes actions in the relationship domain without explicit human authorization.
Rationale: *Enterprise banking sales in KSA are trust-based, relationship-based, and culturally nuanced. A single AI hallucination to a SEVP at SNB could permanently close the door. The cost of one bad message outweighs 100 automated good ones.*

AIHUMAN-FOUNDATION-002
"The 90/10 Resource Model": AI handles approximately 90% of work volume (signal monitoring, scoring, enrichment, research, reasoning, drafting, QC, engagement tracking, analytics). Humans handle approximately 10% (message approval, relationship conversations, strategic overrides, handoff management, outcome classification). But the 10% human contribution carries 90% of the REPUTATIONAL RISK. Therefore: every human decision point must be presented with complete AI-prepared context so the human can make a high-quality judgment in minimum time.
Rationale: *Without AI, a BD person spends 2-3 hours researching per account. With AI, they spend 5 minutes reviewing a strategically reasoned draft.*

AIHUMAN-CONTEXT-001
"Human Decision Context Package" — Every human decision point must present complete context: (1) Message Approval: full reasoning chain, triggering signal with source, account summary, persona profile, chosen strategy with rationale, Golden Window status, conflict flags, ICS score, Effective Opportunity. (2) Sequence Strategy: contacts in order, channel reasoning, timing rationale, competitive landscape, relationship paths, committee coverage, blocker/veto flags. (3) Handoff Response: full engagement history, reply text with sentiment, meeting brief, talking points, likely objections, comparable wins. (4) Override Decision: system recommendation with rationale, what changes if overridden, historical override outcomes, risk assessment. (5) Outcome Classification: sequence timeline, engagement events, contact behavior, signal quality assessment, AI suggested classification.
Rationale: *"Approve t...*

AIHUMAN-DISAGREE-001
"Disagreement Resolution Protocol": The human ALWAYS wins during MVP. But disagreement must be CAPTURED: (a) record AI output, human change, human stated reason (required, min 10 chars), (b) data feeds three mechanisms — prompt improvement (humans change ANGLE → persona prompts need updating), knowledge gap (humans add facts AI didn't know → enrichment incomplete), style mismatch (humans change tone/length → generation constraints need adjusting), (c) monthly report: "Top 10 reasons humans overrode AI" — single most valuable input for AI improvement, (d) CRITICAL: system must NEVER train AI to simply mimic human edits without understanding WHY. A human might edit for situation-specific reasons that shouldn't generalize.

AIHUMAN-DISAGREE-002
"AI Guardrails on Human Actions" — AI doesn't override humans but CAN warn: SOFT WARNINGS (overridable, logged): message contains unverifiable claims ("Source not found in enrichment"), edit introduces banned phrase or exceeds length, human force-promotes account with ICS <40. HARD BLOCKS (require escalation): message would send outside KSA business hours or during Ramadan/Eid, message targets DISQUALIFIED or EXHAUSTED contact, message to DORMANT account in cooling-off period.
Rationale: *Humans make mistakes too — especially tired reviewers at 5pm Thursday. DEEP DRILL: Explainability Layer (5 Rules)*

AILEARN Rules (5)

AILEARN-001
"Minimum Sample Size for Learning" — No parameter changed with fewer than: Scoring weights: 10 meetings across 3+ signal types. Persona prompts: 5 completed sequences to that persona type. Signal prioritization: 15 signals of that type with tracked outcomes. Timing rules: 20 messages in proposed window. QC thresholds: 30 messages through pipeline. Sequence structure: 10 completed sequences. Below minimum = logged as OBSERVATION, not acted upon. Observations accumulate until threshold met.
Rationale: *Overfitting to 3 data points is worse than not learning. "CTO at Bank X liked compliance" is anecdote. "7 of 10 CTOs at Islamic banks responded to compliance" is pattern.*

AILEARN-002
"Learning-to-Change Pipeline" — 7 stages: Stage 1 OBSERVATION: AI detects pattern. Stage 2 HYPOTHESIS: formulated as testable statement. Stage 3 VALIDATION: check sample size, confounding variables, survivorship bias. Stage 4 PROPOSAL: specific parameter change + expected impact + rollback criteria. Stage 5 HUMAN APPROVAL: BD leadership reviews. No change without sign-off. Stage 6 A/B DEPLOYMENT: 30% gets new approach, 70% existing. Compare 2+ weeks. Stage 7 FULL DEPLOYMENT or ROLLBACK: improvement → deploy 100%. Neutral/negative → rollback.
Rationale: *Prevents reckless changes (too fast, too little data) and stagnation (no process to change).*

AILEARN-003
"Prompt Version Control": Every prompt carries: version_id (semantic major.minor.patch), effective_date, change_description, change_rationale (linked to AILEARN-002 proposal), performance_baseline. Every AI_GENERATION record stores prompt_version_id. Enables: "In March, v2.3.1 produced 4.2% reply rate. April v2.4.0 produced 5.8%." Or: "v3.0.0 doubled hallucination rate, rolled back to v2.4.0 in 3 days." No prompt change without version increment. Direct database editing without versioning = SYSTEM VIOLATION.
Rationale: *Without version control, you can't correlate AI quality changes with the prompt changes that caused them.*

AILEARN-004
"Chain Coherence Testing": When ANY agent prompt modified, full 7-agent chain regression-tested: run against 10 historical accounts with known outcomes. Compare to original output, actual sent message, and human-edited version. Pass: equal or better on 7+ of 10 (human-judged, not automated). Fail: worse on 4+ → reject change, investigate downstream effects.
Rationale: *The chain is a system, not independent components. Testing in isolation misses interaction effects.*

AILEARN-005
"AI Capability Gap Registry": Maintained backlog of things AI cannot do well but should: Arabic message generation (HIGH priority, 6mo target), competitive positioning when competitor unknown (MEDIUM, 9mo), optimal send time prediction (LOW, 18mo), Arabic news signal detection (HIGH, 6mo), organizational politics from public data (LOW, likely never fully automatable). Reviewed quarterly. Top gaps get engineering resources.
Rationale: *Knowing what you CAN'T do is as important as improving what you CAN do.*

AIMODEL Rules (3)

AIMODEL-001
"Model Change Protocol": Step 1 EVALUATION: new model against 10 historical test cases. Step 2 SHADOW MODE: 2 weeks parallel with existing model (only existing used for outreach, new logged for comparison). Step 3 COMPARATIVE ANALYSIS: QC rates, output length, hallucination flags, blind human side-by-side. Step 4 CUTOVER: equal or better → gradual 30%→60%→100% over 2 weeks. Worse on any critical dimension → reject. Step 5 MONITORING: 30-day enhanced monitoring.
Rationale: *"AI got worse" is extremely hard to diagnose after the fact. Shadow mode prevents production quality drops.*

AIMODEL-002
"Model Fallback Architecture": Always TWO operational models: PRIMARY (main) and FALLBACK (tested, proven, ready to deploy). Fallback activation: automatic for availability (primary down 30+ min), manual for quality (QC rejection spikes above 40%, latency >30s consistently — requires human confirmation of actual degradation).
Rationale: *Single-model dependency is a catastrophic risk. Outage at the model provider = complete system stoppage without fallback.*

AIMODEL-003
"Model Cost Management": Track per-operation costs: signal classification, account research, full 7-agent chain, QC validation. Monthly budget ceiling defined by leadership. Costs at 80% of ceiling → alert. Exceed ceiling → essential-only (P1/P2 + HOT accounts; P3/P4 + WARM deferred). Optimization: cheaper model for low-stakes ops, cache enrichment intelligence, batch P3/P4 classifications.
Rationale: *AI costs scale with volume. Surge month without tracking can blow budget.*

AIQUALITY Rules (2)

AIQUALITY-001
"Quality Degradation Early Warning" — 6 indicators tracked monthly: (1) Personalization depth: avg account-specific sentences/message, threshold drops below 2.0 from ~3.5 baseline. (2) Reasoning chain length: avg word count of agent outputs, threshold 30%+ below baseline. (3) Vocabulary diversity: unique/total words per week, threshold below 0.4 from ~0.55. (4) Signal reference specificity: % referencing specific signal, threshold below 70% from ~90%. (5) CTA variation: unique CTAs per month, threshold below 4 from ~8. (6) Strategy diversity: unique angles per month, threshold below 3. Any threshold crossed → "QUALITY_DRIFT" alert. Fix investigated within 7 days.
Rationale: *Quality degradation is invisible reviewing individual messages. Only systematic measurement catches gradual decline.*

AIQUALITY-002

"Gold Standard Library": 20 exemplary messages that were sent, received positive replies, led to meetings. Monthly: generate 5 new messages for SAME accounts/contacts using current AI. Human evaluator scores: BETTER/EQUAL/WORSE vs gold standard on personalization, strategic soundness, writing quality. >40% rated WORSE → quality degraded, trigger full AI review. Library updated quarterly: add new excellent messages, retire outdated ones.

Rationale: *Benchmarking against your own best work prevents drift that no metric catches. ITERATION 2 — HUMAN CAPABILITY & KNOWLEDGE LAYER What happens to the HUMANS over time? The AI reshapes how they work, think, and develop skills. If humans atrophy, the entire system becomes fragile.*

AITRANSPARENCY Rules (2)

AITRANSPARENCY-001

"AI Literacy Program": Level 1 (ALL users): what AI does and doesn't do. Level 2 (Reviewers + Owners): how 7-agent chain works, how to read Reasoning Passport, how to identify AI errors, common failure patterns. Level 3 (BD Leadership): scoring formula, three-score model, feedback loops, analytics interpretation, AI change approval. Assessment required before system operation. Annual refresher.

Rationale: *Users who don't understand the AI can't evaluate its output.*

AITRANSPARENCY-002

"AI Limitation Visibility": System proactively surfaces limitations where decisions are made: (a) score with missing data: "Based on incomplete data. Digital maturity: unknown." (b) message with low-confidence steps: "Pain inference confidence: LOW. Consider accuracy." (c) recommendation with close call: "Recommended: CTO first (65%). Alternative: CDO (55%). Close call — your judgment should decide."

Rationale: *Presenting output without uncertainty markers encourages blind trust.*

AITRUST Rules (6)

AITRUST-001

"Trust Capital Ledger" — Each AI subsystem maintains Trust Capital (0-100, starting at 50): Signal Classification: +1 per correct, -5 per false positive triggering wasted outreach, -15 per false positive reaching prospect. Scoring: +2 per correct tier prediction, -3 per significant misrank, -10 per missed deal. Message Generation: +1 per no-edit approval, +3 per positive reply, -2 per MAJOR_REWRITE, -10 per hallucination passing QC, -25 per hallucination reaching prospect. QC: +1 per correct catch, +5 per caught hallucination, -5 per hallucination passing QC but caught by human, -20 per reaching prospect. Enrichment: +1 per verified-accurate data, -3 per wrong contact, -8 per wrong company data. Persona/Pain: +2 per confirmed-relevant pain, -3 per "that's not our priority" response. Thresholds: 70-100 TRUSTED (standard oversight). 40-69 CAUTIOUS (increased review). <40 DISTRUSTED (double...

AITRUST-002

"Override Severity Classification": COSMETIC (formatting, punctuation): 0 trust impact, 0 learning value. STYLISTIC (tone, formality): 0 trust impact, track as style preference. TACTICAL (swap case study, change CTA): -1 message generation trust, moderate learning value. STRATEGIC (change outreach angle, pain inference wrong): -3 strategy trust, -2 pain inference trust, HIGH learning value. SAFETY (block hallucination, prevent policy violation): -10 relevant subsystem, -5 QC if should have caught, CRITICAL learning value. Healthy monthly distribution: 60%+ cosmetic/stylistic, 25% tactical, 10% strategic, 5% → AI quality system failing. Strategic >25% → strategic reasoning needs recalibration.

AITRUST-003

"Recommendation Verifiability Score": FULLY VERIFIABLE (90-100): all claims link to checkable sources. MOSTLY (60-89): most verifiable, some inferences. PARTIALLY (30-59): significant inferences, limited sources. LOW (<30): mostly inference-based. Displayed alongside messages. Low verifiability = reviewer applies MORE independent judgment. High = can rely on AI with spot-checking.

Rationale: *Telling a reviewer "this is well-researched" without showing HOW well-researched asks for blind trust.*

AITRUST-004

"Per-Reviewer Trust Calibration": Track per reviewer: total reviews, override rate by type, canary catch rate, average review time, correlation between approvals and outcomes. High-experience reviewer (200+, good canary, overrides improve outcomes) → expanded autonomy: larger batches, extended SLAs for low-risk. Low-experience/declining → tighter controls: smaller batches, shorter SLAs, spot-checks.

Rationale: *Reward demonstrated judgment with efficiency. Support struggling reviewers.*

AITRUST-005

"Failure Predictability Classification": EXPECTED FAILURE (system failed in known limitation scenario): x1 trust impact — confirms we understand limits. UNEXPECTED FAILURE (system failed where it should have worked): x5 trust impact — our reliability model is wrong. BLACK SWAN (entirely novel failure mode): x3 trust impact — serious but recoverable, expands understanding, triggers new prevention rule.

Rationale: *A system that fails predictably is TRUSTWORTHY. A system that fails unpredictably is DANGEROUS. ITERATION 5 — DECISION SCIENCE LAYER How does a human actually decide? Most systems never ask. The same message can be approved by one reviewer and rejected by another. Understanding WHY creates the deepest learning.*

AITRUST-001a

"Trust Capital Gates Progression": To advance to ESTABLISHED TRUST: ALL subsystems >= 60. To MATURE: ALL >= 70. To FULL PARTNERSHIP: ALL >= 80. ANY subsystem drops below 40 → system-wide trust regresses one level, cannot advance until recovery above 50.

Rationale: *Excellent message generation with terrible signal classification = sending good messages to wrong accounts. Trust must be holistic.*

AUTO Rules (1)

AUTO-CEILING-001

CONFLICT RESOLUTION (FIN-C06): No autonomy level, including A5, ever removes human review from C-suite and P1 Domino Account messages. The RELATIONSHIP domain is permanently human (AIHUMAN-FOUNDATION-001, MP-19). Autonomy progression applies only to the data/execution domain below the C-suite relationship tier. Rationale: aligns the A3-A5 Autonomy Ladder with the permanent-C-suite-review rules so they cannot be read as eventually automating C-suite sends.

CATASTROPHE Rules (3)

CATASTROPHE-001

"Catastrophic Failure Classification": SEV-1 MARKET-THREATENING: grossly offensive message to senior executive, data breach exposing contacts, 500 messages sent due to bug. Response: KILL SWITCH within 5 min, CEO notified, external communication within 24h, post-mortem within 72h. SEV-2 ACCOUNT-THREATENING: hallucination to C-suite despite QC+human, same person contacted from 3 sequences, confidential info leaked cross-account. Response: pause affected account, owner contacts within 4h, honest acknowledgment, post-mortem within 1 week. SEV-3 OPERATIONAL: AI garbage for 2 hours unnoticed, scoring miscalculation for batch, enrichment returns wrong data. Response: scope impact within 2h, fix within 24h, audit failure-window outputs. SEV-4 MINOR: single typo, wrong formatting, one account scored incorrectly with no outreach. Response: fix and document.

CATASTROPHE-002

"Kill Switch": One-click emergency stop available to BD leadership + system admin: (a) halts ALL outreach (email + LinkedIn), (b) pauses all pending messages in queue, (c) stops all AI generation, (d) continues signal ingestion and scoring (intelligence doesn't stop), (e) alerts all operators: "SYSTEM HALTED by [who] at [time]. Reason: [required]." Activation requires reason. Deactivation requires BD leadership approval + root cause resolved.

Rationale: *Every minute of continued operation during catastrophe makes it worse.*

CATASTROPHE-003

"Post-Mortem Framework" — Every SEV-1/SEV-2: WHAT happened (exact events + timestamps). WHO affected (accounts, contacts, messages).

Rationale: *it happened (root cause — which layer failed). WHY NOT CAUGHT (which safety layers should have prevented and didn't). HOW TO PREVENT (specific changes, assigned owner, deadline). BUSINESS IMPACT (relationship damage, recovery possibility). Completed within 7 days. Findings shared with team. Changes within 30 days. Repeat items from previous post-mortems = escalation. WHY: Structured learning from failures prevents recurrence.*

CHANNEL Rules (1)

CHANNEL-GAP-002

CONFLICT RESOLUTION (FIN-C03): Minimum gap between two different-channel touches to the SAME contact is 24 hours, unless the contact engaged on the first channel (open, click, accept), in which case the second channel may follow after 4 hours. This is DISTINCT from ACCT-COORD-001, which governs spacing across different contacts at one account (48h stagger). Rationale: closes the coverage seam between across-contact and same-contact channel coordination.

ETHICS Rules (3)

ETHICS-001

"Ethical Absolutes — What System Must Never Do": (a) NEVER exploit personal vulnerability — mark contact "PERSONAL_SENSITIVITY" if intelligence reveals personal difficulty. (b) NEVER misrepresent identity — messages sent on behalf of named Decimal employees, no fake personas. (c) NEVER manufacture urgency — reference REAL urgency only, no "offer expires Friday" when it doesn't. (d) NEVER disparage competitors by name in automated outreach. (e) NEVER use engagement data punitively — don't escalate to boss or cc colleagues. (f) NEVER ignore opt-out — ABSOLUTE and PERMANENT unless explicitly reversed by contact.

Rationale: *The system manipulates human attention. That power requires ethical constraints.*

ETHICS-002
"Data Ethics": (a) Proportional collection: only data serving defined business purpose. (b) Data minimization in prompts: agents receive only needed data. (c) Retention limits: ARCHIVE accounts 12+ months → purge/anonymize personal contact details. Organizational intelligence retained. (d) Transparency on request: if contact asks "how did you get my info?" answer honestly. (e) PDPL compliance: quarterly review against Saudi data protection requirements.
Rationale: *Legal compliance and moral responsibility for personal data handling.*

ETHICS-003
"Personalization vs Manipulation Line": ACCEPTABLE: "Given your focus on digital lending growth..." (real business priority). "Banks that modernized saw 40% improvement..." (factual, relevant). UNACCEPTABLE: "Your competitors are already ahead" (manufactured fear, may not be true). "You can't afford to wait" (pressure not grounded in deadline). "As a CTO who cares about their legacy..." (identity manipulation). "Other executives at your bank are interested" (internal social proof, may be false and politically dangerous). THE TEST: "Would recipient feel RESPECTED seeing the AI reasoning?" If they'd feel analyzed/profiled/manipulated → crosses the line. Agent 4 produces BUSINESS pains, never insecurities. Agent 5 selects VALUE angles, never fear. Agent 6 writes with RESPECT, never urgency manufacturing.

EVENT Rules (4)

EVENT-RESP-001
"Immediate" for P1 signals means scoring recalculation must BEGIN within 15 minutes of signal ingestion. For P2, within 1 hour. The entire cascade (scoring, tier check, enrichment trigger) must complete within 2 hours for P1 and 6 hours for P2.
Rationale: *P1 signals have a golden window. A new CTO spends weeks 1-4 evaluating the landscape. 15 minutes is fast enough to stay same-day. 1 hour for P2 is sufficient for strategic-but-not-urgent signals.*

EVENT-RESP-002
Signal ingestion and scoring run 24/7/365 regardless of business hours. Only OUTREACH EXECUTION respects business hours and KSA weekends.
Rationale: *A P1 signal at 11pm Thursday should have score updated, enrichment triggered, AI reasoning complete, and message drafted by Friday morning, ready to send Sunday 9am.*

EVENT-RESP-003
The system maintains a "Ready-to-Fire Queue" for messages that have passed AI QC and await either human review or next valid send window. Queue ordered by signal priority (P1 first) then signal age (oldest first within same priority).
Rationale: *Ensures highest-priority messages send first when business hours resume.*

EVENT-RESP-004
Every event carries a "trace_id" linking the original trigger through every downstream action: signal, scoring record, enrichment record, AI generation, message, and engagement.
Rationale: *If anyone asks "why did this person receive this message?" the trace_id answers in one lookup. Full audit trail. Account Coordination*

EXPLAIN Rules (5)

EXPLAIN-001
"Reasoning Chain Transparency" — Every AI-generated message carries a "Reasoning Passport": SIGNAL: text, date, source, confidence. INTERPRETATION: what it means for buying intent (Agent 1). ACCOUNT CONTEXT: organization summary (Agent 2). PERSONA ANALYSIS: role motivations, decision criteria (Agent 3). PAIN INFERENCE: likely operational pains with reasoning (Agent 4). STRATEGY: chosen angle, value proposition, CTA rationale (Agent 5). QC RESULT: all checks passed, confidence score (Agent 7). Stored permanently alongside message. Visible to reviewer. Part of handoff package.
Rationale: *If you can't explain why you sent a message, you shouldn't send it.*

EXPLAIN-002
"Score Decomposition" — Every score breakable into contributing factors: Dynamic Score 82 = Signal Strength 75 (weight 35%, contribution 26.25) + Regulatory Pressure 90 (30%, 27) + Reachability 70 (20%, 14) + Warmth 60 (15%, 9) + intent multiplier + criticality bonus. ICS decomposed similarly. Effective Opportunity shows each multiplier. Every number has traceable source. No black boxes.
Rationale: *When human asks "why is this account #3 not #1?" system shows exactly which dimensions differ.*

EXPLAIN-003
"Negative Decision Logging" — Every time system decides NOT to act, it logs: "Account X not promoted to HOT because ICS = 35 (below 40 per GOV-001)." "Message for Contact Y blocked: OUTREACH_EXHAUSTED (per FAIL-FATIGUE-002)." "Outreach deferred: Golden Window expired, needs regeneration (per TIMING-WIN-002)." These logs are queryable. Owner asks "Why hasn't anything happened with Bank X?" and gets a specific answer.
Rationale: *Silent inaction is the hardest thing to debug. Negative decisions must be as visible as positive ones.*

EXPLAIN-004
"Human Decision Audit Trail" — Every human action logged with: WHO (which human), WHAT (approved/edited/rejected/overrode/escalated), WHEN (timestamp),
Rationale: *(stated reason — required for edits/rejections/overrides), AGAINST (what AI recommended if different), OUTCOME (what happened as result). Creates complete audit trail where BOTH AI reasoning and human judgment are visible. WHY: If a message causes a problem, the trail shows: AI recommended X, human changed to Y, for reason Z. Accountability is clear.*

EXPLAIN-005
"Accountability Framework" — 4 layers: Layer 1 — AI QC Agent: accountable for catching hallucinations, tone violations, factual errors. Layer 2 — Human Reviewer: accountable for strategic judgment (right angle? right timing? right tone for relationship?). Layer 3 — Account Owner: accountable for relationship outcomes. Layer 4 — System Designer (BD leadership): accountable for rules, thresholds, workflows, guardrails. Post-mortem identifies WHICH layer failed. No finger-pointing between AI and humans.
Rationale: *Clear accountability prevents blame-shifting and drives targeted fixes. DEEP DRILL: AI Behavioral Boundaries (3 Rules)*

FAIL Rules (78)

FAIL-SILENT-001
"Signal Health Score" per source. Track: last_successful_poll, signals_returned_count, error_count, consecutive_zero_result_polls. If any source returns zero results for 3 consecutive poll cycles, trigger alert: "SOURCE_POSSIBLY_DOWN."
Rationale: *The system cannot distinguish between "no news today" and "our connection is broken" without monitoring.*

FAIL-SILENT-002
Baseline signal volume thresholds: Google News: zero signals for 24h = alert. SAMA RSS: zero for 7 days = alert. LinkedIn: zero for 48h = alert. Overall: total signals <50% of 7-day average for 2 consecutive days = "SIGNAL_DROUGHT" alert.
Rationale: *Each source has different natural frequency. Thresholds must be source-specific.*

FAIL-SILENT-003
Every poll cycle must log: source, start_time, end_time, http_status, signals_found, errors. Even "zero results" is a valid log. Missing log = poll didn't run (different failure).
Rationale: *Distinguishes "polled and found nothing" from "poll never executed."*

FAIL-SILENT-004
"Stagnation Detection" — If a HIGH-FIT account (Static Fit Score >= 60) has been WATCHING with zero signals for 45 days, flag as "SIGNAL_DARK_ACCOUNT" for weekly human review. Human can: (a) manually research, (b) broaden keywords, (c) accept silence, (d) force-promote.
Rationale: *Good accounts can go quiet. The system must surface them rather than letting them rot.*

FAIL-SILENT-005
Every account has TWO scores: (1) DYNAMIC SCORE (0-100) — signal-driven, changes with events, drives tier assignment. (2) STATIC FIT SCORE (0-100) — ICP match regardless of signals. Calculated from: segment fit, size fit, geography fit, technology gap, regulatory exposure. Set at account creation, updated during enrichment.
Rationale: *Intent (dynamic) and fit (static) are orthogonal. A perfect-fit account with no signals is still worth monitoring.*

FAIL-SILENT-006
Static Fit Score governs: (a) account entry threshold (only Fit >= 40 enters system), (b) tie-breaking priority, (c) stagnation detection threshold (only >= 60), (d) enrichment depth (Fit >= 80 gets FULL enrichment even at WARM).
Rationale: *Not all accounts deserve equal system resources. Fit determines baseline investment.*

FAIL-SILENT-007
"Proactive Signal Generation" — For SIGNAL_DARK accounts with Static Fit >= 70, weekly expanded search: company name + "technology"/"digital"/"innovation", LinkedIn company page posts, job board scraping.
Rationale: *Standard polling may miss signals that broader searches catch. Weekly frequency avoids API overuse. — 1C. Failure Mode 2: False Signals (5 Rules) The system detects a signal that LOOKS like buying intent but isn't. Example: 'Digital transformation' article about an ATM upgrade, not a platform overhaul.*

FAIL-FALSE-001
Two-stage signal classification: Stage 1 — Keyword Detection (automated, catches potential signals). Stage 2 — Contextual Validation (AI reads full content and answers: "Does this indicate technology evaluation in categories Decimal serves?") Output: YES (proceed), MAYBE (proceed, confidence -0.2), NO (discard with log). Both stages must pass.
Rationale: *Keywords match context-free. AI validates actual meaning.*

FAIL-FALSE-002
Confidence score thresholds: 0.8-1.0 HIGH: full processing. 0.6-0.79 MODERATE: full processing, flag VERIFY_IF_POSSIBLE. 0.4-0.59 LOW: scoring only (reduced weight), NOT primary outreach trigger. 0.0-0.39 INSUFFICIENT: store for history, exclude from scoring.
Rationale: *Different confidence levels require different treatment.*

FAIL-FALSE-003
"Signal Retraction" workflow: A signal can be marked FALSE_POSITIVE by a human at any time. When retracted: (1) mark is_false_positive, (2) recalculate score without it, (3) if tier drops, downgrade, (4) if outreach was active, flag for human review, (5) if message already sent referencing this signal, prepare recovery brief for account_owner, (6) feed into AI classifier training.
Rationale: *False signals that trigger outreach can cause reputation damage. Recovery must be structured.*

FAIL-FALSE-004
Track "Signal Accuracy Rate" per source and signal type: (Total - false_positives - low_conf) / Total. Targets: >80% Google News, >90% SAMA RSS, >75% LinkedIn. Source dropping below target for 2 consecutive weeks = "SOURCE_QUALITY_REVIEW." Monthly report: accuracy breakdown matrix.
Rationale: *Quantitative tracking prevents gradual quality degradation.*

FAIL-FALSE-005
"False Signal Pattern Library" — After 90 days, analyze confirmed false positives and identify patterns. Initial seed: app redesigns vs platform-level change, marketing partnerships vs tech deals, branch staff hiring vs tech hiring, sanctions/AML vs banking tech mandates, competitor press releases disguised as industry news. Patterns become negative rules in AI classifier (confidence -0.3).
Rationale: *Systematic pattern recognition prevents repeated mistakes. — 1D. Failure Mode 3: Signal Detected Too Late (6 Rules) The signal was real and high-intent, but the system found it too late. The Golden Window closed. Competitors got there first.*

FAIL-LATE-001
"Time-to-Detection" metric: time between event's ACTUAL date and system's detected_at timestamp. Target: median 48h for P1 in any 30-day period, trigger polling frequency review and source expansion evaluation.
Rationale: *Detection delay directly reduces outreach effectiveness.*

FAIL-LATE-002
"Manual Signal Fast Lane" — Human-entered manual signals are PRE-VALIDATED at confidence 0.9, skip AI classification Stage 2, enter scoring immediately. 24-hour processing SLA.
Rationale: *The human IS the classifier. Manual intelligence from BD team and connectors is high-reliability and time-sensitive.*

FAIL-LATE-003
"Signal Source Expansion Roadmap" — Maintain prioritized backlog of new sources. MVP: Google News, SAMA, LinkedIn, Funding DBs, Manual. Phase 2: Job boards (Bayt.com), Saudi Gazette, Arab News, fintech publications, conference lists. Phase 3: Etimad tenders, App Store reviews, Glassdoor. Each evaluated on: unique signal coverage, false positive rate, API cost, implementation effort.
Rationale: *No single source catches everything. Systematic expansion reduces blind spots.*

FAIL-LATE-004
"End-to-End Pipeline SLA" — Total elapsed time from signal detection to message sent: P1 target <= 24h to review queue, <= 48h to sent. P2: <= 48h to queue, <= 72h to sent. P3: <= 5 business days. P4: no SLA. Weekly bottleneck report.
Rationale: *Each stage adds latency. Without end-to-end SLA, no one owns the total delay.*

FAIL-LATE-005
"Emergency Enrichment" for P1 signals on accounts with no existing contacts: Abbreviated 3-step process: (1) find top 2 contacts by seniority via Apollo, (2) verify emails, (3) generate minimal account summary. Target: under 2 hours. Full enrichment runs in background.
Rationale: *P1 signals on unenriched accounts would otherwise lose their Golden Window waiting for full 7-step enrichment.*

FAIL-LATE-006
"Stale Signal Outreach Decision" — If signal detected AFTER Golden Window closed: P1: still process, but AI must NOT reference as "recent" — frame as ongoing initiative. P2: process only if another FRESH signal exists. P3/P4: context only, no outreach trigger.
Rationale: *Late detection doesn't invalidate the signal, but messaging must be reframed to avoid looking out of touch. — 1E. Failure Mode 4: Signal Without Buying Intent (5 Rules) The signal is real and timely, but doesn't actually indicate technology procurement. 'Record profits' doesn't mean they're buying a lending platform.*

FAIL-INTENT-001
Signals classified on TWO dimensions: RELEVANCE (P1-P4, already exists) and INTENT: ACTIVE_PROCUREMENT (direct vendor evaluation evidence), BUILDING (internal tech build evidence), STRATEGIC_SHIFT (change preceding procurement), CONTEXTUAL (relevant info, no procurement intent). Only ACTIVE_PROCUREMENT and BUILDING can be P1. CONTEXTUAL is P3/P4 max.
Rationale: *Priority without intent creates false urgency.*

FAIL-INTENT-002
Intent-based scoring multiplier: ACTIVE_PROCUREMENT x1.5, BUILDING x1.2, STRATEGIC_SHIFT x1.0, CONTEXTUAL x0.5. A P2 ACTIVE_PROCUREMENT signal (25 x 1.5 = 37.5 pts) scores HIGHER than a P1 CONTEXTUAL (40 x 0.5 = 20 pts).
Rationale: *Intent quality matters more than signal category.*

FAIL-INTENT-003
"Account-Relative Intent" — Same signal gets different intent based on account readiness. Open banking mandate: account readiness 1-3 = ACTIVE_PROCUREMENT (must buy). Readiness 4-6 = BUILDING. Readiness 7-10 = CONTEXTUAL (already compliant). If readiness unknown, default to STRATEGIC_SHIFT.
Rationale: *Same regulation creates different urgency depending on the bank's current capabilities.*

FAIL-INTENT-004
"Intent-to-Conversion Analysis" every 90 days: for each intent level, what % led to meetings? Expected: ACTIVE_PROCUREMENT > BUILDING > STRATEGIC_SHIFT > CONTEXTUAL. If CONTEXTUAL converts within 50% of STRATEGIC_SHIFT, classification is too generous — recalibrate.
Rationale: *Intent labels must map to real-world outcomes or they're meaningless.*

FAIL-INTENT-005
"Signal Accumulation Rule" — 3+ CONTEXTUAL signals for same account within 30 days = auto-upgrade ONE to STRATEGIC_SHIFT.
Rationale: *Multiple contextual signals (profits + hires + conference + press) in short window suggest expansionary phase. Individually none implies buying. Together they suggest growth investment. Cap: once per 30-day window. — 1F. Failure Mode 5: Signal Overload (6 Rules) Too many signals. 20 accounts score HOT. Daily cap is 5-7. Human reviewer has 50 messages. Everything is urgent. Nothing gets attention.*

FAIL-OVERLOAD-001
"Surge Detection" — If P1+P2 signals in any 24h window exceed 3x the trailing 7-day daily average, declare "SIGNAL_SURGE": (a) increase daily cap from 5 to 10, (b) assign second reviewer, (c) log surge event, (d) revert when volume <1.5x average for 3 days.
Rationale: *SAMA regulatory announcements can trigger 15+ accounts simultaneously.*

FAIL-OVERLOAD-002
"Surge Prioritization Stack Rank" — When more HOT accounts than capacity: Rank 1: ACTIVE_PROCUREMENT intent. Rank 2: relationship_warmth >= WARM. Rank 3: highest Static Fit. Rank 4: highest Dynamic Score. Rank 5: earliest Golden Window expiry. Serve top of stack until capacity reached.
Rationale: *Not all HOT accounts are equally likely to convert.*

FAIL-OVERLOAD-003
"Reviewer Load Balancing" — Max 20 messages/reviewer/day. Beyond this, quality degrades. Queue exceeds (reviewers x 20) = "REVIEWER_OVERLOAD" alert. Options: activate backup, extend P3/P4 SLA from 4h to 8h, auto-approve P3/P4 if Phase 2+ automation active. P1/P2 and C-suite: NEVER skip human review.
Rationale: *Reviewer fatigue produces approval rubber-stamping.*

FAIL-OVERLOAD-004
"Surge Differentiation Rule" — When 5+ accounts triggered by SAME signal: AI receives explicit instruction to differentiate. QC Agent checks: "Does this message contain at least 2 sentences specific to THIS account?" Fail = regenerate with stronger account-specific context.
Rationale: *15 banks getting the same SAMA-triggered message with only the name changed looks like mass marketing.*

FAIL-OVERLOAD-005
"Capacity State Machine" — System operates in: NORMAL (standard caps), SURGE (elevated caps, triggered by FAIL-OVERLOAD-001), DROUGHT (expanded signal search, triggered when signals NORMAL requires 3 days sub-threshold.
Rationale: *Different operating modes require different system configurations.*

FAIL-OVERLOAD-006
"Surge Retrospective" — Within 7 days of surge ending: How many accounts served vs queued? Average queue wait time? Golden Windows missed? Reviewer turnaround? Were surge-period messages lower quality (lower reply rate)?
Rationale: *Post-surge analysis feeds capacity planning improvements. — 1G. Failure Mode 6: Signal Manipulation (4 Rules) Companies publish fake signals. A CEO's conference speech about 'digital transformation' may be PR with no budget, no project, no real initiative. Public relations is not buying intent.*

FAIL-MANIP-001
"Observability Score" — Every signal gets evidence-based corroboration assessment: Press release ONLY = 30. Press + hiring = 60. Press + hiring + leadership hire = 80. Press + hiring + leader + RFP = 95. Job postings WITHOUT press release = 70 (stealth build, often more genuine). SAMA mandate + bank below compliance = 90. Conference presentation only = 25.
Rationale: *Single-source PR is unreliable. Corroborating evidence from independent sources validates intent.*

FAIL-MANIP-002
If average Observability Score across active signals <40, ICS Verification dimension is capped at 30 regardless of email verification quality.
Rationale: *Verified email to the right person is worthless if the REASON for reaching out (the signal) is unverifiable PR.*

FAIL-MANIP-003
"Signal Maturation Tracking" — Low-observability signals (<=40) enter "WATCH_FOR_CORROBORATION" state. If corroborating signal from different source arrives within 60 days: upgrade observability to average of both, recalculate ICS, log maturation.
Rationale: *The system doesn't reject PR signals. It withholds judgment and waits for evidence.*

FAIL-MANIP-004
"Account Credibility Index" (after 12 months of data) — Track whether signals at each account lead to actual activity: (Signals leading to verified activity within 6mo) / (Total signals). Credibility >= 70%: +0.1 confidence bonus. 40-69%: neutral. <40%: -0.15 confidence penalty — "this bank talks more than it acts."
Rationale: *Some organizations are chronic PR-without-substance offenders. — 1H. Failure Mode 7: Dark Intent — Buying Without Signals (5 Rules) The most dangerous blind spot. A bank internally approved $5M for lending modernization. Nothing public. System sees intent = 0. Reality: intent = 100.*

FAIL-DARK-001
"Dark Intent Detection Framework" — 4 source categories: (1) Relationship Intelligence: manual entry by BD, confidence 0.80 (anonymous tips 0.50). (2) Conference Intelligence: manual with event/speaker/quote, confidence 0.75 (specific quotes 0.90). (3) Ecosystem/Subsidiary Signals: automated detection + manual linkage, confidence 0.65. (4) First-Party Behavioral: website analytics from identified contacts, confidence 0.70 (anonymous 0.40).
Rationale: *Dark intent can only be detected through human networks and behavioral signals, not public news.*

FAIL-DARK-002
Relationship intelligence gets FULL scoring weight (same as P1 ACTIVE_PROCUREMENT) BUT caps ICS Verification at 50 unless corroborating public signal arrives within 30 days.
Rationale: *Connector who says "they're buying" is usually right, but we can't independently verify a private conversation. High priority, moderate confidence.*

FAIL-DARK-003
"Manual Intelligence Incentive System": (a) every manual signal timestamped and attributed, (b) weekly dashboard showing entries per BD member and conversion rate, (c) attribution credit when manual signal leads to meeting within 60 days, (d) if total team entries <5/month, alert sales leadership, (e) post-conference/call Slack prompt within 24h: "Any intelligence to log?"
Rationale: *If humans forget to log dark intent, the system is blind.*

FAIL-DARK-004
"Connector Intelligence Protocol": (a) designated Decimal relationship manager per connector org, (b) monthly structured check-in: "Any lending/origination/onboarding initiatives at your Saudi clients?", (c) connector intelligence entered as CONNECTOR:[name] source, (d) +0.10 confidence boost over standard manual signals, (e) track connector reliability over time.
Rationale: *Connectors (ITC, Backbase, Tarabut, Geidea, KPMG, Mastercard) are Decimal's eyes inside banks.*

FAIL-DARK-005
"Post-Mortem: Lost Before We Knew" — When system discovers a bank selected a competitor for a deal never detected: log as MISSED_OPPORTUNITY with account, competitor, category, value, how_discovered. Root cause categories: No_Public_Signals / Signals_Missed / Detected_Not_Acted / Relationship_Gap. Quarterly review: how many missed? Estimated revenue lost? Most common root cause?
Rationale: *You can't fix what you don't measure. Missed opportunities are the most expensive failures. — 1I. Failure Mode 8: Signal Conflict (4 Rules) Multiple signals contradict each other: New CTO hired (positive) + Competitor signed (negative) + Funding round (positive) + Layoffs (negative). Current engine has no conflict resolution.*

FAIL-CONFLICT-001
3 conflict types: DIRECT_CONTRADICTION (same topic, opposite conclusions — most recent wins by default, flag for human verification), MIXED_SIGNALS (both opportunity and risk simultaneously — not a contradiction, AI must reason about Decimal-specific relevance), TEMPORAL_SHIFT (earlier signal suggested opportunity, later signal closed it — later signal takes priority).
Rationale: *Different conflict types require different resolution strategies.*

FAIL-CONFLICT-002
"Signal Arbitration" for MIXED_SIGNALS: Step 1: Tag each signal by Decimal product relevance (ORIGINATION/LENDING/ONBOARDING/WORKFLOW/OPEN_BANKING/GENERAL). Step 2: If negative signal specifically names competitor winning in Decimal's category = "BLOCKED." If competitor won different category = potential complementary opportunity. Step 3: Net positive signals >= 1 in Decimal's category = proceed. Net negative = "CONFLICTED_ACCOUNT" for human review.
Rationale: *A competitor winning core banking doesn't block Decimal's origination play.*

FAIL-CONFLICT-003
AI reasoning chain must acknowledge ALL signals for conflicted accounts. Agent 4 (Pain) must reason about how conflicting signals create different pains. Agent 5 (Strategy) must choose angle that acknowledges complexity. Agent 7 (QC) must verify: "Does this message reflect the FULL signal picture?" Cherry-picked messaging = QC REJECTS.
Rationale: *Pretending the competitor deal doesn't exist is worse than addressing it.*

FAIL-BIAS-001
"Outcome Attribution Framework" — When sequence completes without meeting, classify into: SIGNAL_MISS (signal was wrong), TIMING_MISS (too early/late), CONTACT_MISS (wrong person), MESSAGE_MISS (right everything, message didn't resonate), DELIVERY_MISS (never reached contact), EXTERNAL_BLOCK (everything right, external factor prevented response), UNKNOWN. Each feeds its OWN feedback loop. SIGNAL_MISS -> signal weights. CONTACT_MISS -> persona targeting. MESSAGE_MISS -> prompts. DELIVERY_MISS -> infrastructure. EXTERNAL_BLOCK -> NO negative feedback to any engine.
Rationale: *Each failure type has a different cause and solution. Contaminating feedback loops produces wrong learnings.*

FAIL-BIAS-002
"Automated Outcome Inference": No opens + no LinkedIn accept = DELIVERY_MISS or CONTACT_MISS. Opens but no clicks/replies = MESSAGE_MISS. Opens + clicks, no reply = EXTERNAL_BLOCK or MESSAGE_MISS. Negative reply "not interested" = SIGNAL_MISS or TIMING_MISS. "Not the right person" = CONTACT_MISS. "Bad timing, try later" = TIMING_MISS. Auto-reply + no follow-up = EXTERNAL_BLOCK. Only >=0.7 confidence classifications feed feedback loops. Lower = UNKNOWN for human review.
Rationale: *Automated inference covers the clear cases, human review handles the ambiguous.*

FAIL-BIAS-003
"Success Attribution Honesty" — When meeting IS booked, classify
Rationale: *SIGNAL_DRIVEN (signal was the reason), RELATIONSHIP_DRIVEN (warm intro was primary driver), TIMING_LUCK (contacted during existing evaluation by coincidence), PERSISTENCE_DRIVEN (no response to touches 1-3, responded to touch 4). WHY: Prevents the system from taking credit for luck and over-optimizing for non-causal factors.*

FAIL-BIAS-004
Revised Weight Recalibration: Only SIGNAL_DRIVEN meetings count for signal weight adjustments. RELATIONSHIP_DRIVEN meetings excluded from signal quality analysis. Each failure category feeds its own loop — CONTACT_MISS -> persona targeting, MESSAGE_MISS -> prompts, DELIVERY_MISS -> infrastructure. Loops never contaminate each other.
Rationale: *The key insight is that each failure type has its OWN feedback loop. — 1K. Failure Mode 10: Human Bottleneck (5 Rules) The system has 10+ human touchpoints. At each one, a human can become a bottleneck that delays or breaks the pipeline.*

FAIL-HUMAN-001
"Complete Human Dependency Map" — 10 touchpoints: Message Review (4h SLA), Signal Triage (24h), Stagnation Review (weekly), Conflict Resolution (48h), Handoff Response (2h), Manual Signal Entry (ongoing), QC Override (24h), Outcome Classification (monthly), Scoring Recalibration (quarterly), Surge Management (2h). Each has SLA, backup person, and documented consequence of failure.
Rationale: *You can't manage bottlenecks you haven't mapped.*

FAIL-HUMAN-002
"Human Resilience Principles": (a) No single point of human failure — every role has a backup, (b) escalation timers on EVERY touchpoint, (c) graceful degradation — if human fails SLA, system escalates/queues/proceeds with guardrails, doesn't stop, (d) "Human Health Score" — % of SLAs met across all touchpoints weekly. Target >= 85%. Below 70% = operational review.
Rationale: *The system must be designed to survive any single human failure.*

FAIL-HUMAN-003
"Reviewer Pool Architecture": Minimum 2 trained reviewers always. Training: 3 days (Day 1 = understanding AI chain, Day 2 = practicing with historical examples, Day 3 = live review with supervision). Certification: first 20 reviews audited, >= 90% agreement = certified. Vacation protocol: no reviewer goes on leave without backup confirmed active.
Rationale: *Reviewer quality directly determines outreach quality.*

FAIL-HUMAN-004
"Progressive Automation Roadmap": Phase 1 (MVP): all touchpoints human. Phase 2 (months 4-6): message review optional for P3/P4 non-C-suite, signal triage auto-resolves 90%+ fuzzy match, outcome classification auto-inferred for clear cases. Phase 3 (months 7-12): message review optional for P2 Director-level if thresholds met, conflict resolution gets AI recommendation with human confirm. NEVER automated: C-suite message review, handoff response, scoring recalibration, manual signal entry.
Rationale: *Gradual automation builds trust while reducing bottleneck risk.*

FAIL-HUMAN-005
"Bottleneck Early Warning System": Leading indicators: review queue depth >10 = AMBER, >20 = RED + activate backup. Average review time trending 50%+ above 30-day average = reviewer fatigued. Reviewer no reviews in 6+ hours during business hours = availability check. Manual signal entries declining 2+ months = intelligence flow drying up. Lagging indicators: messages expired from queue, Golden Windows missed, handoff SLA breaches.
Rationale: *Detecting bottlenecks early prevents pipeline stalls. — 1L. Module 0: Intelligence Governance Engine (11 Rules) Module 0 sits ABOVE all other modules. Every other module asks 'What should we DO?' Module 0 asks 'Can we TRUST what we think we know?' This is the meta-layer that prevents the system from acting...*

FAIL-MISATTR-002
Multi-Entity Signals: When signal mentions 2+ accounts, create ONE record per account with shared group_id. Signal type determined independently per account context. Partner mentions stored as metadata, not separate signals.
Rationale: *"SNB and Riyad Bank announce joint initiative" creates two signals, one per bank.*

FAIL-MISATTR-003
Misattribution Recovery: Re-map to correct account, recalculate BOTH accounts, check if outreach referenced the signal — if message said "your digital initiative" and it was actually the other bank's initiative = reputation-critical error, immediate human intervention. Log as ENTITY_RESOLUTION_ERROR, update aliases.
Rationale: *Sending SNB's initiative to Riyad Bank's CTO is permanently trust-destroying.*

FAIL-MISATTR-004
Pre-Outreach Entity Verification: QC CHECK ZERO (before all other QC checks): company name in message matches account name, signal referenced belongs to this account, no MULTI_ENTITY flags. Fail = QC REJECTS with "ENTITY_VERIFICATION_FAILED." Hard block, no override without human confirmation.
Rationale: *This is the most expensive possible error.*

FAIL-SCOPE-001
Account Relevance Scope: Every account has defined relevance_scope — business divisions Decimal is relevant to. Example: SNB = ["Retail Banking", "SME Lending", "Digital Banking", "Open Banking"]. Out-of-scope signals stored but EXCLUDED from scoring.
Rationale: *SNB's insurance technology investment shouldn't inflate scoring for Decimal's lending pitch.*

FAIL-SCOPE-002
Contact Relevance Filter: Only contacts in scope divisions join active buying committee. "Head of Treasury Technology" at a retail-banking-scoped account = stored but OUT_OF_SCOPE for outreach. Exception: C-suite always in scope.
Rationale: *Treasury technology contacts receiving lending platform outreach is irrelevant and unprofessional.*

FAIL-SCOPE-003
Scope Expansion Protocol: Changed only through: (a) manual update with documented reason, (b) product launch batch-update for all applicable accounts. Changes logged, trigger: re-evaluation of OUT_OF_SCOPE signals/contacts, score recalculation.
Rationale: *Scope expansion is strategic, not automatic.*

FAIL-OWNER-001
Owner Load Limits: Max 10 HOT accounts per owner (require active management). Max 20 WARM. Owner at capacity + new HOT account = assign to next available. All owners at capacity = "CAPACITY_OVERFLOW" alert, max 48h wait for HOT accounts.
Rationale: *Overloaded owners become bottlenecks across all their accounts.*

FAIL-OWNER-002
Owner Transition Protocol: Owner leaves/reassigned → 24h TRANSITION_HOLD (no new outreach, active sequences continue). Leadership reassigns within 24h based on expertise + load + relationship continuity. New owner gets Transition Brief per account. ENGAGED contacts flagged: "ACTIVE CONVERSATION — review immediately."
Rationale: *Orphaned accounts with active conversations lose prospects.*

FAIL-OWNER-003
Owner Override Rules: CAN: delay outreach up to 14 days with reason, reprioritize contacts, edit messages, add manual signals. CANNOT: permanently suppress HOT account without leadership approval, change scoring formula, override QC rejections, delete records. All overrides logged with type, reason, duration, approved_by.
Rationale: *Owners need flexibility — but not unchecked power.*

FAIL-FATIGUE-001
Account Fatigue Score = (completed sequences with no positive reply) x 10 over rolling 12 months. 0-20: normal. 30-40: CAUTION, next outreach requires owner justification. 50+: FATIGUED, 180-day reset, re-engagement only through different channel or connector introduction. Score resets to 0 only after successful engagement.
Rationale: *Repeated unsuccessful outreach annoys prospects and damages reputation.*

FAIL-FATIGUE-002
Contact Fatigue: Max 3 lifetime sequences per contact. After 3 with no positive reply = "OUTREACH_EXHAUSTED," permanently removed from outreach pool. Exception: contact initiates contact with Decimal (inbound) = fatigue cleared, one new sequence allowed.
Rationale: *Enterprise professionals talk. A 4th campaign will create active anti-Decimal sentiment.*

FAIL-FATIGUE-003
Outreach Freshness Requirement: New sequence requires at least ONE of: (a) NEW signal since last sequence, (b) DIFFERENT contact targeted, (c) DIFFERENT outreach angle (AI confirms substantially different), (d) connector warm introduction. None met = sequence blocked.
Rationale: *"We have no new reason to reach out" means don't reach out.*

FAIL-FATIGUE-004
Nurture Fatigue: WARM accounts max 1 nurture touch/month. 6 consecutive zero-engagement touches = "NURTURE_EXHAUSTED," removed from nurture. Re-enters only if promoted to HOT then back to WARM.
Rationale: *Monthly emails to someone who never opens is spam with extra steps.*

FAIL-DECAY-001
Data Decay Rates: Contact role/title: 12-18 months. Email: 6-12 months (re-verify before each sequence). Strategic priorities: 3-6 months. Digital maturity: 6-12 months. Technology stack: 6-12 months. Buying committee composition: 3-6 months. Competitor landscape: 3-6 months.
Rationale: *KSA banking executives change roles every 1-3 years. Data without refresh becomes fiction.*

FAIL-DECAY-002
Pre-Sequence Data Validation: Before launching ANY new sequence: (a) verify contact still holds same role (LinkedIn check), role changed = STOP + re-enrich, left company = DISQUALIFIED + find replacement, (b) re-verify email, INVALID = LinkedIn-only, (c) referenced signal not contradicted by newer info.
Rationale: *"Pre-flight check" takes minutes but prevents messaging people who no longer work there.*

FAIL-DECAY-003
Data Staleness Threshold: If most recent enrichment >120 days AND no new signal in 60 days: ICS Data Completeness decays to 50%, Verification decays to 40%. Combined effect: ICS drops significantly, potentially triggering GOV-001 investigate-only gate.
Rationale: *System says "we used to know about this account, but our knowledge is too old to trust." — 2L. Reply-Based Account Pause Logic (4 Rules)*

FAIL-RUBBER-001
"Anti-Rubber-Stamping Mechanisms": (a) Minimum review time tracking: <30 seconds for email or <15 for LinkedIn = "SPEED_REVIEW" flag (meaningful review takes 60-90 seconds). (b) Spot-check canaries: weekly, system inserts deliberately flawed message (wrong company name, non-existent signal). Approved canary = "CANARY_MISSED" alert. Flaw caught before sending. Reviewer notified. (c) Review Quality Score per reviewer: canary catches, edit rate, time per review, agreement with post-hoc audits. (d) Rotation: reviewers don't review same account repeatedly. Monthly rotation.
Rationale: *After 50 good messages, reviewer assumes #51 is fine. Canaries test vigilance.*

FAIL-RUBBER-002
"Review Quality Culture": Canary system framed as QUALITY TOOL not surveillance: (a) all reviewers told upfront during training, (b) results private, not shared with team, (c) missed canary = coaching not discipline, (d) Review Quality Score for training needs, not performance evaluation, (e) canary frequency decreases as reviewer quality improves.
Rationale: *Surveillance-framed systems create resentment and gaming. Quality-framed systems create improvement. FM2 — AI Hallucination (2 Rules)*

FAIL-HALLUC-001
"Three-Layer Hallucination Prevention": Layer 1 — Generation Constraint: Agent 6 explicitly instructed to use ONLY facts from enrichment, signals, and Approved Claims Library. Unverifiable = "CLAIM_UNVERIFIED" placeholder. Layer 2 — Automated Fact-Check: before Agent 7, cross-reference every entity name, number, date, claim against databases. Unmatched claims flagged with specific text highlighted. Layer 3 — Human Review: flagged claims shown in RED. Unflagged in normal text. Visual hierarchy focuses human attention on highest-risk parts.
Rationale: *Triple-layer means hallucination must fool generator, fact-checker, AND human.*

FAIL-HALLUC-002
"Hallucination Incident Response": If hallucinated claim IS sent: Step 1: account owner notified within 1 hour. Step 2: assess — did prospect notice? Step 3: if noticed, owner contacts personally, acknowledges error honestly, provides correct info. Step 4: incident logged as "HALLUCINATION_BREACH" with which layer(s) failed + root cause. Step 5: add hallucination pattern to fact-checker negative rules.
Rationale: *Honest correction builds more trust than silence. FM3 — AI Staleness (2 Rules)*

FAIL-STALE-001
"AI Currency Maintenance Schedule": Approved Claims Library: monthly (triggered by product launch, client win, certification, pricing change). Persona Prompts: quarterly (triggered by persona mismatch feedback). Strategy Prompts: monthly (triggered by market/competitive changes). Cultural Rules: quarterly (triggered by policy changes, lessons learned). Competitive Intelligence: continuously. Signal Classification: monthly (triggered by false positive thresholds). QC Criteria: quarterly (triggered by systematic failures). All version-controlled. Rollback within 1 hour if degradation.
Rationale: *AI that doesn't evolve becomes irrelevant as markets change.*

FAIL-STALE-002
"AI Performance Degradation Detection": Track monthly — thresholds for "STALE_AI" trigger: QC approval rate drops 10+ points from 90-day average, human override rate increases 10+ points, reply rate declines 30%+ (controlling for seasonality), average edit distance increasing, MAJOR_REWRITE rate >20% for any category. STALE_AI triggered: emergency prompt review, enrichment audit, competitive check, claims library verification. Fix within 7 business days.
Rationale: *Gradual degradation is invisible reviewing individual messages. Systematic measurement catches decline. FM4 — Human Over-Reliance (1 Rule)*

FAIL-OVERRELY-001
"Human Independent Judgment Preservation": (a) Monthly "Red Team": owners independently rank top 5 accounts WITHOUT seeing system ranking first, then compare. Divergence is GOOD. Perfect alignment after 6+ months = possible over-reliance. (b) AI Confidence Display: always show confidence level with recommendations. Never present as certainties. (c) Override encouragement: if owner NEVER overrides in 90 days, manager conversation: genuinely agreeing or defaulting? (d) Market Sensing Reports: monthly BD team report based on THEIR conversations, independent of AI analysis. Compare "what market tells us" vs "what AI tells us."
Rationale: *An AI-dependent team cannot function during an outage. FM5 — Wrong Metric Optimization (2 Rules)*

FAIL-WRONGMETRIC-001
"Metric Hierarchy": NORTH STAR: pipeline revenue from ABM-sourced opportunities. PRIMARY: meetings booked, positive reply rate. SECONDARY: total reply rate, signal-to-meeting conversion. OPERATIONAL: open rate, click rate, LinkedIn accept rate, QC pass rate. AI must NEVER optimize operational at expense of primary. A change that increases open rates but decreases positive reply rates is BAD. Feedback loops ONLY recalibrate based on primary/north star. Operational metrics for diagnostics only.
Rationale: *Optimizing vanity metrics produces impressive dashboards and empty pipeline.*

FAIL-WRONGMETRIC-002
"Metric Manipulation Detection": Divergence between operational and primary = warning: open rate up but reply flat = misleading subjects. QC pass up but override also up = QC weakened. LinkedIn accept up but DM reply flat = connection good, follow-up bad. Monthly check required.
Rationale: *Metrics that improve independently of outcomes indicate gaming or miscalibration. FM6 — AI Memory Leakage (2 Rules)*

FAIL-LEAK-001
"Account Isolation Principle": Each reasoning chain receives ONLY that account's data: enrichment, signals, contact personas, plus universal Approved Claims Library and general strategy guidelines. Must NOT receive other accounts' data, engagement outcomes, or response patterns.
Rationale: *What works at SNB may fail catastrophically at Al Rajhi. Each bank has its own culture, priorities, and politics.*

FAIL-LEAK-002
"Cross-Account Learning Rules": Patterns CAN be learned across accounts if: (a) derived from 5+ accounts, (b) about PERSONA TYPES not specific people, (c) about MESSAGE CHARACTERISTICS not specific content, (d) applied as TENDENCY not hard rule. ALLOWED: "CTOs tend to prefer messages under 100 words." NOT ALLOWED: "CTO at Bank A liked compliance pitch — use for Bank B."
Rationale: *Aggregate behavioral patterns are useful. Specific account strategies are not transferable. DEEP DRILL: Trust Progression Model (1 Rule)*

FAIL-HUMAN-004-REV
CONFLICT RESOLUTION (FIN-C07): The Progressive Automation Roadmap (FAIL-HUMAN-004) is subordinate to TRUST-001 regression. Automation phase is a function of current Trust level, not elapsed calendar time. Any trust-regression event immediately reverts the corresponding automation phase by one level for the affected message categories, regardless of how many months have elapsed. Calendar months are the earliest-possible schedule, never a guarantee. Rationale: gives the forward-only roadmap an explicit reverse gear tied to trust.

GOVERNANCE Rules (4)

GOVERNANCE-001
"System Governance Structure": Three layers: OPERATIONAL (Daily): owners + reviewers — message quality, engagement, day-to-day. Report to BD leadership. STRATEGIC (Monthly): BD leadership — portfolio performance, AI quality, learning pipeline, competitive landscape, capacity. Decides AI changes, account universe, resources. ETHICAL (Quarterly): BD leadership + legal + external advisor — system behavior vs ethical boundaries, PDPL, data practices, AI transparency, incidents.
Rationale: *Different oversight frequencies for different risk horizons.*

GOVERNANCE-002
"Quarterly System Audit": 6 dimensions: (a) AI quality: QC rates, hallucination incidents, override patterns. (b) Outreach ethics: complaints, opt-out compliance, data handling. (c) Human operations: reviewer quality, fatigue indicators, skill preservation. (d) System integrity: kill switch TESTED (yes, actually test quarterly), backup restoration verified, data externalization current. (e) Financial: AI costs vs budget, cost per opportunity, ROI. (f) Competitive: market position changes, competitor movements, reference pipeline. Results documented. Action items tracked. Repeat items = escalation.
Rationale: *Comprehensive quarterly health check prevents gradual system degradation.*

GOVERNANCE-003
"Bias Monitoring": Quarterly analysis: (a) Gender: response rates by contact gender significantly different? Messaging inadvertently gendered? (b) Segment: systematically favoring certain bank types? Islamic banks getting less attention despite similar fit? (c) Seniority: appropriately distributing outreach or always defaulting to C-suite? (d) Recency: overly weighting recent signals while ignoring established accounts? Statistically significant bias detected → investigate root cause (weights? prompts? enrichment?) and correct.
Rationale: *Undetected bias in an automated system scales bias to every outreach the system sends.*

GOVERNANCE-004
"Cross-Principle Consistency Check": Quarterly, verify that rules from Principle 3 don't conflict with Principles 1, 2, 4, or 5. As the rule set grows, contradiction risk increases. Example: Principle 1 says "respond immediately to P1 signals" but Principle 3 says "every message needs 4-hour human review" — what happens when P1 arrives at 10pm? Resolution: processing is immediate (P1), human review happens next business morning (P3), message is in Ready-to-Fire queue by Sunday 9am. Both principles satisfied through queue architecture. Document all such tensions and their resolutions.
Rationale: *Principles that contradict each other in practice produce unpredictable system behavior. ITERATION 4 — TRUST ECONOMICS LAYER Trust is not binary. Trust is an economic asset. It accumulates through deposits (successful outcomes) and depletes through withdrawals (failures). The withdrawal rate is always ...*

HUMAN Rules (4)

HUMAN-AUTO-001
Three-phase human review reduction: Phase 1 (Days 0-90): 100% human review. Phase 2 (91-180): Optional for P3/P4 + non-C-suite if QC rate >85% and override 92% and override <5%. C-suite messages: ALWAYS human reviewed, permanently.
Rationale: *C-suite errors are reputation-destroying. Lower-risk messages can be automated once trust is established.*

HUMAN-AUTO-002
Escalation chain for message review: Hour 0-4: Primary reviewer (account_owner). Hour 4-8: Backup reviewer (buddy). Hour 8-24: Team lead. Hour 24+: Message expires, must be regenerated with fresh context.
Rationale: *Context goes stale. "This week's announcement" becomes last week's after 48 hours.*

HUMAN-AUTO-003
Track edit distance on human edits: MINOR_TWEAK 50%. If MAJOR_REWRITE rate exceeds 30% for a signal_type or persona_type, trigger prompt optimization review.
Rationale: *High rewrite rates mean the AI is failing for that category — fix the prompts, not the individual messages.*

HUMAN-AUTO-004
Human-added factual claims must be logged separately from AI-generated claims.
Rationale: *The system validates AI claims against enrichment data. Human additions are not validated but must be tracked for audit purposes. Timing Windows & Volume Controls*

HUMANFATIGUE Rules (2)

HUMANFATIGUE-001
"Decision Load Balancing": (a) Max 10 sequential decisions without break — system prompts 5-minute break reminder. (b) Highest-stakes decisions in morning (9-11am): P1, C-suite, handoffs. P3/P4 in afternoon. (c) Complexity variation: don't present 10 conflicted accounts in a row. (d) Queue visibility: reviewer always sees queue depth. Surge = load balancing flag.
Rationale: *Decision quality declines measurably throughout the day. System design should account for human biology.*

HUMANFATIGUE-002
"Decision Quality Correlation": Track: time of day vs approval rate (98% after 3pm but 82% before noon → afternoon rubber-stamping), queue depth vs review time (drops when deep → rushing), day of week vs override rate (Friday quality drop → end-of-week fatigue). Used to ADJUST SYSTEM not penalize humans: shift P1/P2 to mornings, add afternoon reviewer, reduce Friday load.
Rationale: *Optimizing the human is as important as optimizing the AI.*

HUMANMEMORY Rules (2)

HUMANMEMORY-001
"Account Strategy Notes": Every owner maintains written notes per HOT account, updated monthly: current strategic assessment (own words, not AI), learnings from direct conversations, relationship dynamics observed, decision rationale for overrides/delays, prediction of what happens next and why. Stored alongside AI intelligence. Part of handoff package. Reviewed in quarterly strategy sessions.
Rationale: *When owner leaves, human intelligence about politics and dynamics leaves too — unless written down.*

HUMANMEMORY-002
"Post-Meeting Intelligence Capture": Within 24 hours of every meeting/significant conversation, owner logs: topics discussed, prospect's stated priorities and concerns, political dynamics observed, next steps agreed, owner's deal assessment, budget signals, competitive signals. Feeds into: enrichment updates, consensus scoring, risk assessment, AI context for future messages.
Rationale: *A 30-minute meeting generates more intelligence than 6 months of signal monitoring. ITERATION 3 — ADVERSARIAL, CATASTROPHIC & ETHICAL LAYER What happens when things go catastrophically wrong? Not 'slightly generic message' but 'reputation crisis,' 'system being gamed,' or 'ethical dilemma.'*

HUMANREVIEW Rules (5)

HUMANREVIEW-001
"Accountability Attachment Record": Every outbound action stores: AI_RECOMMENDED_BY (chain version), REVIEWED_BY (human name + timestamp), ACCOUNT_OWNER (ultimately responsible human), DECISION (approve/edit/reject), REASONING (linked to HUMANDEC-001 taxonomy). Record is IMMUTABLE — cannot be edited after the fact. Answers: "When this message was sent to SNB's CEO, who approved it and why?"
Rationale: *AI cannot appear in a meeting to explain why it sent a message. A human can. Accountability attaches to persons, not systems.*

HUMANREVIEW-002
"Strategic vs Execution Review": STRATEGIC (higher stakes, less frequent): covers account targeting, stakeholder selection, outreach narrative, competitive positioning, timing strategy. Reviewer: owner or BD leadership (Level 2-3). Frequency: once per account per outreach cycle. EXECUTION (lower stakes, more frequent): individual message quality, wording, CTA. Reviewer: any certified (Level 1+). Frequency: per message during MVP, reducible per trust. CRITICAL: Strategic review is NEVER automated — even at FULL PARTNERSHIP, human always approves account STRATEGY. Execution review within approved strategic boundaries can auto-send if trust permits.
Rationale: *Getting strategy wrong means EVERY message is wrong. This is the high-leverage review.*

HUMANREVIEW-003
"Reasoning Checkpoint Access": Reviewer interface allows reviewing ANY intermediate step: expandable Agent 1 → 2 → 3 → 4 → 5 → 6 → 7. DEFAULT view: final message + strategy summary + signal reference (fast routine review). EXPANDED view: full chain (when something feels off). MANDATORY expansion: C-suite contact, SIGNAL_CONFLICT flag, ICS <60, first message to new account, Verifiability Score <50.
Rationale: *Reviewing only the final message is reviewing the symptom. Reviewing reasoning is reviewing the cause.*

HUMANREVIEW-004
"Reviewer Competency Tiers": Level 1 OPERATIONAL: P3/P4 to Director and below. Requires 4-week onboarding + certification (20 audited reviews, >=90% agreement). Level 2 STRATEGIC: P1/P2, VP contacts, strategic reviews, conflicted accounts. Requires Level 1 experience (100+ reviews) + demonstrated strategic judgment + BD experience. Level 3 EXECUTIVE: C-suite messages, Domino Account outreach, market strategy, override approvals. Requires senior BD experience + deep KSA banking knowledge + multi-account relationship context. Messages routed to appropriate level. C-suite WAITS for Level 3 — better to delay than have junior reviewer approve CEO message.
Rationale: *Not all reviews are equal. Not all reviewers are qualified for all reviews.*

HUMANREVIEW-005
"Review Step Sunset Assessment": Every 6 months, per message category (signal priority x contact seniority x trust level): (a) VALUE_ADDED rate from HUMANDEC-005, (b) BLOCKED_HARM rate, (c) cost of this review step. VALUE_ADDED >20% OR BLOCKED_HARM >2% → JUSTIFIED, keep. VALUE_ADDED 5-20% AND BLOCKED_HARM <2% → QUESTIONABLE, consider spot-check (30% random). VALUE_ADDED <5% AND BLOCKED_HARM <1% → NOT JUSTIFIED, candidate for removal per trust progression. Exception: ANY category involving C-suite or P1 → ALWAYS justified regardless of metrics.
Rationale: *Review that never changes outcomes is overhead. But some categories must stay reviewed because the consequence of the ONE error is too high. ITERATION 8 — WHY AI REASONS THE WAY IT DOES Good outcome does not equal good reasoning. Bad outcome does not equal bad reasoning. An AI that produces a meeting through lucky reasoning will produce a dis...*

HUMANSKILL Rules (3)

HUMANSKILL-001
"Manual Capability Preservation": Every system operator must demonstrate WITHOUT AI: (a) research account from public sources, write 1-page intelligence brief, (b) write personalized outreach email from scratch, (c) manually calculate account score, (d) prioritize 10 accounts for outreach, (e) conduct meeting preparation without AI brief. Skills assessment: every 6 months, real accounts.
Rationale: *AI is a capability AMPLIFIER. If humans can't function independently, the amplifier produces zero when it breaks.*

HUMANSKILL-002

"Manual-First Onboarding": Week 1: operate manually — research without AI summaries, write without AI drafts, score by hand. Week 2: shadow AI — see AI outputs, compare to manual work, understand strengths/weaknesses. Week 3: operate WITH AI — review, approve, manage accounts. Week 4: certified reviewer. Skip weeks 1-2 for experienced enterprise sales hires, but certification still required.

Rationale: *Must understand what AI does before judging whether it does it well.*

HUMANSKILL-003

"Knowledge Externalization": Critical intelligence must NOT exist only in AI system: (a) account summaries exportable to PDF/DOCX, (b) contact data synced to CRM (HubSpot is backup), (c) signal history exportable to CSV/JSON, (d) scoring history exportable, (e) prompt library in version-controlled docs, (f) Approved Claims Library as standalone document. Monthly backup verification: can we reconstruct top 10 accounts from external backups alone? If NO → externalization insufficient.

Rationale: *AI systems aren't permanent. Organizational knowledge must survive any single system failure.*

ICS Rules (8)

ICS-CALC-001

ICS (0-100) calculated from 6 weighted dimensions: Signal Diversity 20%, Signal Recency 15%, Data Completeness 20%, Contact Coverage 15%, Verification Level 15%, Source Reliability 15%.

Rationale: *Intelligence reliability is multi-dimensional. Strong in one area doesn't compensate for weakness in another.*

ICS-DIV-001

Signal Diversity scoring: 0 sources = 0pts. 1 source = 20pts (single source = unreliable). 2 sources = 50pts. 3 sources = 75pts. 4+ sources = 100pts.

Rationale: *Multi-source triangulation is fundamentally more reliable than single-source intelligence.*

ICS-REC-001

Signal Recency scoring: within 7 days = 100pts. 8-21 days = 70pts. 22-45 days = 40pts. 46-90 days = 15pts. No signal or 90+ days = 0pts.

Rationale: *Old intelligence is unreliable. A bank's priorities can shift in weeks.*

ICS-COMP-001

Data Completeness checklist (each item adds points): Company name+segment+country: +10. AI account summary: +15. Digital maturity assessed: +10. Open banking readiness: +10. Strategic initiatives (>=2): +15. Technology stack known: +10. Procurement complexity: +10. Competitive landscape: +10. Recent news within 30 days: +10. Total: 100.

Rationale: *Every missing data point is a gap the AI has to guess around.*

ICS-CONTACT-001

Contact Coverage: 0 contacts = 0pts. 1 = 15. 2 = 30. 3 = 50. 4-5 spanning 2+ departments = 75. 6+ spanning 3+ departments including C-level = 100.

Rationale: *Enterprise deals require multi-threaded engagement. Single contact = single point of failure.*

ICS-VERIF-001

Verification from: verified emails (up to 40pts), signals from high-reliability sources (up to 30pts), enrichment from premium sources (up to 30pts). Formula: (verified_email% x 0.4 + verified_signal% x 0.3 + premium_enrichment% x 0.3) x 100.

Rationale: *Unverified data is unreliable data.*

ICS-SOURCE-001

Source reliability ratings: SAMA Official 0.95, established news 0.85, Google News general 0.70, LinkedIn company page 0.75, LinkedIn personal 0.60, funding DBs 0.80, verified manual BD 0.85, unverified rumor 0.40. Account ICS Source — Reliability = average reliability across all signals x 100.

Rationale: *Not all sources are equally trustworthy.*

ICS-DECAY-001

ICS decays by 2 points per week after last enrichment or signal ingestion if no new data arrives.

Rationale: *Stale confidence is false confidence. Even if nothing has been contradicted, understanding degrades over time.*

LAW2 Rules (2)

LAW2-EXCEPTION-001

Decision Complexity Classification: SINGLE_DECISION_MAKER (<100 employees, CEO decides) = 1 contact sufficient. SMALL_COMMITTEE (100-500, 2-3 deciders) = 2 contacts min. STANDARD_COMMITTEE (500-5000, 5-7 deciders) = standard rules. COMPLEX_COMMITTEE (5000+, 10+ stakeholders, procurement-driven) = 3+ contacts, human-led early.

Rationale: *Treating a 30-person fintech like a 30,000-person bank wastes resources.*

LAW2-EXCEPTION-002
Lightweight Pipeline for SINGLE_DECISION_MAKER: abbreviated enrichment (CEO only), abbreviated AI chain (skip persona psychology + multi-angle strategy), shorter sequence (3 touches), binary reachability (can reach CEO or can't).
Rationale: *Speed matters more for startups. CEO who raised funding makes vendor decisions in 30 days.*

LAW4 Rules (2)

LAW4-EXCEPTION-001
Personal vs Organizational Signal Scope: Individual-origin signals (LinkedIn post, conference quote) tagged "PERSONAL." ORGANIZATIONAL signals drive outreach to any contact. PERSONAL signals primarily to the person who generated it. DON'T tell CDO "I saw your CTO's LinkedIn post."
Rationale: *Outing colleague's social media to another executive violates internal politics. —*

LAW4-EXCEPTION-002
2+ personal signals on same topic from different individuals within 30 days = auto-upgrade to ORGANIZATIONAL.
Rationale: *Multiple people independently signaling = organizational-level activity.*

LAW5 Rules (2)

LAW5-EXCEPTION-001
Contact Engagement Heatmap: Per-contact engagement breakdown maintained alongside account-level score. Shows which contacts most responsive. Feeds human strategy and persona performance analytics. Does NOT independently trigger state changes.
Rationale: *Contact data INFORMS humans but doesn't drive system actions.*

LAW5-EXCEPTION-002
Contact Velocity Alert: Single contact with rapidly escalating engagement (open -> click -> website -> reply in 72h) creates CONTACT_VELOCITY_ALERT for account_owner. Does NOT change account score/state.
Rationale: *Operational signal for human to prioritize.*

LAW6 Rules (1)

LAW6-EXCEPTION-001
Contact-Focal Moments: Temporary zoom-in events when one contact becomes primary unit: meeting booked (AI brief focuses on this person), referral received (outreach crafted for referred person), connector introduction (messaging optimized for this person), procurement RFP (response formatted for procurement contact). Temporary, not architectural.
Rationale: *Strategy is account-level. Execution has moments when one person is the game.*

QUEUE Rules (1)

QUEUE-ARCH-001
CONFLICT RESOLUTION (FIN-C05): P1 processing (scoring, enrichment, message generation, QC) is immediate and runs 24/7 regardless of business hours. Human review and SEND are decoupled from processing: a QC-passed message enters the Ready-to-Fire Queue and waits for (a) human review per the active Trust level and (b) the next valid KSA send window. A P1 signal at 11pm Thursday is fully processed by ~1am, reviewed Sunday morning, and sent in Sunday 9am-12pm. EVENT-RESP-001 and HUMAN-AUTO-002 both defer to this queue. Rationale: encodes GOVERNANCE-004 so immediate-processing and mandatory-review no longer collide.

TEMPORAL Rules (5)

TEMPORAL-DECISION-001
"Timing Paradox Rule": For timing-sensitive decisions calculate: COST_OF_ACTING_NOW (risk of premature × impact of TOO_EARLY) vs COST_OF_WAITING (risk of missing window × impact of TOO_LATE + daily information decay). Waiting > Acting → ACT NOW, accept uncertainty. Acting > Waiting → WAIT with maximum wait period. P1: cost of waiting very high, default ACT NOW. P2: can wait 48-72h for better intelligence. P3/P4: can wait for full enrichment.

TEMPORAL-DECISION-002
"Window State Confidence": HIGH >=80% (multiple corroborating signals, act on it). MODERATE 50-79% (some signals, act with lighter touch — positioning or exploratory). LOW <50% (ambiguous, do NOT launch full outreach, investigate first). Display confidence alongside state: "Window: OPENING (62%). Basis: new CTO + Q1 timing. Missing: budget confirmation."

TEMPORAL-DECISION-003
"Cross-Account Timing Coordination": Peer group / shared board accounts: (a) sequence strategically — stronger opportunity first, positive signal becomes social proof for second. (b) Domino Account timing takes priority. (c) NEVER same-signal outreach to two connected accounts on same day — stagger 3+ business days. (d) Referral from Account A to Account B: approach B within 48h while endorsement is fresh.

TEMPORAL-DECISION-004
"Internal Readiness Gates": External timing (window OPEN) must align with internal readiness across 6 dimensions: enrichment sufficient (ICS >=40), message quality (QC passed), reviewer available, account owner assigned, competitive intelligence known, relevant collateral exists. Each not-ready has specific response: emergency enrichment, fix message, queue for reviewer, assign owner, flag blind spot, proceed anyway for P1/P2 but prepare collateral in parallel.

TEMPORAL-DECISION-005
"Timing Irreversibility Assessment": LOW (LinkedIn connection — ignorable, recoverable): confidence >=50% sufficient. MEDIUM (email to Director — they'll delete and forget): confidence >=65%. HIGH (email to C-suite — they REMEMBER): — confidence >=80%. EXTREME (connector introduction — damages both relationships if wrong): confidence >=85% AND Level 2+ reviewer. Higher cost of mistake → higher confidence required. Iteration 3: Timing Failure Cascades (6 Rules)

TIMING Rules (18)

TIMING-WIN-001
Each signal type has a "Golden Window" — the period where outreach has maximum impact: LEADERSHIP_HIRE: Days 7-30. REGULATORY: Days 1-14. DIGITAL_INITIATIVE: Days 1-21. FUNDING: Days 14-45. HIRING_SURGE: Days 1-30.
Rationale: *Each event type has a natural lifecycle. Too early = premature. Too late = the window closed.*

TIMING-WIN-002
If a message cannot be sent within the Golden Window (stuck in review, holiday delay), it must be REGENERATED with updated framing.
Rationale: *A message saying "your recent announcement" 40 days later is embarrassing.*

TIMING-VOL-001
The 5-accounts/day outreach limit is based on: (a) human review capacity (~10-15 messages/day), (b) email deliverability (new domain warmup), (c) MVP quality assurance. Review monthly. Increase by 2/day/month up to max 15/day.
Rationale: *Gradual scaling prevents quality collapse and deliverability damage.*

TIMING-VOL-002
When >5 accounts qualify for outreach on the same day, rank by: (1) Signal priority P1 first, (2) Account score highest first, (3) Relationship warmth warmest first. Top 5 enter outreach. Remainder go to PRIORITY_QUEUE, max 3 business days wait.
Rationale: *Prioritization ensures best opportunities get served first.*

TIMING-VOL-003
P1 Golden Window expiry OVERRIDES daily volume cap via "EMERGENCY_SLOT" — up to 2 emergency slots/day on top of regular 5. P2/P3/P4 do not get emergency slots. Max total: 7 accounts/day.
Rationale: *Missing a P1 Golden Window is a higher cost than slightly exceeding volume limits. AI Reasoning Chain*

TIMING-PHILOSOPHY-001
"Window-Based Timing Model" — 5 states per account: WINDOW_CLOSED (no evaluation, outreach wasted), WINDOW_OPENING (early signals, PREPARE — enrich, identify committee, draft messages, but do NOT outreach yet), WINDOW_OPEN (active evaluation, EXECUTE — every day matters, launch outreach), WINDOW_CLOSING (shortlist formed, URGENT ENTRY — if not already in conversation, attempt with high-value differentiator), WINDOW_CLOSED_AGAIN (decision made, STOP — log outcome, begin cooldown). Every signal interpreted for where it places account in window cycle. KEY INSIGHT: prepare during OPENING, execute during OPEN. Most competitors detect OPEN and start preparing then — Decimal should already be in conversation.

TIMING-PHILOSOPHY-002
"Window State Inference" — Signal patterns mapped to states: CLOSED = no signals, stable. OPENING = first change signal (new leader, initiative, regulatory). OPEN = multiple corroborating signals (budget confirmed, hiring, vendor meetings). CLOSING = advanced signals (shortlist, procurement). CLOSED_AGAIN = decision signal. System response per state: CLOSED → monitor. OPENING → prepare. OPEN → execute immediately. CLOSING → urgent entry if viable. CLOSED_AGAIN → stop.
Rationale: *The system that prepares during OPENING and executes during OPEN beats the system that starts preparing during OPEN.*

TIMING-FAIL-001
"Four Timing Failure Modes": TOO_EARLY (outreach during CLOSED — contact burned, "didn't you hear me last time?"). TOO_LATE (outreach during CLOSING — "we've already chosen," wasted effort). RIGHT_TIME_WRONG_SPEED (window OPEN but 5-day pipeline vs 30-day window — competitors arrive first, frame evaluation). RIGHT_TIME_WRONG_INTENSITY (IMMEDIATE urgency but standard 14-day sequence — prospect met 3 vendors by our Touch 2). Each has different root cause: signal interpretation, detection speed, pipeline velocity, urgency calibration.

TIMING-METRICS-001
"Timing Quality Metrics": Window Detection Lead Time (days from OPENING signal to first outreach, target 80%). Window Stage Accuracy (% correct state inference, target >70%). First-Contact Advantage (% where Decimal was first/second vendor, target >50%). Timing-Attributed Failures (% failed outreach where root cause was timing, target <20%). Signal-to-Outreach Velocity (median days P1 to sent, target <=5 business days).

TIMING-PORTFOLIO-001
"Window Urgency Prioritization": CLOSING window > OPEN window > OPENING window. Within same state, by Effective Opportunity. Account with EO 60 whose window CLOSING in 2 weeks MORE URGENT than EO 80 whose window just OPENED for 3 months. Dynamic priority changes daily as windows progress. System re-evaluates whenever any window state changes.

TIMING-PORTFOLIO-002
"Window Collision Management": Too many windows open at once: rank by urgency x opportunity. Accounts that can't be served: perform preparation (enrichment, connector activation) even if outreach can't launch. Accounts that WILL miss window: log "WINDOW_MISSED_CAPACITY" — different from detection failure. Feeds capacity planning.

TIMING-INTEL-001
— "Contact-Level Timing Patterns": Learn over time: email open time per contact, response latency per persona (CTOs 2-3 days, procurement 5-7, C-suite 7-14), day-of-week patterns. After 5+ data points: suggest optimal send times. MVP: use defaults (Tue-Thu 9-11am). Personalize as patterns emerge.

TIMING-INTEL-002
"Event-Relative Timing": Leadership change: outreach Day 7-14 (not Day 1 onboarding, not Day 30+ opinions formed). SAMA announcement: Day 1-3 (urgency highest immediately). Funding: Day 14-21 (week 1-2 celebration, weeks 3-4 "what do we build?"). After negative reply: minimum 45-day gap, but P1 regulatory can compress to 21 days. After sequence completion: 45-day cooldown, but OPENING window during cooldown can shorten to 30 days.

TIMING-INTEL-003
"Competitor Timing Intelligence": Track when competitors approach banks. Competitors typically 2-3 weeks after SAMA → Decimal approaches within 1 week. Competitors dominate conference-season → Decimal goes pre/post-conference. Competitor annual sales push (Q1 blitz, year-end discount) → Decimal positions BEFORE.

TIMING-RELATIONSHIP-001
"Timing as Signal to Prospect": 2 days after announcement = "paying attention." 3 weeks after = "mass-mailing from list." During Ramadan = "don't understand culture." During crisis = "care about selling not about you." 1 day after competitor loss = "ambulance-chasing." 30 days after = "respect the process." Every timing decision must pass: "Would recipient perceive this as thoughtful and relevant, or tone-deaf?"

TIMING-RELATIONSHIP-002
"Timing Blackout Events" beyond Ramadan/Eid: Active M&A; (full blackout per ORG-002). Contact personal difficulty (30-day delay for that contact). Account public failure (14-day delay, no reference to failure unless human-approved). National mourning (government-declared, all outreach halts).

TIMING-FAIL-002
"Timing Drift Detection": 3 drift patterns: (1) sequences get slower (review SLAs creep, 48h target becomes 5 days — weekly velocity check). (2) Send windows become rigid (always Tue-Thu 9am despite data suggesting Wednesday 2pm — monthly A/B test). (3) Cooldowns become universal (45 days for everyone despite context-based exceptions — audit exceptions applied, zero = logic broken). Monthly timing audit: are we as fast as we claim?

TIMING-FAIL-003
"Timing Cannibalization": Poorly timed touches relative to each other: email at 9am + LinkedIn at 9:05 = "automated system." Touch 2 arrives while prospect composing reply to Touch 1. Three contacts get Touch 1 same day despite 48h stagger. Detection: validate actual send times match planned. Flag if two touches <24h apart. Prevention: pre-launch timing simulation maps ALL touches across contacts/channels on timeline. Conflicts adjusted before sending.

TRUST Rules (2)

TRUST-001
"Trust Progression Framework" — 6 levels: ZERO TRUST (Day 0): AI drafts only, cannot queue. 100% review. Advance after 30 days no incidents. INITIAL TRUST (Day 30+): AI queues for review. Auto-enrichment, auto-scoring. QC >60% to advance. DEVELOPING TRUST (Day 60+): QC >80%. Review SLA can extend to 8h for P3/P4. ESTABLISHED TRUST (Day 90+): QC >85%, override 92%, override <5%. P2 Director optional review. C-suite ALWAYS human. FULL PARTNERSHIP (Day 360+): AI handles routine. Humans focus on relationships/strategy. C-suite/P1 always human. Quarterly full audit. CRITICAL: Trust can REGRESS. Single hallucination = drop one level. Reputation damage = drop to INITIAL.
Rationale: *Trust earned incrementally, lost instantly — mirrors human trust in enterprise relationships. ITERATION 1 — AI LEARNING & EVOLUTION ARCHITECTURE How does the AI actually get smarter without getting dumber? Learning in a h...*

TRUST-AUTHORITY-001
CONFLICT RESOLUTION (FIN-C01): The Trust Progression Framework (TRUST-001) is the single source of truth for autonomy and review-reduction. The day-based phases in HUMAN-AUTO-001 and FAIL-HUMAN-004 are minimums that cannot fire before the corresponding Trust level is reached. Where the three rules disagree on timing or gate, the LATER date and the STRICTER gate always win. Rationale: three independent schedules for the same transition created ambiguity; a single authority with conservative tie-breaking removes it.

Section 2: Master Entity Model (137 rules)

ACCT Rules (2)

ACCT-STATE-001
Account maintains a denormalized current_state block (tier, window_state, readiness, stage, last_decision_ts) updated by the scoring engine on every recalculation. Read-optimized cache; authoritative history lives in the intelligence entities.
Rationale: *#4 — Why does that denormalization not create an integrity risk? Because the cache is write-controlled: only the scoring engine writes current_state, and every write emits a System_Event. If cache and source disagree, the event log reconstructs truth. The Account never accepts a direct human edit to current_state.*

ACCT-STATE-002
current_state fields are system-writable only. Human changes must target the underlying intelligence entity, which retriggers recalculation. Direct writes to current_state are rejected.
Rationale: *#5 — What is the deepest reason Account exists? Because the unit of selling is the organization, not the person — Principle 2 made physical. The Account is the relationship container and decision anchor: the object against which Effective Opportunity and Decision Score resolve. Without it the system is a contact spammer, not an ABM engine.*

ATTR Rules (5)

ATTR-RECORD-001
Each meaningful outcome (reply, meeting, pilot, win/loss) is linked to the contributing Reasoning_Chain, Prompt_Version, timing/window, and connector path that produced it.
Rationale: *#2 — Why model multi-touch attribution rather than last-touch? Enterprise wins come from many touches over time. Last-touch attribution misleads; a fair model credits the sequence that built the outcome.*

ATTR-MULTITOUCH-001
Attribution uses a multi-touch model crediting the contributing touches/decisions across the journey, not only the final action. Credit weights are explicit and auditable.
Rationale: *#3 — Why attribute failures as well as successes? Avoiding mistakes is as valuable as repeating wins. Loss attribution reveals which decisions, timings, or angles destroy opportunities.*

ATTR-FAILURE-001
Negative outcomes (opt-outs, negative sentiment, losses) are attributed with equal rigor. Failure attribution feeds Decision_Pattern decay and Rule_Registry review.
Rationale: *#4 — Why feed attribution into scoring weights and patterns? Attribution is only useful if it changes behavior. Its output must recalibrate scoring weights, timing models, and decision patterns.*

ATTR-FEEDBACK-001
Attribution results feed Feedback_Loop_Entries that recalibrate scoring weights, timing models, persona inferences, and pattern confidence. Attribution without feedback is prohibited as wasted learning.

ATTR-PRIMACY-001
Every significant outcome must produce an Outcome_Attribution. Claims that a tactic 'works' must be backed by attribution evidence, not assertion. OUTCOME_ATTRIBUTION — Field Specification Field Type Business Rule Source attribution_id UUID Immutable ATTR-RECORD- 001 outcome_type Enum REPLY / MEETING / PILOT / WIN / LOSS / OPT_OUT ATTR-RECORD- 001 account_id / contact_id UUID Subject — contributing_touches Array[Object] Multi-touch credit weights ATTR-MULTITO UCH-001 linked_chain/prompt/wind Array[UUID] Causal contributors ATTR-RECORD- ow/connector 001 feedback_entry_id UUID Link to recalibration ATTR-FEEDBAC K-001 ENTITY 31: TIMING_RECORD The analytics ledger of timing decisions and their results — when the system acted, why then, and whet...

BASELINE Rules (5)

BASELINE-RECORD-001
The system maintains Performance_Baselines per key metric (response rate, conversion, hallucination rate, timing accuracy, capital efficiency). All experiments are judged against the relevant baseline.
Rationale: *#2 — Why baseline per metric and per segment? A global baseline hides where the system is weak. Per-segment baselines (by account type, signal type) reveal targeted strengths and gaps.*

BASELINE-SEGMENT-001
Baselines are maintained per metric AND per segment. Regressions in any segment are detectable even when the global average looks healthy.
Rationale: *#3 — Why require regression detection against baselines? Silent regressions erode performance over time. Active comparison to baseline catches degradation early — including from self-improvement gone wrong.*

BASELINE-REGRESSION-001
Monitored metrics are continuously compared to baseline. Significant regression triggers alerts and can auto-pause the responsible change (feedback rollback, prompt/model rollback).
Rationale: *#4 — Why re-baseline deliberately rather than continuously? If the baseline silently tracks current performance, you can't detect drift. Baselines must be updated deliberately and recorded, preserving the ability to see long-term change.*

BASELINE-REBASE-001
Re-baselining is an explicit, logged governance action, not automatic. Prior baselines are retained so long-term trajectory (improvement or decay) remains visible.

BASELINE-PRIMACY-001
Claims of improvement require beating the recorded baseline with significance. The baseline is the authoritative reference for all performance judgments and experiment evaluation. PERFORMANCE_BASELINE — Field Specification Field Type Business Rule Source baseline_id UUID Immutable version node BASELINE-REC ORD-001 metric / segment Enum/Object What + where measured BASELINE-SEG MENT-001 baseline_value / Object Reference performance BASELINE-REC confidence_interval ORD-001 regression_thresholds Object Alert/auto-pause triggers BASELINE-REG RESSION-001 established_at / Timestamp Re-baseline history BASELINE-REB superseded_at ASE-001

BCM Rules (2)

BCM-RECORD-001
A Buying_Committee_Member links one Contact to one Account-deal context with influence_weight, stance, and role-in-decision. A Contact may have zero or many committee memberships.
Rationale: *#2 — Why track influence weight explicitly? Consensus scoring (CONSENSUS-002) weights stakeholder positions by influence. Treating all members equally mis-models how enterprise decisions actually form (a skeptical CFO outweighs an enthusiastic analyst).*

BCM-NETWORK-001
Committee members link to Relationship_Edge entries capturing reporting lines and influence flows. Consensus feasibility is assessed over the network, not the member list in isolation.
Rationale: *#5 — Deepest reason this entity exists? Enterprise deals are won or lost in the committee, not the inbox. This entity makes the invisible politics of the buying group explicit, trackable, and reasoned-about — the core of Account-Centric Design.*

CONFLICT Rules (5)

CONFLICT-RECORD-001
Each rule conflict is recorded with the conflicting rules, context, the tier-hierarchy resolution applied, and the outcome. Silent resolution is prohibited.
Rationale: *#2 — Why resolve via the 6-tier hierarchy rather than ad hoc? Ad hoc resolution is unpredictable and unauditable. The Constraint Hierarchy (Legal > Safety > Relationship > Timing > Revenue > Optimization) gives a stable, explainable ordering.*

CONFLICT-HIERARCHY-001
Conflicts resolve by the 6-tier Constraint Hierarchy. The winning tier and rule are recorded. Conflicts unresolvable within time limits (4h/24h) escalate to human review.
Rationale: *#3 — Why build a precedent library from resolutions? Repeated identical conflicts shouldn't be re-litigated. After enough consistent resolutions, the pattern becomes a standing rule, speeding future decisions.*

CONFLICT-PRECEDENT-001
Three-or-more consistent resolutions of the same conflict type promote to a standing rule in Rule_Registry. The precedent library is queried before escalating new conflicts.
Rationale: *#4 — Why time-box conflict resolution? An unresolved conflict stalls action and can miss windows. Hard limits force either automated resolution or human escalation, preventing decision paralysis.*

CONFLICT-TIMEBOX-001
Conflicts must resolve within tiered limits (4h automated / 24h with human). Breaching the limit auto-escalates; timing-critical conflicts get the shortest limit.
Rationale: *#5 — Deepest reason this entity exists? It makes the system's value hierarchy explicit and consistent — ensuring that when principles collide, safety and relationships reliably outrank short-term revenue. It exists so the system's character is stable under pressure.*

CONFLICT-PRIMACY-001
Any decision arising from a resolved conflict carries a link to its Conflict_Resolution record, making the trade-off auditable. Revenue may never silently override Legal, Safety, or Relationship tiers. CONFLICT_RESOLUTION — Field Specification Field Type Business Rule Source conflict_id UUID Immutable CONFLICT-REC ORD-001

CONNECTOR Rules (5)

CONNECTOR-RECORD-001
Each Connector is stored with reach (which accounts/people they access), relationship strength to Decimal, and a per-connector capital budget. Connectors are reusable access assets.
Rationale: *#2 — Why give each connector a capital budget? Connector capital is finite (SCARCITY-003). Over-asking burns the relationship. A per-connector quarterly budget with reciprocity tracking prevents fatigue and preserves the asset.*

CONNECTOR-BUDGET-001
Each Connector has a quarterly ask-budget and reciprocity ledger. Exceeding the budget requires giving value back first; fatigue signals pause asks to that connector.
Rationale: *#3 — Why track connector reach explicitly? The value of a connector is which doors they open. Tarabut reaching SNB/SAB/Alinma simultaneously is worth more than a single-account intro. Reach must be mapped to route requests optimally.*

CONNECTOR-REACH-001
Connector reach maps to accessible accounts and committee members via Relationship_Edge. High-reach connectors are reserved for high-value, multi-account openings.
Rationale: *#4 — Why track reciprocity and value-given? Connectors are relationships, not vending machines. Sustainable access requires giving before taking. Tracking the give/take balance keeps connector relationships healthy long-term.*

CONNECTOR-RECIPROCITY-001
Each ask debits, each value-delivered credits the reciprocity ledger. Negative reciprocity (taking without giving) blocks further asks until rebalanced.
Rationale: *#5 — Deepest reason Connector exists? Warm beats cold by an order of magnitude in KSA enterprise selling. The connector is the embodiment of relationship-first GTM — the asset that converts intelligence into access. It exists because trust is transferable through people.*

CONNECTOR-PRIMACY-001
For maximum-tier targets, the system attempts connector-mediated warm entry before any cold approach, subject to connector budget and reciprocity health. CONNECTOR — Field Specification Field Type Business Rule Source connector_id UUID Immutable CONNECTOR-R ECORD-001 connector_name / type String/Enum PERSON / FIRM / ALUMNI_NETWORK / PARTNER CONNECTOR-R ECORD-001 reach Object Accounts/people accessible CONNECTOR-R EACH-001

CONTACT Rules (3)

CONTACT-ROLE-001
Contact stores identity and persona; deal-specific positioning lives on Buying_Committee_Member. A Contact with no active deal positioning is still valid.
Rationale: *#4 — Why must persona live on Contact, not be inferred per message? Persona (motivations, risk posture, decision style) is stable and expensive to derive. Re-inferring per message wastes capital and breeds inconsistency. Storing it makes reasoning Step 4 deterministic and auditable.*

CONTACT-PERSONA-001

Contact stores structured persona_profile (role_motivations, decision_style, risk_posture, communication_preference, evidence_orientation) with provenance + confidence. Stale personas (>180 days or post-role-change) flagged for re-enrichment.

Rationale: *#5 — Deepest reason Contact exists? Trust is built person-to-person; the system must never treat a human as a disposable address. Contact embodies the Relationship and Market Memory principles — forcing the system to remember who it spoke to and what it owes them.*

CONTACT-GOVERNANCE-001

No outreach may be generated to a Contact without checking interaction_lineage for prior negative sentiment, do-not-contact flags, or active fatigue. A burned relationship is a hard gate, not a soft penalty. CONTACT — Field Specification

DEP Rules (4)

DEP-RECORD-001

The Market_Dependency_Graph stores directed dependency edges between accounts: how a defined outcome at account A (win, pilot, public reference) shifts win-probability or readiness at account B. Edges carry an effect type (REFERENCE / REPUTATION / NETWORK / COMPETITIVE) and a magnitude.

Rationale: *#2 — Why does this turn allocation into actual portfolio theory rather than a nicer ranking? Because portfolio theory optimizes over COVARIANCE between bets, not just individual expected values. A moderate-EROA account that unlocks three high-value references can beat a high-EROA isolated account. Dependency edges let Portfolio_Allocation value an account partly by what winning it does to the rest of the portfolio — sequencing 'keystone' accounts first.*

DEP-CONFIDENCE-001

Dependency magnitudes carry a confidence and a source (ASSUMED / ANALOG / EMPIRICAL). Early-stage edges are explicitly ASSUMED and capped in how strongly they influence allocation. Edges earn EMPIRICAL status only after observed reference/network effects, at which point their influence cap lifts.

Rationale: *#4 — Why model competitive (negative) dependencies, not just positive reference lift? Because outcomes can hurt as well as help. Losing publicly at one account, or being seen to fail, can depress others (negative reputation effect); and winning an incumbent's flagship can harden a competitor's defense elsewhere. A graph that only models positive lift is dangerously optimistic.*

DEP-NEGATIVE-001

Dependency edges may be negative: a public loss or failure can lower dependent accounts' probability (reputation contagion), and some wins trigger competitive hardening elsewhere. Allocation and Risk_Assessment consider negative dependencies; high negative blast-radius raises the risk weight of an action.

Rationale: *#5 — Deepest reason Market_Dependency_Graph exists? Because it lets the system reason about the market as a connected system rather than a list of isolated targets — understanding that its moves ripple. It is judgment about second-order effects, the most advanced form of strategic reasoning in the architecture, and the final piece that makes the system a portfolio strategist rather than an account-by-account optimizer.*

DEP-PRIMACY-001

Account sequencing and capital allocation must consider dependency effects where edges exist. The system does not treat accounts as independent when known reference, reputation, network, or competitive dependencies link them; isolated-bet allocation is permitted only where no dependency is known. MARKET_DEPENDENCY_GRAPH — Field Specification Field Type Business Rule Source dependency_id UUID Immutable directed edge DEP-RECORD-0 01 from_account_id / UUID A influences B DEP-RECORD-0 to_account_id 01 effect_type Enum REFERENCE / REPUTATION / NETWORK / COMPETITIVE DEP-RECORD-0 01 trigger_outcome Enum WIN / PILOT / PUBLIC_REFERENCE / LOSS / FAILURE DEP-NEGATIVE -001 magnitude / direction Float/Sign Prob shift (may be negative) DEP-NEGATIVE -001 ...

ENGAGE Rules (5)

ENGAGE-SOURCE-001

Engagement_Event records recipient-side or third-party-observed actions, distinct from outbound Messages. Each records actor, action_type, and source-of-observation.

Rationale: *#2 — Why feed scoring in near-real-time? Engagement is a signal — an inbound buying-intent indicator. A reply or repeated profile views should retrigger scoring (Principle 1).*

ENGAGE-TRIGGER-001

High-intent Engagement_Events (reply, meeting-accept, repeated content interaction) retrigger account scoring and may open/widen a Buying_Window. Engagement is a first-class signal class.

ENGAGE-GRANULARITY-001
Engagement is stored as discrete timestamped events, never pre-aggregated. Aggregations computed at read time so velocity, clustering, and recency remain recoverable.
Rationale: *#4 — Why link to the specific Message that provoked it? Attribution requires the cause-effect pair. To learn which message earned the reply, the event must point back to the triggering Message and its Reasoning_Chain.*

ENGAGE-ATTRIBUTION-001
Where traceable, an Engagement_Event stores triggering_message_id, enabling Outcome_Attribution to credit the specific reasoning and prompt version.
Rationale: *#5 — Deepest reason Engagement_Event exists? It is the system's only feedback from reality about whether its strategy worked. It makes the system a learning loop rather than a one-way broadcaster — proof this is intelligence, not automation.*

ENGAGE-LEARNING-001
Every Engagement_Event contributes to a Feedback_Loop_Entry. Absence of engagement after N touches is itself a learning signal (triggers sequence pause / strategy review). ENGAGEMENT_EVENT — Field Specification Field Type Business Rule Source event_id UUID Immutable ENGAGE-SOUR CE-001 account_id / contact_id UUID FKs — action_type Enum OPEN / CLICK / REPLY / PROFILE_VIEW / ENGAGE-SOUR CONTENT_INTERACTION / MEETING_ACCEPT CE-001 triggering_message_id UUID FK to Message (if traceable) ENGAGE-ATTRI BUTION-001 observed_via Enum Source of observation ENGAGE-SOUR CE-001 intent_class Enum HIGH / MEDIUM / LOW (drives retrigger) ENGAGE-TRIG GER-001 occurred_at Timestamp Discrete, never aggregated ENGAGE-GRAN ULARITY-001 TIER 2 — INTELLIGENCE ENTITIE...

ENRICH Rules (5)

ENRICH-RECORD-001
Each enrichment run is stored with source, query, cost, raw response, and timestamp. Derived facts written to Account/Contact carry a pointer back to the Enrichment record that produced them.
Rationale: *#2 — Why track source and cost per enrichment? Enrichment consumes system capital (API spend, rate limits). Tracking cost per source enables EROA accounting and prevents over-enriching low-value accounts.*

ENRICH-COST-001
Enrichment records source_cost and counts against the account's capital budget. Enrichment of accounts below a fit threshold requires justification; speculative enrichment of COLD accounts is rate-limited.
Rationale: *#3 — Why store freshness/expiry on enrichment? Stale enrichment is a named failure mode. Firmographics, headcount, and roles change; acting on year-old data damages credibility. Each fact needs a confidence-decay and re-fetch trigger.*

ENRICH-FRESHNESS-001
Each enriched fact carries source_date and a type-specific staleness window. Stale facts are flagged and must be re-verified before use in reasoning; staleness lowers Intelligence Confidence.
Rationale: *#4 — Why reconcile conflicting sources rather than overwrite? Apollo and Clearbit may disagree. Last-write-wins silently destroys signal. The system must record both and resolve by source-reliability, preserving the conflict for audit.*

ENRICH-RECONCILE-001
When sources conflict, both values are retained with source-reliability weighting; the resolved value records its rationale. Unresolved high-stakes conflicts lower confidence and may trigger human review.
Rationale: *#5 — Deepest reason Enrichment exists? It is how the system converts the unknown into the known — the sensory apparatus that turns a bare account name into an actionable intelligence profile. Without provenance and freshness, knowledge silently rots.*

ENRICH-PRIMACY-001
No persona inference, pain hypothesis, or buying-committee map may be built on facts lacking an Enrichment provenance link. Unsourced 'facts' are treated as hypotheses, not evidence. ENRICHMENT — Field Specification Field Type Business Rule Source enrichment_id UUID Immutable ENRICH-RECO RD-001 account_id / contact_id UUID Target of enrichment — source Enum APOLLO / CLEARBIT / CLAY / LINKEDIN / OTHER ENRICH-RECO RD-001 query / raw_response Object Provenance of derived facts ENRICH-RECO RD-001 source_cost Number Capital accounting ENRICH-COST- 001 source_date / Date Freshness + re-fetch trigger ENRICH-FRESH staleness_window NESS-001 conflict_resolution Object Retained conflicts + rationale ENRICH-RECO NCILE-001 fetched_at Timestamp Run time ...

FEEDBACK Rules (4)

FEEDBACK-RECORD-001
Each system adjustment (weight change, threshold shift, pattern update, prompt promotion) is recorded as a Feedback_Loop_Entry citing the evidence that justified it.
Rationale: *#2 — Why require evidence linkage for every adjustment? Unjustified self-modification is dangerous. Requiring evidence prevents the system from changing itself on noise and enables rollback of bad changes.*

FEEDBACK-EVIDENCE-001
No adjustment may occur without a linked evidence basis (Outcome_Attribution, Timing_Record, review patterns). Adjustments failing significance thresholds are rejected.
Rationale: *#3 — Why guard against feedback-loop instability? Naive feedback can oscillate or overfit recent data (chasing noise). The system must damp changes and protect immutable principles from erosion.*

FEEDBACK-REVERSIBLE-001
Every Feedback_Loop_Entry is reversible; a degradation in monitored performance can roll back recent adjustments. Rollbacks are logged as System_Events.

FEEDBACK-PRIMACY-001
All self-improvement flows through Feedback_Loop_Entries. Silent, unlogged, or unbounded self-modification is prohibited. FEEDBACK_LOOP_ENTRY — Field Specification Field Type Business Rule Source feedback_id UUID Immutable FEEDBACK-RE CORD-001 adjustment_type Enum WEIGHT / THRESHOLD / PATTERN / PROMPT / FEEDBACK-RE MODEL_ROUTING CORD-001 evidence_links Array[UUID] Attribution/Timing/Review basis FEEDBACK-EVI DENCE-001 change_delta Object Before/after values FEEDBACK-RE CORD-001 stability_guard Object Rate-limit/damping applied FEEDBACK-STA BILITY-001 reversible_to UUID Rollback pointer FEEDBACK-RE VERSIBLE-001 ENTITY 33: PERFORMANCE_BASELINE The reference benchmark of how the system performs, against which improvements, regressions, and exp...

GAP Rules (5)

GAP-RECORD-001
When the system encounters a decision it cannot make confidently (missing data, unsupported case, low model competence), it records a Capability_Gap instead of guessing.
Rationale: *#2 — Why prefer recording a gap over a low-confidence guess? A confident-sounding wrong answer damages trust and brand. Declaring a gap and escalating is safer than fabricating (anti-hallucination).*

GAP-ESCALATE-001
Recognized gaps route the decision to human review or hold, never to a fabricated answer. The system explicitly surfaces 'I don't know / can't yet' rather than hallucinating.
Rationale: *#3 — Why aggregate gaps into a roadmap? Recurring gaps reveal where to invest (new data source, new capability, readiness gap to close). Aggregation turns individual misses into prioritized improvement.*

GAP-ROADMAP-001
Recurring Capability_Gaps aggregate into an improvement/readiness roadmap with frequency and business-impact ranking, feeding investment and Strategic Readiness planning.
Rationale: *#4 — Why link gaps to readiness and patience? Many gaps are readiness gaps (no reference, no compliance brief). Linking them connects 'we can't act well' to 'we're not ready to engage,' enforcing strategic patience.*

GAP-READINESS-001
Capability_Gaps tied to engagement readiness feed the Strategic Readiness Score. Unclosed critical gaps hold engagement until addressed (the Patience Test).
Rationale: *#5 — Deepest reason this entity exists? It institutionalizes humility — the system's willingness to admit limits, which is the foundation of trust and safety. It exists so the system improves deliberately instead of failing silently.*

GAP-PRIMACY-001
Declaring a Capability_Gap is always permitted and never penalized. Acting confidently outside demonstrated capability is the failure mode the system is built to prevent. CAPABILITY_GAP — Field Specification

INIT Rules (5)

INIT-RECORD-001
Each known Initiative is stored with type, sponsor, maturity stage, and criticality. Outreach reasoning must map Decimal's value to a specific initiative where one exists.
Rationale: *#2 — Why track initiative maturity stage? A just-announced initiative buys differently from one mid-build. Maturity determines whether Decimal is early (shaping) or late (displacing an incumbent), which changes the entire strategy.*

INIT-MATURITY-001
Initiative maturity (ANNOUNCED/PLANNING/BUILDING/LIVE/SCALING) is tracked. Project-maturity models set the engagement angle; entering at BUILDING vs LIVE triggers different playbooks.
Rationale: *#3 — Why track initiative criticality? Not all initiatives are equal. A CEO-sponsored transformation outranks a departmental experiment. Criticality drives how much capital the opportunity justifies.*

INIT-CRITICALITY-001
Each Initiative carries a criticality rating (strategic weight, executive sponsorship, budget size). Criticality feeds the opportunity equation; low-criticality initiatives cap allocated capital.
Rationale: *#4 — Why link initiatives to sponsors and windows? An initiative without a sponsor is a rumor; an initiative whose sponsor just left is dying. Linking sponsor (a committee member) and any buying window keeps initiative intelligence live, not static.*

INIT-SPONSOR-001
Each Initiative links its executive sponsor (Buying_Committee_Member) and any associated Buying_Window. Sponsor departure triggers re-assessment of initiative viability.
Rationale: *#5 — Deepest reason Initiative exists? It is the bridge between Decimal's capability and the account's actual spending. It exists because money flows to initiatives — selling to an account without an initiative is selling to no one.*

INIT-PRIMACY-001
Where no initiative can be identified, the account is treated as lower-readiness and outreach shifts to discovery/education rather than solution-selling. INITIATIVE — Field Specification Field Type Business Rule Source initiative_id UUID Immutable INIT-RECORD-0 01 account_id UUID FK to Account — initiative_type Enum DIGITAL_ONBOARDING / SME_LENDING / NEO_EXPANSION INIT-RECORD-0 / OPEN_BANKING / POS_FINANCE / OTHER 01 maturity_stage Enum ANNOUNCED / PLANNING / BUILDING / LIVE / SCALING INIT-MATURITY- 001

MARKETMEM Rules (5)

MARKETMEM-RECORD-001
Market_Interaction_History aggregates all interactions across accounts and time, keyed to persistent Contact identity, forming Decimal's market reputation memory.
Rationale: *#2 — Why track interactions across career transitions? MARKETMEM-002 requires knowing if Decimal engaged this PERSON before, anywhere. Career moves are where relationship capital transfers; losing this history loses the relationship.*

MARKETMEM-CAREER-001
When a Contact changes employer, their full interaction history is preserved and surfaced in the new account context. Prior rapport (or damage) carries forward.
Rationale: *#3 — Why compute a reputation blast-radius per contact? High-influence contacts shape how a whole segment perceives Decimal. A mistake with them propagates. Blast-radius sizing tells the system where maximum care is required.*

MARKETMEM-BLAST-001
Each interaction estimates a reputation blast_radius based on contact influence and network position. High-blast contacts mandate maximum-tier handling and conservative outreach.
Rationale: *#4 — Why track brand-capital depletion signals here? Repeated negative interactions are an early warning of reputation damage. Aggregating them market-wide surfaces depletion before it becomes a crisis (the Brand Capital Depletion Alert).*

MARKETMEM-DEPLETION-001
Market_Interaction_History monitors depletion signals (rising negative sentiment, ignored outreach, complaints). Two-or-more triggers pause market expansion pending review and repair.
Rationale: *#5 — Deepest reason this entity exists? It is the system's conscience and long memory — the reason it behaves like a trusted market participant rather than a spam engine. It exists because reputation, once spent, is the hardest capital to rebuild.*

MARKETMEM-PRIMACY-001
Every outreach decision consults Market_Interaction_History for the target person. Prior unresolved damage is a hard gate; the system never re-burns a known-damaged relationship. MARKET_INTERACTION_HISTORY — Field Specification Field Type Business Rule Source history_id UUID Immutable ledger entry MARKETMEM-R ECORD-001 contact_id UUID Persistent person key MARKETMEM-C AREER-001 account_context_id UUID Account at time of interaction MARKETMEM-R ECORD-001

MESSAGE Rules (5)

MESSAGE-RECORD-001
Every outbound message — sent, queued, or QC-rejected — is stored with full content, channel, sequence position, and the Reasoning_Chain that produced it. Rejected drafts retained for failure analysis.
Rationale: *#2 — Why link every Message to its Reasoning_Chain? Principle 5 forbids generating messages from raw data — every message must have a traceable thought process. Without the link you can't diagnose whether a bad outcome came from bad reasoning or bad execution.*

MESSAGE-TRACE-001
No Message may exist without a reasoning_chain_id. A message with no reasoning lineage is rejected before send.
Rationale: *#3 — Why store prompt_version and AI metadata on Message? Messages are generated by evolving prompts. To optimize and enable rollback, each message records which prompt version and model produced it — tying outcomes to specific generation logic.*

MESSAGE-AI-PROVENANCE-001
Each AI-generated Message stores prompt_version, model_id, ai_generated, and hallucination_score from QC. Links to Prompt_Version and AI_Model_Registry.

MESSAGE-OUTCOME-001
Each Message records response_received, response_sentiment, response_latency. Feeds Outcome_Attribution and Feedback_Loop_Entry; sentiment also updates Contact.interaction_lineage.
Rationale: *#5 — Deepest reason Message exists? Message is where strategy meets reality and the system is most accountable — the point of brand-capital expenditure. Storing it fully keeps the system honest and prevents repeating a mistake on the same person.*

MESSAGE-CAPITAL-001
Every sent Message debits the relevant capital ledgers (brand, relationship, connector) and links to the Trust_Capital_Ledger entry it generated, making the cost of each touch explicit. MESSAGE — Field Specification Field Type Business Rule Source message_id UUID Immutable MESSAGE-REC ORD-001 account_id / contact_id UUID FKs — reasoning_chain_id UUID Mandatory link MESSAGE-TRA CE-001 sequence_id / UUID/Int Workflow position — sequence_step channel Enum EMAIL / LINKEDIN — content Text Full message body MESSAGE-REC ORD-001 status Enum DRAFT / QC_REJECTED / QUEUED / SENT MESSAGE-REC ORD-001 ai_generated / Mixed AI provenance MESSAGE-AI-P prompt_version / ROVENANCE-0 model_id / 01 hallucination_score response_received / Mixed Outcome capture ME...

MODELREG Rules (5)

MODELREG-RECORD-001
Each AI model and version is registered with capabilities, intended decision types, and activation status. Generated artifacts record the model_id that produced them.
Rationale: *#2 — Why track per-model performance? A model upgrade can silently degrade quality. Per-model performance metrics catch regressions and inform which model handles which task.*

MODELREG-PERF-001
Performance metrics (quality, hallucination rate, outcome correlation) are tracked per model/version. Regression beyond thresholds blocks promotion and can trigger rollback.
Rationale: *#3 — Why route decision types to specific models? Different tasks suit different models (cheap/fast for parsing, strong for strategy). Explicit routing optimizes cost and quality.*

MODELREG-ROUTING-001
Decision types route to designated models per the registry. Routing changes are versioned and auditable; a model may only handle decision types it is approved for.
Rationale: *#4 — Why require rollback capability? If a new model misbehaves in production, the system must revert instantly. Rollback is a safety requirement, not a nicety.*

MODELREG-ROLLBACK-001
Every model activation is reversible. A one-action rollback restores the prior approved model for affected decision types; rollback events are logged as System_Events.

MODELREG-PRIMACY-001
No AI generation may use an unregistered or unapproved model for a given decision type. Reasoning_Chains and Messages must reference a valid AI_Model_Registry entry. AI_MODEL_REGISTRY — Field Specification Field Type Business Rule Source model_id String Model + version identifier MODELREG-RE CORD-001 capabilities / approved_d Object/Array What it may do MODELREG-RO ecision_types UTING-001 performance_metrics Object Quality/hallucination/outcome MODELREG-PE RF-001 active / rollback_target Bool/ID Status + revert pointer MODELREG-RO LLBACK-001 activated_at Timestamp Activation time — ENTITY 28: PROMPT_VERSION A versioned prompt template used to generate AI outputs, enabling optimization, A/B testing, and rollback.
Rationale: *#1 — Why version prompts...*

PATTERN Rules (5)

PATTERN-RECORD-001
Recurring decision contexts and their outcomes are abstracted into Decision_Pattern entities (situation signature, decision taken, outcome distribution, confidence).
Rationale: *#2 — Why require an outcome distribution, not a single result? One success can be luck. A pattern is trustworthy only across many instances. Storing the distribution captures reliability, not anecdote.*

PATTERN-EVIDENCE-001
Each pattern carries the count and outcome distribution of instances. Patterns below a minimum instance threshold are advisory only and cannot auto-drive decisions.
Rationale: *#3 — Why link patterns to the rules they support or challenge? Patterns are how rules are validated empirically. A pattern that contradicts a rule is a signal the rule needs revisiting; one that confirms it strengthens confidence.*

PATTERN-RULE-LINK-001
Patterns link to related Rule_Registry entries. A pattern persistently contradicting a rule flags that rule for review; confirming patterns raise rule confidence.
Rationale: *#4 — Why decay or retire stale patterns? Markets change; yesterday's winning pattern can become today's mistake. Patterns must age out or be re-validated to avoid the system fighting the last war.*

PATTERN-DECAY-001
Pattern confidence decays without fresh confirming instances. Stale patterns are demoted to advisory and re-validated before reuse in automated decisions.
Rationale: *#5 — Deepest reason this entity exists? It is how the system gets wiser over time rather than merely busier — the accumulation of strategic experience. It exists to turn outcomes into reusable intelligence, the essence of a learning system.*

PATTERN-PRIMACY-001
Decision patterns inform but never silently override explicit rules or human judgment. Auto-application requires sufficient evidence and current confidence; otherwise patterns are presented as recommendations. DECISION_PATTERN — Field Specification

PORT Rules (6)

PORT-RECORD-001
Each allocation cycle produces a Portfolio_Allocation distributing finite capital (attention slots, connector asks, daily account starts) across accounts ranked by Decision Score and EROA.
Rationale: *#2 — Why rank by Expected Return on Attention, not raw score? A high score that costs enormous capital may be worse than a moderate score that's cheap. EROA normalizes opportunity by cost, maximizing total return under constraint.*

PORT-EROA-001
Allocation ranks accounts by EROA = Effective Opportunity / Total Capital Cost. Negative-EROA accounts are excluded this cycle regardless of raw attractiveness.
Rationale: *#3 — Why enforce diversification / concentration limits? All capital on one whale risks total loss if it stalls. Some spread protects the portfolio and the brand. Concentration caps manage that risk.*

PORT-DIVERSIFY-001
Allocation enforces concentration limits so no single account or connector consumes beyond a cap of cycle capital. Over-concentration requires explicit human approval.
Rationale: *#4 — Why reconcile allocation against actual spend each cycle? Plans drift from reality. Comparing planned vs actual capital spend catches overspend and feeds better future allocation.*

PORT-RECONCILE-001
Each cycle reconciles planned vs actual capital spend per account/connector. Systematic overspend triggers recalibration of cost models.
Rationale: *#5 — Deepest reason this entity exists? It embodies scarcity economics — the recognition that attention is the binding constraint and must be invested, not sprayed. It exists so the system behaves like a disciplined investor of finite capital.*

PORT-PRIMACY-001
No account may consume system capital outside the active Portfolio_Allocation. Sequence starts and connector asks draw down allocated budgets; exhausted budgets queue work to the next cycle. PORTFOLIO_ALLOCATION — Field Specification

PORT-EROA-00
1 concentration_limits Object Caps per account/connector PORT-DIVERSI FY-001 reconciliation Object Planned vs actual spend PORT-RECONC ILE-001 created_at

PROMPT Rules (6)

PROMPT-RECORD-001
Each prompt template version is stored with its text, target decision type, and model compatibility. Generated artifacts record prompt_version for traceability.
Rationale: *#2 — Why tie outcomes back to prompt versions? To know which phrasing converts, outcomes must attribute to the prompt that produced them. Without this link, prompt optimization is guesswork.*

PROMPT-OUTCOME-001
Outcomes (response rate, sentiment, conversion) attribute to the prompt_version via Message and Outcome_Attribution, enabling evidence-based prompt optimization.
Rationale: *#3 — Why support A/B testing of prompts? Improvement requires controlled comparison. The entity must allow concurrent versions with traffic splits and significance tracking.*

PROMPT-AB-001
Prompt versions support controlled A/B allocation with sample-size and significance tracking. Winning variants promote only on statistically meaningful improvement, not noise.
Rationale: *#4 — Why require rollback and approval for prompt changes? A bad prompt can damage brand at scale instantly. Changes need approval and instant revert, like any production change touching customers.*

PROMPT-GOVERNANCE-001
Prompt promotion to production requires approval and is reversible. Prompts affecting maximum-tier contacts undergo human review before activation.
Rationale: *#5 — Deepest reason this entity exists? It treats the system's words — its actual voice to the market — as a governed, improvable, accountable asset. It exists because in this system, language is the product surface, and it must be controlled.*

PROMPT-PRIMACY-001
No Message may be generated by an unregistered prompt. Every AI-generated artifact links to a valid Prompt_Version; untracked prompt use is prohibited.

PROMPT-AB-00
1 active / approved_by Bool/UUID Governance status PROMPT-GOVE RNANCE-001 ENTITY 29: CAPABILITY_GAP A recorded instance where the system recognized it lacked the data, skill, or authority to act well — institutionalized humility.
Rationale: *#1 — Why record what the system CANNOT do as an entity? A system that hides its blind spot*

REVIEW Rules (5)

REVIEW-RECORD-001
Each human review is stored with the artifact reviewed, the decision (approve/edit/reject), edits made, rationale, and reviewer identity. Reviews are immutable once submitted.
Rationale: *#2 — Why capture the rationale, not just the verdict? A bare reject teaches the system nothing. The rationale is the training signal that improves future AI proposals and detects rubber-stamping.*

REVIEW-RATIONALE-001
Approvals and rejections require a rationale. Rationale-free verdicts are flagged; high rates of rationale-free approvals signal rubber-stamping and trigger oversight review.
Rationale: *#3 — Why monitor for rubber-stamping explicitly? Rubber-stamping is a named failure mode that hollows out oversight. The system must detect reviewers who approve everything instantly without engagement.*

REVIEW-RUBBERSTAMP-001
Review patterns (approval rate, review latency, edit frequency) are monitored. Anomalous rubber-stamping patterns reduce the effective trust granted to that reviewer's approvals and alert governance.
Rationale: *#4 — Why feed reviews back into AI confidence and autonomy? Trust must be earned and calibrated. Consistent human agreement should expand AI autonomy on that decision type; disagreement should contract it.*

REVIEW-TRUST-001
Review outcomes update the AI's earned-autonomy level per decision type via the Trust_Capital_Ledger. Sustained agreement raises autonomy; disagreement lowers it and may reinstate mandatory review.
Rationale: *#5 — Deepest reason this entity exists? It is the hinge of human-AI collaboration — the mechanism by which authority is shared, trust is calibrated, and accountability is preserved. It exists so the human remains meaningfully in command.*

REVIEW-PRIMACY-001
Decisions above defined risk/autonomy thresholds require a Human_Review_Record before execution. The system never self-grants authority beyond its earned, reviewed autonomy level. HUMAN_REVIEW_RECORD — Field Specification

RULEREG Rules (1)

RULEREG-VERSION-001
Rule changes create new versions; prior versions are retained. Decisions reference the rule version in force at decision time, enabling valid historical audit and rollback.
Rationale: *#3 — Why classify each rule by constraint tier? The 6-tier Constraint Hierarchy resolves conflicts. Each rule must carry its tier so conflict resolution is deterministic.*

SCORE Rules (5)

SCORE-RECORD-001
Each scoring run writes an immutable Three_Score_Record snapshot (fit/ICS, intent/dynamic, intelligence-confidence) with per-dimension contributions and the weight-set version used. The latest snapshot updates Account.current_state.

SCORE-TRIAD-001
Scores are stored as three independent dimensions and never collapsed into a single number at storage time. The Effective Opportunity equation consumes them separately; ICS gates intent so low-confidence cannot inflate priority.
Rationale: *#3 — Why store per-dimension contributions, not just totals? Black-box scoring has no explainability (a named pitfall). To justify outreach to a human reviewer and to debug mis-scoring, each contributing dimension's value and weight must be recoverable.*

SCORE-EXPLAIN-001
Every Three_Score_Record stores dimension-level breakdown: each input, its raw value, applied weight, and contribution. A score with no breakdown is invalid.
Rationale: *#4 — Why version the weight-set on each record? Weights are recalibrated monthly using engagement data. Comparing scores across time is meaningless unless you know which weight-set produced each. Versioning enables apples-to-apples trend analysis and safe rollback.*

SCORE-VERSION-001
Each record references a weight_set_version from Rule_Registry. Recalibration creates a new version; historical records keep their original version for valid longitudinal comparison.
Rationale: *#5 — Deepest reason this entity exists? It is the quantified judgment of the system — the bridge between raw intelligence and prioritized action. It exists so that prioritization is explainable, auditable, and improvable rather than a magic number.*

SCORE-PRIMACY-001
No prioritization, sequencing, or capital allocation may reference a score not backed by a stored Three_Score_Record. Ranking on ephemeral or unstored scores is prohibited. THREE_SCORE_RECORD — Field Specification Field Type Business Rule Source score_record_id UUID Immutable snapshot SCORE-RECOR D-001 account_id UUID FK to Account — fit_score_ics Int 0-100 ICP match / Ideal Customer Score SCORE-TRIAD- 001 intent_score_dynamic Int 0-100 Current buying-behavior score SCORE-TRIAD- 001 intelligence_confidence Int 0-100 Data-quality confidence; gates intent SCORE-TRIAD- 001 dimension_breakdown Array[Object] Per-input value, weight, contribution SCORE-EXPLAI N-001 weight_set_version String FK to Rule_Registry weight version SCORE-VERSIO N-001 co...

SEQ Rules (5)

SEQ-RECORD-001
Each Outreach_Sequence stores ordered steps, channel per step, timing gaps, and stop conditions. It enforces hard limits: max touches per channel, minimum gaps, and the 14/10-day spreads.
Rationale: *#2 — Why encode stop/exit conditions in the sequence? A sequence that ignores responses is spam. It must halt on reply, on negative sentiment, or on window closure — exit logic is core, not optional.*

SEQ-EXIT-001
Each sequence defines exit conditions (reply received, negative sentiment, window closed, do-not-contact). Triggering any exit immediately pauses/terminates remaining steps.
Rationale: *#3 — Why bind sequence pacing to system-wide rate limits? Per-contact limits aren't enough; the system has global throughput caps (max new accounts/day in MVP). The sequence must respect both contact-level and system-level pacing.*

SEQ-RATELIMIT-001
Sequence scheduling respects contact-level limits AND global throughput caps. When global capacity is exhausted, sequence starts queue by priority (Decision Score), not first-come.
Rationale: *#4 — Why make each step re-evaluate before firing? State changes between planning and execution (window closes, contact leaves, sentiment drops). A step must re-check conditions at fire-time, not blindly execute the plan.*

SEQ-REEVAL-001
Before each step fires, the system re-evaluates current account/contact/window state and the governing Reasoning_Chain freshness. Stale or invalidated steps are regenerated or cancelled.
Rationale: *#5 — Deepest reason this entity exists? It operationalizes timing and restraint — turning strategy into paced, condition-aware action instead of a burst of messages. It exists because in enterprise, cadence and patience are as important as content.*

SEQ-PRIMACY-001
All outbound activity flows through an Outreach_Sequence governed by these constraints. Ad-hoc out-of-sequence sends are prohibited except explicit human-initiated one-offs, which are still logged. OUTREACH_SEQUENCE — Field Specification

SIGNAL Rules (5)

SIGNAL-PERSISTENCE-001
Every signal is stored immutably with raw source payload, detection timestamp, and source attribution — even after scoring/action. Signals decay in weight, not existence; never deleted.
Rationale: *#2 — Why store the raw payload, not just the parsed signal? Parsing logic evolves. A signal parsed under v1 may need reinterpretation under v2. Raw payload allows reprocessing and protects against parser bugs corrupting evidence.*

SIGNAL-PROVENANCE-001
Signal stores raw_payload (immutable) and parsed_interpretation (versioned, parser_version). Reparsing creates a new interpretation linked to the same raw payload; original retained.
Rationale: *#3 — Why a decay model on the entity itself? Timing is the primary competitive weapon. A CTO-hiring signal is P1-hot at week 1 and noise at month 6. Current weight is a function of age, so the decay curve is a property of signal type, computed at read time.*

SIGNAL-DECAY-001
Each Signal carries a signal_type whose decay function (half-life, hard-expiry) sets current weight = base_priority x decay(age). Expired signals remain stored, contribute zero weight, and cannot independently trigger outreach.
Rationale: *#4 — Why must Signal carry deduplication identity? The same event arrives via multiple sources (funding round on Crunchbase, LinkedIn, news). Counting it thrice corrupts scoring. The entity needs an event-identity key to collapse duplicates while preserving corroboration.*

SIGNAL-DEDUP-001
Each Signal computes an event_fingerprint (entity + event_type + time-window). Signals sharing a fingerprint collapse into one logical signal with multiple corroborating_sources; corroboration raises confidence but does NOT multiply weight.

SIGNAL-PRIMACY-001
No account may enter or advance in outreach without at least one qualifying, non-expired, deduplicated signal. The triggering signal is permanently linked to the resulting Reasoning_Chain. SIGNAL — Field Specification Field Type Business Rule Source signal_id UUID Immutable SIGNAL-PERSIS TENCE-001 account_id UUID FK to Account (or Unresolved_Intelligence) ACCT-IDENTITY -005 signal_type Enum Drives priority + decay curve SIGNAL-DECAY -001 priority Enum P1 / P2 / P3 base weight SIGNAL-DECAY -001 raw_payload Object Immutable source data SIGNAL-PROVE NANCE-001 parsed_interpretation Object Versioned parse (parser_version) SIGNAL-PROVE NANCE-001 event_fingerprint String Dedup key SIGNAL-DEDUP -001 corroborating_sources Array Multi-source confirma...

STRAT Rules (5)

STRAT-RECORD-001
Each engaged account has one active Account_Strategy stating the value angle, competitive posture, target committee members, and connector path. Sequences must align to it.
Rationale: *#2 — Why tie strategy to a specific initiative and window? Strategy without a concrete initiative and timing is abstract. Anchoring it makes the plan actionable and time-aware.*

STRAT-ANCHOR-001
Account_Strategy anchors to a target Initiative and any Buying_Window. Absent an initiative, strategy defaults to discovery/relationship-building, not solution-selling.
Rationale: *#3 — Why record the readiness gates in the strategy? Strategic Patience requires not engaging before Decimal is ready (references, collateral, compliance). The strategy must hold the gates that must clear before active selling.*

STRAT-READINESS-001
Account_Strategy records readiness gates (e.g. SAMA CSF brief, data residency, Islamic-finance support) that must clear before solution-selling. Unmet hard gates hold the strategy in preparation mode.
Rationale: *#4 — Why version strategy and require review on major state change? Accounts change (sponsor leaves, incumbent renews). A strategy frozen against old reality fails. Versioning and triggered review keep strategy live.*

STRAT-VERSION-001
Strategy is versioned; major events (sponsor departure, window close, competitive renewal) trigger mandatory strategy review. Superseded versions are retained for learning.
Rationale: *#5 — Deepest reason this entity exists? It is the account-level intelligence that makes Decimal feel like one coherent, strategic partner rather than a stream of disconnected messages. It exists because account-centric selling needs an account-centric plan.*

STRAT-PRIMACY-001
Every Outreach_Sequence and Message must reference and conform to the active Account_Strategy. Conflicts between a tactic and the strategy are resolved in favor of the strategy or escalate to review. ACCOUNT_STRATEGY — Field Specification

SYSEVENT Rules (4)

SYSEVENT-RECORD-001
Every significant state change, decision, and action emits an immutable, append-only System_Event with actor, type, payload, and timestamp. The log is never edited or deleted.
Rationale: *#2 — Why make it append-only and immutable? Audit integrity demands tamper-evidence. If events can be altered, the audit trail is worthless for compliance and debugging.*

SYSEVENT-IMMUTABLE-001
System_Events are write-once. Corrections are new compensating events referencing the original, never edits. This preserves a tamper-evident history for enterprise/compliance audit.
Rationale: *#3 — Why use events to coordinate, not just to log? In an event-driven system, events are the coordination mechanism — scoring retriggers, window openings, and sequence advances are all event-driven. The log doubles as the nervous system.*

SYSEVENT-COORD-001
Designated System_Events trigger downstream workflows (scoring recalculation, window evaluation, sequence re-eval). Coordination flows through events, ensuring reactions are traceable to causes.
Rationale: *#4 — Why include causal linkage between events? To debug and explain, you must follow cause to effect. Linking each event to its triggering event reconstructs full decision chains across the system.*

SYSEVENT-CAUSAL-001
Each event records its triggering_event_id where applicable, forming causal chains. Any action can be traced backward to the originating signal and forward to its consequences.

TIMING Rules (5)

TIMING-RECORD-001
Each timing-sensitive decision records when action was taken, the window state at the time, the rationale for that timing, and the eventual result.
Rationale: *#2 — Why capture window-state-at-action? To learn the optimal moment, the system must correlate results with where in the window it acted (early/peak/closing). Without window state, timing learning is impossible.*

TIMING-WINDOWSTATE-001
Each Timing_Record stores the Buying_Window state at action time. Correlating state with outcomes calibrates the ideal action point within windows.
Rationale: *#3 — Why record missed-timing (acted too early/late)? Both prematurity and lateness are failures (timing failure cascades). Recording near-misses and misses teaches the system to correct its timing model.*

TIMING-MISS-001
Acted-too-early and acted-too-late outcomes are explicitly recorded, including windows missed entirely. These feed window-prediction and urgency models to reduce future mistiming.
Rationale: *#4 — Why feed timing analytics back into window prediction? Timing learning must improve future forecasts. Records should recalibrate decay curves and closure predictions.*

TIMING-CALIBRATE-001
Timing_Records recalibrate signal decay curves and Buying_Window closure predictions. Persistent mistiming on a signal type adjusts that type's decay function.

TIMING-PRIMACY-001
Timing decisions of material consequence must produce a Timing_Record. Timing models may only be adjusted on the basis of recorded timing evidence. TIMING_RECORD — Field Specification Field Type Business Rule Source timing_record_id UUID Immutable TIMING-RECOR D-001 account_id / decision_ref UUID Subject decision — action_time / Timestamp/Enum When + where in window TIMING-WINDO window_state_at_action WSTATE-001 timing_rationale Text Why then TIMING-RECOR D-001 timing_outcome Enum OPTIMAL / TOO_EARLY / TOO_LATE / MISSED TIMING-MISS-0 01 calibration_impact Object Feeds decay/closure models TIMING-CALIBR ATE-001 ENTITY 32: FEEDBACK_LOOP_ENTRY A single learning event: an observation that adjusts a weight, threshold, pattern, or model — the uni...

TRUST Rules (5)

TRUST-RECORD-001
The Trust_Capital_Ledger tracks earned-autonomy balance per decision type, credited by successful reviewed decisions and debited by failures/overrides. Autonomy levels derive from this balance.
Rationale: *#2 — Why per decision type rather than one global trust level? The AI may be reliable at signal interpretation but weak at competitive strategy. One global number over- or under-grants autonomy. Per-type trust calibrates correctly.*

TRUST-TYPED-001
Trust is tracked per decision type. High trust in one type does not transfer to another. Each type has its own threshold for moving from mandatory-review to autonomous action.
Rationale: *#3 — Why debit trust on failures and overrides? Trust must be responsive to evidence. A failure or a human override is evidence the AI was wrong; trust (and autonomy) should drop accordingly.*

TRUST-DEBIT-001
Failures, human rejections, and bad outcomes debit trust for that decision type. A sharp debit can instantly reinstate mandatory human review (autonomy circuit-breaker).
Rationale: *#4 — Why guard against trust gaming / metric manipulation? If the AI could inflate its own trust (e.g. by routing only easy decisions for review), oversight collapses. The ledger must resist manipulation (a named failure mode).*

TRUST-INTEGRITY-001
Trust crediting uses outcome-weighted, sampled, and difficulty-adjusted decisions; the AI cannot self-select which decisions count toward trust. Manipulation patterns freeze autonomy and alert governance.

TRUST-PRIMACY-001
No autonomous action may exceed the earned-autonomy level recorded in the Trust_Capital_Ledger for that decision type. The ledger is the single source of truth for AI authority. TRUST_CAPITAL_LEDGER — Field Specification Field Type Business Rule Source ledger_id UUID Immutable entries TRUST-RECOR D-001 decision_type Enum Per-type trust bucket TRUST-TYPED- 001 balance / autonomy_level Float/Enum Earned trust + granted autonomy TRUST-RECOR D-001 credit_debit_events Array[Object] Outcome-weighted changes TRUST-DEBIT-0 01 integrity_flags Array Manipulation detection TRUST-INTEGR ITY-001 updated_at Timestamp System-managed — ENTITY 25: SYSTEM_EVENT The immutable, append-only log of everything the system does — the spine of auditability and event...

VENDOR Rules (5)

VENDOR-RECORD-001
Each known incumbent/competitor in an account is stored with product footprint, contract status, and entrenchment level. Strategy must account for incumbency where present.
Rationale: *#2 — Why track entrenchment level specifically? The Entrenchment Modifier in the opportunity equation depends on it. Deep lock-in (integrated, multi-year contract) makes near-term displacement unrealistic and should suppress opportunity scoring.*

VENDOR-ENTRENCH-001
Entrenchment (integration depth, contract length remaining, switching cost) feeds the Entrenchment Modifier. High entrenchment suppresses opportunity unless a renewal window or dissatisfaction signal is present.
Rationale: *#3 — Why track incumbent satisfaction/dissatisfaction? Entrenchment plus dissatisfaction is the prime displacement opening. A renewal approaching with complaints is a window; the system must capture satisfaction signals to find these.*

VENDOR-SATISFACTION-001
Dissatisfaction signals (complaints, failed projects, public RFPs) are tracked against incumbents. Dissatisfaction + approaching renewal opens a displacement Buying_Window.
Rationale: *#4 — Why track contract/renewal timing? Displacement is only realistic near renewal. Knowing renewal timing converts a hopeless 'rip and replace' into a well-timed competitive entry.*

VENDOR-RENEWAL-001
Known contract/renewal dates are stored and feed window prediction. Outreach ramps as renewal approaches; mid-contract displacement attempts are de-prioritized absent strong dissatisfaction.
Rationale: *#5 — Deepest reason this entity exists? It is the system's competitive awareness — the difference between selling into reality and selling into a fantasy of empty space. It exists because every enterprise account already has incumbents.*

VENDOR-PRIMACY-001
Account strategy must state the incumbency posture (greenfield / coexist / displace / wait) backed by Vendor_Intelligence. An unstated competitive posture lowers strategy confidence. VENDOR_INTELLIGENCE — Field Specification Field Type Business Rule Source vendor_intel_id UUID Immutable VENDOR-RECO RD-001 account_id UUID FK to Account — incumbent_name / String/Object What they run today VENDOR-RECO product_footprint RD-001

WINDOW Rules (5)

WINDOW-RECORD-001
A Buying_Window is created when window-opening signals fire (new license, exec hire, funding, mandate). It records open_state (OPENING/PEAK/CLOSING/CLOSED), trigger signals, and a Window Multiplier consumed by the opportunity equation.
Rationale: *#2 — Why track window state transitions rather than just 'open/closed'? The optimal moment is near peak, not at first detection (acting too early wastes capital; too late misses it). The state machine encodes WHEN within the window to act.*

WINDOW-STATE-001
Window state transitions are timestamped and event-driven. The Window Multiplier is highest at PEAK and decays toward CLOSING. Outreach timing decisions read window state, not merely window existence.
Rationale: *#3 — Why attach the causing signals to the window? A window without a known cause can't be reasoned about or defended. Linking the trigger signals lets the message reference the real catalyst and lets the system learn which signal types open real windows.*

WINDOW-CAUSE-001
Each Buying_Window links the signals that opened it. Windows with no qualifying signal cause are invalid. Closed-window signals are retained to train window-prediction models.
Rationale: *#4 — Why predict window closure rather than only observe it? Reacting after closure is too late. The system must forecast decay to schedule action before the window shuts — this is the difference between timing as weapon and timing as luck.*

WINDOW-PREDICT-001
Each window carries an estimated_closure with confidence, updated as signals age. Approaching closure raises execution urgency in the decision sequence and may pre-empt lower-priority work.
Rationale: *#5 — Deepest reason Buying_Window exists? It is the temporal heart of the system — the object that turns 'who to contact' into 'who to contact NOW.' It exists because in enterprise sales, being right at the wrong time is being wrong.*

WINDOW-PRIMACY-001
Timing decisions in the Universal Decision Sequence (step 5) read Buying_Window state. No 'now vs later' decision may be made without consulting the account's active window, if any. BUYING_WINDOW — Field Specification Field Type Business Rule Source window_id UUID Immutable WINDOW-RECO RD-001 account_id UUID FK to Account — open_state Enum OPENING / PEAK / CLOSING / CLOSED WINDOW-STAT E-001 window_multiplier Float Consumed by Effective Opportunity eq. WINDOW-RECO RD-001 trigger_signal_ids Array[UUID] Causing signals WINDOW-CAUS E-001 opened_at / Timestamp Lifecycle + forecast WINDOW-PRED estimated_closure ICT-001

Section 3: Modules and Workflows (97 rules)

M0 Rules (5)

M0-GATE-001
No intelligence (signal, enrichment, engagement) may be consumed by a downstream module until M0 has assigned it an Intelligence Confidence Score and passed it through freshness, dedup, and provenance checks. Unblessed data is quarantined, never silently used.
Rationale: *#2 — Why compute confidence as a SCORE rather than a pass/fail flag? Because data quality is a gradient, not a binary. A two-week-old firmographic fact is usable but weaker than a fresh one. A score lets downstream modules weight intelligence proportionally (ICS gates intent in the opportunity equation) instead of crudely accepting or rejecting it.*

M0-CONF-001
M0 computes a three-dimensional Intelligence Confidence Score (source reliability x freshness x corroboration) per intelligence item. The score flows into Three_Score_Record as the confidence dimension and proportionally weights the item everywhere it is used.
Rationale: *#3 — Why route conflicts and gaps through M0 rather than each module handling its own? Because consistency requires one authority. If every module resolves conflicting enrichment its own way, the system contradicts itself. M0 centralizes conflict resolution and gap declaration so the whole system reasons from one reconciled truth and one honest list of what it does not know.*

M0-RECONCILE-001
M0 owns intelligence-conflict reconciliation (via ENRICH-RECONCILE-001) and Capability_Gap declaration. When sources disagree or data is missing, M0 records the conflict/gap and emits a confidence-lowered or quarantined result rather than letting modules guess independently.
Rationale: *#4 — Why must M0 be event-driven rather than a periodic batch cleaner? Because timing is the primary weapon — a hot signal blessed six hours late is a missed window. M0 must evaluate intelligence the moment it arrives so downstream reaction is immediate, while still catching the staleness of older items at read time.*

M0-EVENT-001
M0 evaluates each intelligence item on arrival (event-driven), emitting a System_Event on bless/quarantine. It also re-evaluates confidence at read time so decay since ingestion is always reflected. No fixed-interval batch gate is permitted for hot signals.
Rationale: *#5 — Deepest reason M0 exists? Because the system's single greatest risk is acting confidently on bad information — the hallucination/staleness/metric-manipulation failure family. M0 is the institutional discipline that the system never mistakes noise for knowledge. It is the foundation of trust for everything above it.*

M0-PRIMACY-001
M0 is the first module in the pipeline and an immutable governance authority. Its confidence scores and quarantine decisions cannot be overridden by downstream modules; only a governed human action can release quarantined intelligence. ① Workflow Logic (step-by-step) 1. Intelligence item arrives (signal / enrichment / engagement) → emit ingest System_Event 2. Provenance check: is raw_payload + source attribution present? (SIGNAL-PROVENANCE / ENRICH-RECORD) 3. Dedup check: compute event_fingerprint; collapse to corroborating_sources if matched 4. Freshness check: apply decay/staleness window for the item type

M1 Rules (5)

M1-CONTINUOUS-001
M1 polls configured sources on type-specific schedules (high-priority sources more frequently) and ingests push sources in real time. Polling cadence per source is configurable in Rule_Registry; hot signal types get the shortest intervals.
Rationale: *#2 — Why normalize signals into one schema rather than store source-native formats? Because downstream modules must reason uniformly. A LinkedIn hire, a SAMA regulatory notice, and a funding round are different shapes but must all become comparable Signal entities with type, priority, and fingerprint. Normalization is what makes cross-source intelligence possible.*

M1-NORMALIZE-001
M1 normalizes every raw source payload into the Signal schema (signal_type, priority, raw_payload, parsed_interpretation, fingerprint) with parser_version stamped. Raw payload is always retained for reparsing under future parser versions.

M1-RESOLVE-001
M1 performs entity resolution against Account.aliases. Resolved signals attach to the Account; unresolved signals go to Unresolved_Intelligence for later resolution (per ACCT-IDENTITY-005). Misattribution is worse than non-attribution — ambiguous matches stay unresolved pending confidence.
Rationale: *#4 — Why does M1 hand off to M0 rather than directly trigger scoring? Because raw detection is not blessed intelligence. M1 finds and structures; M0 judges quality. Separating detection from governance prevents a noisy or duplicate signal from triggering a scoring cascade before its confidence is established.*

M1-HANDOFF-001
M1 emits each normalized signal to M0 for governance, not directly to M2 scoring. Scoring is triggered only after M0 blesses the signal. M1 never assigns confidence or triggers downstream action itself.
Rationale: *#5 — Deepest reason M1 exists? Because the system can only be as aware as its senses. M1 is the sensory organ — without continuous, structured, attributed perception of the market, every other module is reasoning in the dark. It is the physical embodiment of event-driven architecture.*

M1-PRIMACY-001
Every account entry and advancement traces to a Signal produced by M1 (per SIGNAL-PRIMACY-001). M1 is the sole authorized producer of externally-sourced Signal entities; other modules may produce engagement-derived signals but not external ones. ① Workflow Logic (step-by-step) 1. Cron/push trigger fires for a configured source (cadence from Rule_Registry) 2. Fetch external data (News, LinkedIn, SAMA RSS, funding, hiring, job boards) 3. Parse & normalize into Signal schema; stamp parser_version 4. Entity resolution against Account.aliases → Account or Unresolved_Intelligence 5. Compute event_fingerprint for dedup 6. Persist Signal (immutable raw_payload + parsed_interpretation) 7. Emit Signal to M0 for governance (NOT directly to scoring) ② ...

M10 Rules (5)

M10-RECORD-001
M10 records every human review as an immutable Human_Review_Record (artifact, decision approve/edit/reject, edits, rationale, reviewer, latency). It routes qualified engagement to the right human role (SDR/AE/technical) by trigger type (REVIEW-RECORD-001).
Rationale: *#2 — Why require rationale on every review decision, not just the verdict? Because a bare approve/reject teaches the system nothing and hides rubber-stamping. Rationale is the training signal that improves future AI proposals and lets the system detect oversight that has gone hollow.*

M10-RATIONALE-001
M10 requires a rationale for approvals and rejections. Rationale-free verdicts are flagged; high rates of instant rationale-free approvals are detected as rubber-stamping and reduce the effective trust granted to that reviewer (REVIEW-RUBBERSTAMP-001).
Rationale: *#3 — Why does M10 feed the Trust_Capital_Ledger? Because autonomy must be earned and calibrated per decision type. Consistent human agreement should expand AI autonomy; disagreement should contract it, even reinstating mandatory review. M10 is how reviews translate into changing AI authority.*

M10-TRUST-001

M10 review outcomes update the Trust_Capital_Ledger per decision type: sustained agreement raises earned autonomy, disagreement lowers it and can trip the autonomy circuit-breaker back to mandatory review (REVIEW-TRUST-001, TRUST-DEBIT-001).

Rationale: *#4 — Why gate high-risk/max-tier actions through M10 before execution? Because some actions (max-tier contacts, high blast-radius, low-confidence reasoning, compliance-sensitive) carry consequences that exceed the AI's earned autonomy. M10 ensures a human reviews these before they reach the market.*

M10-GATE-001

Actions above defined risk/autonomy thresholds (max-tier contact, high blast-radius, compliance-sensitive, low-confidence chain) require an M10 Human_Review_Record before M8 may execute. The system never self-grants authority beyond its ledger-recorded autonomy (REVIEW-PRIMACY-001).

Rationale: *#5 — Deepest reason M10 exists? Because the human must remain meaningfully in command, and trust between human and AI must be built on evidence, not assumption. M10 is the mechanism of shared authority and preserved accountability — the reason the system earns independence honestly rather than seizing it.*

M10-PRIMACY-001

M10 is the sole authority on human-AI authority transfer and review capture. AI autonomy levels are defined only by the Trust_Capital_Ledger as updated through M10; no module may act beyond its earned, M10-reviewed autonomy. ① Workflow Logic (step-by-step) 1. Positive-intent trigger from M9 (reply, meeting request, pricing/security question) → route to role 2. OR high-risk/max-tier action from M4/M7/M8 → enqueue for mandatory review 3. Present artifact (reasoning chain / message / strategy) + context to human 4. Capture decision: approve / edit / reject + mandatory rationale 5. On approve → release action to M8; on edit → revised artifact to M8; on reject → return to M4/M7

M11 Rules (3)

M11-CENTRAL-001

M11 is the sole authority for system self-adjustment. Every change (scoring weight, threshold, timing curve, prompt promotion, pattern update) is a Feedback_Loop_Entry citing evidence, rate-limited and reversible (FEEDBACK-RECORD-001, FEEDBACK-PRIMACY-001).

Rationale: *#2 — Why attribute outcomes multi-touch rather than last-touch? Because enterprise wins come from many touches over time; last-touch attribution misleads and would teach the system the wrong lessons. Multi-touch credit reveals which sequence of decisions actually built the outcome.*

M11-ATTRIBUTE-001

M11 produces Outcome_Attribution with multi-touch credit across contributing Reasoning_Chains, Prompt_Versions, timing, and connector paths (ATTR-MULTITOUCH-001). Failures are attributed with equal rigor (ATTR-FAILURE-001).

M11-BASELINE-001

M11 maintains Performance_Baselines per metric and segment, continuously compares live metrics, and on significant regression alerts and can auto-pause/roll-back the responsible change (BASELINE-REGRESSION-001). Re-baselining is an explicit logged action (BASELINE-REBASE-001).

Rationale: *#5 — Deepest reason M11 exists? Because the system's deepest purpose is to accumulate judgment, not just activity. M11 is the disciplined heartbeat of learning — evidence-based, bounded, auditable, reversible improvement that makes the system measurably wiser over time without betraying its principles.*

M12 Rules (2)

M12-SEPARATE-001

M12 owns the Tier 6 entities and their slow, deliberate update cadence, distinct from M11's operational recalibration. M12 outputs condition upstream modules (M2/M4/M5/M6); it does not itself score, reason, time, or allocate (META-T6 admission test).

Rationale: *#2 — Why does M12 maintain DNA as a slow prior rather than recompute it often? Because character persists while state fluctuates. Recomputing DNA on every signal would let a turbulent quarter erase years of established character. M12 updates DNA only when deviations from the prior persist across multiple assessments.*

M12-DNA-001

M12 maintains Organizational_DNA as a slow prior, updating only when Organizational_Assessment deviations persist across multiple readings (DNA-PRIOR-001). It clusters DNA into reusable buying archetypes and assigns provisional archetypes to sparse accounts (DNA-ARCHETYPE-001).

Rationale: *#3 — Why does M12 own market-level narrative tracking? Because narrative is the only supra-account entity — it shapes the whole segment, not one account. A dedicated owner tracks narrative lifecycle and computes per-account alignment, feeding framing and timing across all accounts simultaneously.*

M13 Rules (5)

M13-VALUE-001
M13 produces an economic profile per Opportunity (estimated_deal_size, budget_probability, procurement_cycle_value, revenue_realization_probability, expected_LTV). Estimated value enters the Effective Opportunity equation as a value term, so magnitude — not just attractiveness — drives ranking.
Rationale: *#2 — Why model budget PROBABILITY, not just budget size? Because a large budget that may never be approved is worth less than a smaller funded one. Procurement reality is probabilistic — mandates get cut, budgets freeze. Modeling the probability that money actually flows prevents the system from chasing impressive-sounding but unfunded opportunities.*

M13-PROBABILITY-001
M13 separates budget SIZE from budget PROBABILITY (likelihood of approval and release). Expected value = size x probability x realization. A large unfunded opportunity is discounted below a smaller funded one; org instability and reorg states (M-org) lower budget probability.
Rationale: *#3 — Why feed economics into M6 allocation, not just M2 scoring? Because capital should flow to expected RETURN, and return requires value. M6's EROA was dividing opportunity by cost without a real value numerator. With M13, allocation becomes genuine return-on-capital: a moderate-fit $5M opportunity can rightly outrank a high-fit $50K one.*

M13-ALLOCATION-001
M13's expected value is the value numerator in M6's EROA. Allocation maximizes expected economic return under capital constraint, not just attractiveness. Negative or trivial expected-value opportunities are de-prioritized regardless of signal strength.

M13-LTV-001
M13 estimates expected lifetime value including expansion potential and reference value (linked to Market_Dependency_Graph). Opportunities that are keystones or platform-entry points carry LTV above their initial deal size; this feeds M6 keystone valuation.
Rationale: *#5 — Deepest reason M13 exists? Because a system that allocates finite capital without modeling economic value is flying blind on the one dimension that ultimately matters — return. M13 gives the system economic gravity: the sense that not all opportunities deserve equal capital because not all are worth the same. It turns prioritization into investment.*

M13-PRIMACY-001
No opportunity may be allocated significant capital without an M13 economic profile (or an explicit unknown-value flag that caps the capital committed). Value estimates carry confidence; low-confidence estimates cap committed capital until refined. ① Workflow Logic (step-by-step) 1. Opportunity created/updated by M19 → request economic profile 2. Estimate deal size from opportunity type, account size, initiative criticality, comparable deals 3. Estimate budget probability from funding signals, sponsorship, org stability 4. Estimate realization probability + expected LTV (expansion + reference value) 5. Compute expected economic value (size x budget-prob x realization) with confidence 6. Emit value term to M2 (opportunity equation) and M6 (E...

M14 Rules (5)

M14-DYNAMICS-001
M14 maintains competitive dynamics per Opportunity: competitor_strength, displacement_probability, incumbent_satisfaction, replacement_triggers, political_protection_index. These are live, evolving assessments distinct from M3's static Vendor_Intelligence facts.
Rationale: *#2 — Why model political protection specifically? Because in KSA banking, an incumbent can be technically weak yet politically immovable (a relationship-based or regulator-favored vendor), or technically strong yet politically exposed. Ignoring political protection makes the system pursue technically-winnable but politically-impossible displacements — burning capital on dead ends.*

M14-POLITICAL-001
M14 maintains a political_protection_index per incumbent (relationship depth, regulatory favor, executive sponsorship of the incumbent). High protection suppresses displacement_probability even when technical vulnerability is high; the system avoids politically-impossible displacements.
Rationale: *#3 — Why track replacement triggers rather than just current state? Because displacement happens at moments — a failed project, a renewal, a leadership change, a public RFP. Knowing the triggers lets the system wait for the opening rather than pushing against a stable incumbent. This is timing-as-weapon applied to competition.*

M14-TRIGGER-001
M14 tracks replacement triggers (incumbent failure, renewal approach, sponsor departure, public RFP, dissatisfaction signals). A trigger firing opens or amplifies a displacement Buying_Window (via M5) and raises displacement_probability. Absent triggers, entrenched incumbents are de-prioritized.
Rationale: *#4 — Why feed competitive dynamics into strategy (M4) and simulation (M15)? Because the right strategy depends on the incumbent's likely response. A vulnerable incumbent invites direct displacement; a protected one demands coexistence or patience. M15 stress-tests strategy against incumbent retaliation — which requires M14's model of how the incumbent will react.*

M14-STRATEGY-001
M14 dynamics condition M4 competitive posture (displace/coexist/wait/greenfield) and feed M15 simulation as the incumbent-response model. A strategy assuming an incumbent will not react, against a strong/protected incumbent, is flagged as unrobust.
Rationale: *#5 — Deepest reason M14 exists? Because enterprise selling is usually a contest against an incumbent, and a system that cannot read the battlefield will pick the wrong fights. M14 gives the system competitive awareness — the difference between selling into a fantasy of empty space and selling into a contested, dynamic, political reality.*

M14-PRIMACY-001
Any displacement strategy must be backed by an M14 competitive assessment. Pursuing displacement without modeling incumbent vulnerability and political protection is prohibited; the default assumption is that every enterprise opportunity is contested. ① Workflow Logic (step-by-step) 1. Opportunity with incumbent identified (from M19/M3) 2. Assess competitor_strength + incumbent_satisfaction from signals/enrichment 3. Compute political_protection_index (relationships, regulatory favor, sponsorship)

M15 Rules (2)

M15-ROBUST-001
M15 selects the strategy that is robust across scenarios (strong expected outcome with bounded downside), not merely the highest best-case outcome. A high-mean / catastrophic-downside strategy is rejected in favor of a robust alternative. Downside weighting reflects brand/relationship capital at risk.

M15-CALIBRATE-001
M15 outputs are ADVISORY (labeled estimates) until M11 calibrates scenario predictions against observed outcomes. Until calibrated, M15 may recommend and flag fragile strategies but may not auto-reject an M4 strategy without human review. Calibration lifts this restriction per decision type.
Rationale: *#5 — Deepest reason M15 exists? Because mature strategy is not choosing the best plan — it is choosing the plan that survives the buyer reacting. M15 moves the system from picking a strategy to pressure-testing it, from optimism to robustness. It is the difference between a planner and a strategist.*

M16 Rules (5)

M16-TRACK-001
M16 processes conversational replies (post-M9) to extract structured signals: objections, commitments, sentiment shifts, and newly-emerged stakeholders. These enrich the Opportunity's committee and feed M10 handoff with conversation context, not just a 'positive reply' flag.
Rationale: *#2 — Why track stakeholder EMERGENCE during conversation? Because buying committees reveal themselves through dialogue — a CTO loops in security, procurement appears, a new sponsor is cc'd. These emergent stakeholders must be added to the opportunity's committee in real time, or the committee map goes stale exactly when the deal heats up.*

M16-STAKEHOLDER-001
Stakeholders emerging in conversation are added to the Opportunity's Buying_Committee (via M19/M3) with their observed stance. The committee map updates live during active conversations; emergence is a first-class signal that can reshape strategy.
Rationale: *#3 — Why must M16 be scoped to TRACKING before AUTONOMOUS handling at MVP? Because the FSD explicitly excludes autonomous negotiation from MVP, and autonomously conducting enterprise conversations carries high brand/relationship risk before the system has earned that trust. M16 starts by extracting and surfacing conversation intelligence FOR the human; autonomous handling is a later, trust-gated expansion.*

M16-SCOPE-001
At first activation M16 TRACKS and surfaces conversation intelligence for human handoff; it does NOT autonomously negotiate. Autonomous qualification/objection-handling/scheduling is a later capability, gated by Trust_Capital_Ledger autonomy for the conversation decision type and explicit human enablement.
Rationale: *#4 — Why feed conversation outcomes into learning (M11/M17)? Because which objections arise and how they resolve is high-value training signal — it teaches the system common objections, effective responses, and where strategies stall. Conversation data sharpens future reasoning, messaging, and the expert heuristics in M17.*

M16-LEARN-001
M16 emits objection/commitment/resolution data to M11 (attribution + pattern learning) and M17 (expert heuristics). Recurring objections inform Decision_Patterns and message strategy; unresolved objection types surface as Capability_Gaps.

M16-PRIMACY-001
Once active, M16 is the authority on conversation-state intelligence between engagement and handoff. It enriches but never bypasses M10 human handoff at MVP; autonomous action requires earned autonomy and is out of MVP scope. ① Workflow Logic (step-by-step) 1. Reply/conversation event from M9 received 2. Extract objections, commitments, questions, sentiment shift 3. Detect emergent stakeholders → add to Opportunity committee (via M19/M3) 4. Update conversation state on the Opportunity 5. Surface structured conversation context to M10 for informed handoff 6. Emit objection/commitment/resolution data to M11 + M17 7. (POST-MVP, trust-gated) handle qualification/objection/scheduling autonomously ② Entities Consumed / Written Reads (consumes) Wr...

M17 Rules (4)

M17-EXTRACT-001
M17 analyzes Human_Review_Records to detect recurring reviewer patterns and abstracts them into named heuristics (e.g. 'Reviewer-X Heuristic #14: reject engagement when readiness < 60'). Heuristics carry the evidence count and the reviewer they derive from.
Rationale: *#2 — Why attribute heuristics to specific reviewers rather than blend them? Because reviewers have different expertise and sometimes disagree. Blending them into one anonymous policy loses the signal of WHO is reliable on WHAT, and hides genuine expert disagreement that itself is informative. Attributed heuristics preserve the source of judgment.*

M17-ATTRIBUTE-001
Heuristics are attributed to their source reviewer and decision domain. Conflicting heuristics between reviewers are retained and surfaced (not silently merged), preserving expert disagreement as signal. Reviewer reliability per domain (from M10 outcomes) weights heuristic confidence.
Rationale: *#3 — Why promote stable heuristics toward Decision_Patterns and candidate rules? Because a heuristic confirmed across many decisions and validated by outcomes is institutional knowledge that should inform the system broadly — not stay locked to one reviewer's queue. Promotion is how individual expertise becomes organizational capability.*

M17-FEEDBACK-001
Validated heuristics inform M4 reasoning (so AI proposals pre-satisfy known reviewer expectations) and M10 review routing (auto-handling cases that established heuristics clearly resolve, escalating genuinely novel ones). This reduces rubber-stamping pressure by removing trivial reviews.
Rationale: *#5 — Deepest reason M17 exists? Because an organization's hardest-won asset is the judgment of its experts, and a system that lets that judgment die in individual decisions wastes it. M17 institutionalizes expertise — turning what a reviewer knows into what the system knows. It is the human-side mirror of strategic memory.*

M17-PRIMACY-001
M17 converts human judgment into reusable institutional knowledge under governance. Extracted heuristics inform but never silently override explicit rules or current human review; promotion to rules is always governed (parallels PATTERN-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Human_Review_Records accumulate from M10 2. Mine rationales for recurring decision patterns per reviewer + domain 3. Abstract stable patterns into named, attributed heuristics with evidence count 4. Validate heuristics against eventual outcomes (reviewer reliability weighting) 5. Surface conflicting heuristics between reviewers (do not merge) 6. Promote outcome-validated heuristics → Decision_Patterns / candidate rules (governed) 7. Feed validated heuristics t...

M18 Rules (3)

M18-OBSERVE-001
M18 ingests and learns from third-party market events (competitor wins, public procurement awards, industry announcements, implementation successes/failures) that do not involve Decimal. These become learning signal alongside Decimal's own outcomes (M11).
Rationale: *#2 — Why does market learning especially strengthen Tier 6 (M12)? Because narrative, dependency, and DNA are market-level judgments best learned from market-wide events. A competitor's win at one bank predicts narrative spread and reference effects across the segment; an implementation failure teaches which approaches fail. M18 is the primary external feed into strategic memory.*

M18-COMPETITIVE-001
M18 routes observed competitor wins/losses/failures to M14, updating competitor strength, displacement probability, and replacement triggers from real market outcomes rather than only account-local signals.
Rationale: *#5 — Deepest reason M18 exists? Because a system that learns only from its own actions learns slowly and narrowly, while the market runs a continuous experiment it could learn from for free. M18 turns the whole market into a teacher — the single biggest accelerant to the system's judgment. It is learning at market scale.*

M18-PRIMACY-001
M18 broadens the learning loop beyond Decimal's actions. All market-derived lessons flow through the same governed, bounded, reversible machinery as internal learning (M11) and inherit the same guardrails; inferred lessons are confidence-discounted and never override observed fact. ① Workflow Logic (step-by-step) 1. Ingest third-party market events (news, procurement awards, announcements, failure reports) via M1/M0 2. Classify event type + extract observed facts (who, what, when) 3. Infer lessons (timing/stakeholder/vendor/narrative) with explicit confidence 4. Route: narrative/dependency/DNA → M12; competitor moves → M14; patterns → M11 5. Apply lessons at confidence-discounted weight through M11's bounded machinery 6. Retain market event...

M19 Rules (5)

M19-GRAPH-001
M19 owns construction and maintenance of the per-account opportunity graph: it creates Opportunity entities, assigns committees/windows/incumbents/budgets to the correct Opportunity, and maintains influence paths between nodes. M3 supplies facts; M19 partitions them into journeys.
Rationale: *#2 — Why model stakeholders, vendors, and budgets as NODES rather than fields? Because the same stakeholder can influence multiple opportunities, the same incumbent can defend several, and budgets can be shared or contested across journeys. A node-and-edge graph captures these many-to-many realities that flat fields on a single Opportunity cannot.*

M19-NODES-001
The opportunity graph contains typed nodes (Opportunity, Stakeholder, Vendor, Budget, Initiative) and influence edges. A stakeholder node may connect to several Opportunity nodes; a budget node may fund or be contested across journeys. Cross-opportunity influence is first-class.

M19-RESOLVE-001
M19 resolves each blessed signal to the correct Opportunity within the account (creating a new Opportunity when none fits). Ambiguous signals attach at account level pending resolution, never mis-assigned to an arbitrary journey. This is opportunity-resolution, parallel to M1 entity-resolution.
Rationale: *#4 — Why does M19 sit upstream of scoring (M2) and strategy (M4) now? Because M2 and M4 now operate per opportunity — they cannot score or strategize until M19 has established WHICH opportunity the intelligence belongs to. M19 becomes part of the mandatory pipeline: detect → bless → resolve-to-opportunity → score → reason.*

M19-PIPELINE-001
M19 runs after M0 blessing and before M2 scoring in the re-anchored pipeline. Scoring, windows, committees, and strategy all key off the Opportunity M19 assigns. No per-opportunity decision may proceed on intelligence M19 has not yet resolved to a journey.
Rationale: *#5 — Deepest reason M19 exists? Because it makes the system see the account the way the account actually is — a portfolio of parallel buying journeys — rather than a single averaged narrative. M19 is what turns 'SNB is one deal' into 'SNB is four deals, each at a different stage against a different incumbent.' It is the structural truth the whole re-anchored architecture rests on.*

M19-PRIMACY-001
M19 is the authority on opportunity structure within an account. Account-level roll-ups are derived from M19's graph; no module may treat an account as a single opportunity when M19 has decomposed it into several. ① Workflow Logic (step-by-step) 1. Blessed signal / enrichment arrives for an account (from M0/M3) 2. Opportunity-resolution: does this belong to an existing Opportunity, or spawn a new one? 3. If new → create Opportunity (type, linked initiatives); if existing → attach 4. Assign/Update nodes: stakeholders → committees, vendors → incumbents, budgets → funding 5. Maintain influence edges (cross-opportunity stakeholder/budget links) 6. Emit per-Opportunity events to M2 (score), M5 (window), M4 (strategy), M13 (economics) 7. Maintain...

M2 Rules (5)

M2-CENTRAL-001
M2 is the sole authority that writes Three_Score_Record. All prioritization across the system reads M2's stored scores; no module computes its own private score. Every run stamps the weight_set_version used.
Rationale: *#2 — Why compute three scores and two equations rather than a single priority number? Because fit, intent, and confidence are orthogonal, and opportunity vs. decision are different questions. The Effective Opportunity equation answers 'how attractive is this?'; the Decision Score answers 'should we spend capital now?'. Collapsing them hides why an account ranks where it does and lets cheap-but-weak beat expensive-but-strong incorrectly.*

M2-EQUATION-001
M2 computes Effective Opportunity = Dynamic x (ICS/100) x Stage x Budget x Entrenchment x Risk x Window, then Decision Score = Effective Opportunity x (Readiness/100) / Capital Cost. Both are stored with full dimension breakdown. ICS gates intent so low-confidence cannot inflate priority.
Rationale: *#3 — Why must M2 retrigger on events rather than run on a schedule? Because a stale ranking misallocates capital. When a new signal is blessed, a window opens, engagement arrives, or readiness changes, the affected account's score must immediately recompute so the priority queue always reflects current reality.*

M2-RETRIGGER-001
M2 subscribes to events (blessed signal, window state change, engagement, readiness/strategy change, org-assessment change) and recomputes the affected account's scores on each. Account.current_state cache is updated only by M2, emitting a System_Event per recompute.
Rationale: *#4 — Why store the full dimension breakdown on every run? Because explainability is non-negotiable — a human reviewer and an audit must be able to see why an account is HOT. The breakdown also enables monthly recalibration: you cannot improve weights you cannot trace.*

M2-EXPLAIN-001
Every M2 run writes per-dimension contributions (input, raw value, weight, contribution) into Three_Score_Record. A score without breakdown is invalid and rejected. Breakdowns feed M11 recalibration.
Rationale: *#5 — Deepest reason M2 exists? Because attention is the binding constraint and must be invested where return is highest. M2 is the system's prioritization conscience — the module that decides, transparently and consistently, what deserves the system's finite capital next. It converts knowledge into ranked intent.*

M2-PRIMACY-001
No sequencing, allocation, or outreach may reference a priority not produced and stored by M2. Rankings on ephemeral or module-local scores are prohibited (per SCORE-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Subscribed event received (blessed signal / window change / engagement / readiness change)

M3 Rules (4)

M3-TRIGGER-001
M3 enriches on M2 HOT-tier detection (or explicit human request). Speculative enrichment of COLD accounts is rate-limited and requires justification. Enrichment scope scales with Decision Score — higher score earns deeper enrichment.

M3-GOVERN-001
M3 submits enriched facts to M0 for provenance, freshness, and conflict reconciliation before they are usable. Each fact carries source + source_date; conflicts are reconciled by source reliability with both values retained (ENRICH-RECONCILE-001).
Rationale: *#4 — Why does M3 feed both scoring (M2) and strategy (M4)? Because enrichment changes both how attractive an account is and how to approach it. New committee data and initiative intelligence retrigger scoring AND inform the reasoning chain. M3 is the bridge that deepens both prioritization and strategy.*

M3-FEED-001
On enrichment completion, M3 emits events that retrigger M2 scoring (new dimensions available) and notify M4 (committee + initiative + vendor intelligence ready for strategy). Enrichment is not complete until these events fire.
Rationale: *#5 — Deepest reason M3 exists? Because the system must convert the unknown into the known before it can act wisely. M3 is the research faculty — the difference between reaching out to a name and reaching out to a understood organization with a mapped committee and a known initiative. It earns the right to personalize.*

M3-PRIMACY-001
No persona inference, pain hypothesis, or committee strategy may be built on facts lacking M3-sourced, M0-blessed provenance (per ENRICH-PRIMACY-001). Unsourced claims are hypotheses, not evidence. ① Workflow Logic (step-by-step) 1. HOT-tier event from M2 (or human request) received 2. Determine enrichment depth from Decision Score (capital-scaled) 3. Fetch company/firmographic data (Apollo, Clearbit, Clay, LinkedIn) 4. Identify decision-makers → build Buying_Committee_Member records 5. Verify contact emails; resolve reachability (email vs LinkedIn-only) 6. Gather initiative, vendor/incumbent, and org-stability intelligence 7. Submit all facts to M0 (provenance, freshness, conflict reconciliation) 8. On bless: persist Enrichment + linked en...

M4 Rules (5)

M4-SEPARATE-001
M4 produces the Reasoning_Chain and Account_Strategy BEFORE M7 generates any message. Generation without a complete, M4-produced reasoning chain is rejected (per REASON-PRIMACY-001 and MESSAGE-TRACE-001).
Rationale: *#2 — Why must M4 maintain a standing Account_Strategy above individual reasoning chains? Because sequences are tactics and the account needs one coherent strategy they all serve. Without a standing strategy, two touches can contradict each other and the account experiences an incoherent Decimal. The strategy holds the angle, posture, committee targets, connector path, and readiness gates.*

M4-STRATEGY-001
M4 maintains one active Account_Strategy per engaged account (value angle, competitive posture, target committee members, connector path, readiness gates, anchor initiative/window). All Reasoning_Chains and downstream sequences must conform to it (per STRAT-PRIMACY-001).
Rationale: *#3 — Why does M4 consume Organizational_DNA and Market_Narrative, not just signals? Because strategy is about character and context, not just events. The same signal demands a different strategy for a slow-consensus, governance-heavy account (SNB pattern) than for a fast-mover, and a different framing inside an accelerating narrative than a vacuum. M4 conditions its reasoning on Tier 6 judgment.*

M4-CONDITION-001
M4 conditions strategy on Organizational_DNA (pacing, consensus difficulty, displacement feasibility) and Market_Narrative alignment/phase (framing, urgency). A strategy contradicting known DNA (e.g. fast-close on a slow-consensus account) is flagged for justification (DNA-PRIMACY-001).

M4-CHAIN-001
M4 executes the mandatory reasoning sequence (signal interpretation → account research → pain inference → persona analysis → strategy decision → [handoff to M7 generation] → QC) storing each step's input/output/confidence. Any skipped or low-confidence step invalidates the chain (REASON-STEPS-001).
Rationale: *#5 — Deepest reason M4 exists? Because the difference between an intelligent system and a template engine is whether there is a defensible thought process behind every action. M4 is the strategic mind — it ensures the system always knows WHY it is about to act, can explain it, and can improve it. It is reasoning made institutional.*

M4-PRIMACY-001
No message, sequence step, or strategic action may exist without an M4-produced, complete, QC-passing Reasoning_Chain conforming to the active Account_Strategy. This is enforced at the data layer, not by convention. ① Workflow Logic (step-by-step) 1. Enrichment-ready event from M3 (or strategy-review trigger) received 2. Step 1 Signal interpretation: what happened, why it matters to Decimal 3. Step 2 Account research: priorities, initiative, incumbency, DNA, narrative alignment 4. Step 3 Pain inference: operational pains implied by signals + research 5. Step 4 Persona analysis: target committee member motivations, decision style, risk posture 6. Step 5 Strategy decision: angle, value prop, ask, connector path, timing intent 7. Build/refresh...

M5 Rules (5)

M5-WINDOW-001
M5 owns Buying_Window creation, state transitions (OPENING/PEAK/CLOSING/CLOSED), and the Window Multiplier. Windows open on qualifying signals intersecting account state (and narrative phase); the multiplier peaks at PEAK and decays toward CLOSING (per WINDOW-STATE-001).
Rationale: *#2 — Why predict window closure rather than just observe it? Because reacting after closure is too late — that is the difference between timing as weapon and timing as luck. M5 must forecast decay so action is scheduled before the window shuts, raising execution urgency as closure approaches.*

M5-PREDICT-001
M5 maintains estimated_closure with confidence per window, updated as signals age and via M11 timing calibration. Approaching closure raises execution urgency in the decision sequence and can pre-empt lower-priority work (WINDOW-PREDICT-001).
Rationale: *#3 — Why does M5 read Market_Narrative phase, not just account signals? Because a window opened by a signal riding an accelerating narrative is stronger than the same signal in a vacuum. Narrative phase amplifies or dampens the window — timing is about the market's tempo, not just the account's.*

M5-NARRATIVE-001
M5 amplifies the Window Multiplier when the opening signal aligns with an ACCELERATING Market_Narrative and dampens it for a DECLINING one (per NARR-PHASE-001). Narrative phase is a first-class input to window strength.
Rationale: *#4 — Why does M5 gate execution rather than just inform scoring? Because the decision sequence has a distinct timing step (step 5) that asks 'now or later?' independently of 'how attractive?'. M5 answers it: even a high-opportunity account may wait if its window is merely opening, and a closing window may jump the queue.*

M5-GATE-001
M5 provides the timing decision in the Universal Decision Sequence. No 'now vs later' decision may be made without consulting the account's active window state (WINDOW-PRIMACY-001). Closing windows can re-prioritize execution order ahead of higher raw scores.
Rationale: *#5 — Deepest reason M5 exists? Because in enterprise selling, being right at the wrong time is being wrong. M5 is the system's sense of timing — the discipline that turns 'who to contact' into 'who to contact NOW.' It is timing-as-weapon made into running code.*

M5-PRIMACY-001
M5 is the authority on engagement timing. Execution (M8) may not start or advance a sequence step against M5's window/timing guidance without explicit override. Timing is enforced, not advisory. ① Workflow Logic (step-by-step)

M6 Rules (5)

M6-ALLOCATE-001
M6 produces a Portfolio_Allocation each cycle, distributing finite capital (attention slots, connector asks, daily account starts) across accounts. No account consumes system capital outside the active allocation (PORT-PRIMACY-001); exhausted budgets queue work to the next cycle.

M6-EROA-001
M6 ranks accounts by EROA and excludes negative-EROA accounts for the cycle (PORT-EROA-001). Capital cost aggregates attention, connector, brand, and system costs per SCARCITY-001.
Rationale: *#3 — Why does M6 consume the Market_Dependency_Graph? Because opportunities are correlated, not independent — winning SNB lifts SAB/Riyad/Alinma. A keystone account that unlocks references can beat a higher-standalone-EROA isolated account. M6 must value an account partly by its effect on the rest of the portfolio (true portfolio theory).*

M6-DEPENDENCY-001
M6 incorporates Market_Dependency_Graph effects: an account's effective value includes expected lift to dependent accounts (keystone value), and negative dependencies (reputation contagion) raise its risk weight. Dependency influence is capped while edges are ASSUMED (DEP-CONFIDENCE-001).
Rationale: *#4 — Why enforce concentration limits and reconcile spend each cycle? Because all-capital-on-one-whale risks total loss and over-concentration on one connector burns the asset. Limits manage portfolio risk; reconciliation of planned vs. actual spend catches overspend and feeds better future allocation.*

M6-DISCIPLINE-001
M6 enforces concentration limits per account/connector (PORT-DIVERSIFY-001) and reconciles planned vs actual capital spend each cycle (PORT-RECONCILE-001). Systematic overspend triggers M11 cost-model recalibration.
Rationale: *#5 — Deepest reason M6 exists? Because the system must behave like a disciplined investor of finite, irreplaceable capital — not a sprayer of attention. M6 embodies scarcity economics: the recognition that attention is the binding constraint and must be invested where total, correlated return is highest. It is portfolio judgment made operational.*

M6-PRIMACY-001
Sequence starts (M8) and connector asks draw down M6-allocated budgets. M6 is the sole authority on cross-account capital distribution; modules may request capital but not self-allocate beyond their cycle budget. ① Workflow Logic (step-by-step) 1. Cycle start (or significant re-rank event) → gather candidate accounts with Decision Scores 2. Compute Total Capital Cost per account (attention/connector/brand/system) 3. Compute EROA; exclude negative-EROA accounts 4. Apply Market_Dependency_Graph: add keystone lift, raise risk for negative blast-radius 5. Apply concentration limits per account/connector 6. Produce Portfolio_Allocation (capital per account, connector ask budgets, daily starts) 7. Release allocated budgets to M8 (execution) and c...

M7 Rules (5)

M7-SEPARATE-001
M7 generates message content strictly FROM an M4 Reasoning_Chain; it may not introduce strategy or facts absent from the chain. Each generated Message links its reasoning_chain_id, prompt_version, and model_id (MESSAGE-AI-PROVENANCE-001).
Rationale: *#2 — Why must every generation pass QC before it can be queued? Because hallucination is a named failure mode and a single fabricated fact can destroy credibility with an auditor-minded buyer. QC (hallucination score, factual grounding, tone, enterprise suitability) is a hard gate, not a suggestion.*

M7-QC-001
M7 runs mandatory QC on every draft: hallucination_score, factual-grounding against blessed enrichment, tone, and enterprise suitability. Drafts failing thresholds are rejected (status QC_REJECTED, retained for analysis) and never queued (REASON-QC-001).
Rationale: *#3 — Why ground every claim against M0-blessed enrichment specifically? Because a message may only assert what the system actually, verifiably knows. Grounding generation against blessed facts (not the model's parametric guesses) is how the system stays truthful with high-stakes buyers.*

M7-GROUND-001
Every factual claim in a generated message must trace to an M0-blessed Enrichment fact or Signal. Ungrounded claims are stripped or the draft is rejected. The model may not introduce unverifiable specifics (numbers, names, dates) absent from the intelligence base.
Rationale: *#4 — Why version prompts and route generation through the registry? Because phrasing performance must be measurable and reversible. Tying each message to a Prompt_Version enables A/B testing, outcome attribution, and instant rollback of a prompt that damages brand at scale.*

M7-PROMPT-001
M7 generates only via registered Prompt_Versions and approved AI_Model_Registry models for the decision type. Outcomes attribute to prompt_version for M11 optimization; prompts affecting max-tier contacts require human approval before activation (PROMPT-GOVERNANCE-001).

M7-PRIMACY-001
No Message exists without an M7 generation linked to a Reasoning_Chain and a registered Prompt_Version, having passed QC. Untracked or un-QC'd message creation is prohibited (MESSAGE-TRACE-001, PROMPT-PRIMACY-001). ① Workflow Logic (step-by-step) 1. Generation request from M4 with completed Reasoning_Chain + target channel/persona/language 2. Select registered Prompt_Version + approved model for the decision type 3. Generate draft content grounded strictly in chain + blessed enrichment 4. QC: hallucination score, factual grounding, tone, enterprise suitability, language correctness 5. Decision: PASS → status QUEUED (hand to M8) / FAIL → status QC_REJECTED (retain, optionally regenerate) 6. Persist Message (content, provenance, QC results, c...

M8 Rules (5)

M8-SEQUENCE-001
M8 executes all outbound through Outreach_Sequence with hard limits: max email/LinkedIn touches per contact, minimum inter-touch gaps, the 14/10-day spreads, and max new accounts/day. Ad-hoc out-of-sequence sends are prohibited except logged human one-offs (SEQ-PRIMACY-001).
Rationale: *#2 — Why re-evaluate each step at fire-time rather than execute the plan blindly? Because state changes between planning and sending — a window closes, a contact leaves, sentiment drops, the strategy is revised. A step must re-check current state and chain freshness before firing, regenerating or cancelling if stale.*

M8-REEVAL-001
Before each step fires, M8 re-evaluates account/contact/window state, Account_Strategy conformance, and Reasoning_Chain freshness. Stale or invalidated steps are regenerated (via M4/M7) or cancelled, never sent blindly (SEQ-REEVAL-001).
Rationale: *#3 — Why bind execution to M5 timing and M6 capital, not just per-contact limits? Because per-contact limits aren't enough — the system has global capacity (daily starts) and timing imperatives (closing windows). Execution must respect both the portfolio budget from M6 and the timing guidance from M5.*

M8-CONSTRAINTS-001
M8 sequence starts draw down M6-allocated budgets and respect M5 timing guidance. When global capacity is exhausted, starts queue by Decision Score (not FIFO). Closing-window urgency from M5 can reorder the queue.
Rationale: *#4 — Why does M8 enforce exit conditions as first-class logic? Because a sequence that ignores responses is spam and burns brand. Execution must halt immediately on reply, negative sentiment, window closure, or do-not-contact — exit logic is core to the module, not an afterthought.*

M8-EXIT-001
M8 enforces sequence exit conditions (reply, negative sentiment, window closed, do-not-contact, fatigue). Any exit immediately pauses/terminates remaining steps. Contact-level fatigue and Market_Interaction_History gates are checked before every send (CONTACT-GOVERNANCE-001).
Rationale: *#5 — Deepest reason M8 exists? Because execution is where strategy meets reality and brand capital is actually spent. M8 operationalizes timing and restraint — paced, condition-aware contact rather than a burst of messages. It is the discipline that the system always sends the right touch, to the right person, at the right cadence, or not at all.*

M8-PRIMACY-001
All market contact flows through M8 under these constraints. No module may dispatch outbound messages directly; M8 is the sole execution authority and the sole writer of SENT status. ① Workflow Logic (step-by-step) 1. Queued message + Outreach_Sequence plan received; sequence start checked against M6 budget + M5 timing 2. Pre-send gate: re-evaluate state, strategy conformance, chain freshness, contact fatigue/DNC/history 3. If stale → route back to M4/M7 for regeneration; if exit condition met → terminate 4. Dispatch via channel tool (email: Instantly/Smartlead; LinkedIn: Heyreach) respecting gaps 5. Record SENT Message; debit capital; emit sent System_Event 6. Schedule next step with required gap; await engagement (M9) or exit 7. On engage...

M9 Rules (5)

M9-CAPTURE-001
M9 captures recipient/third-party-observed actions as Engagement_Event entities (action_type, observed_via, occurred_at), distinct from M8's outbound Messages. Where traceable, each links triggering_message_id (ENGAGE-ATTRIBUTION-001).
Rationale: *#2 — Why treat engagement as a first-class signal that retriggers scoring? Because engagement IS inbound buying intent — a reply or repeated profile views should immediately update the account's score and may open or widen a window. Treating engagement as passive analytics would waste the strongest real-time intent signal the system has.*

M9-SIGNAL-001
High-intent Engagement_Events (reply, meeting-accept, repeated content interaction) retrigger M2 scoring and notify M5 to open/widen a Buying_Window. Engagement is a first-class signal class, processed near-real-time, not batched (ENGAGE-TRIGGER-001).
Rationale: *#3 — Why store engagement as discrete events rather than aggregate counts? Because timing and sequence carry meaning — three opens in an hour signal different intent than three over three weeks. Discrete, timestamped events preserve the temporal pattern (velocity, clustering, recency) that aggregates destroy.*

M9-GRANULAR-001
M9 stores discrete timestamped Engagement_Events, never pre-aggregated counters. Velocity/clustering/recency are computed at read time (ENGAGE-GRANULARITY-001). Sentiment on replies updates Contact.interaction_lineage and Market_Interaction_History.

M9-FEED-001
M9 emits engagement to M8 (sequence exit/advance), M2 (rescore), M5 (window), and M11 (Feedback_Loop_Entry). Absence of engagement after N touches is itself emitted as a learning signal (triggers sequence pause / strategy review per ENGAGE-LEARNING-001).
Rationale: *#5 — Deepest reason M9 exists? Because it is the system's only feedback from reality about whether its strategy worked — the sensory return path that makes the system a learning loop, not a one-way broadcaster. M9 is what makes this intelligence rather than automation.*

M9-PRIMACY-001
Every recipient action that the system can observe is captured by M9. No other module infers engagement independently; M9 is the sole authority on engagement state, feeding all reactive and learning modules. ① Workflow Logic (step-by-step) 1. Engagement webhook/poll received from channel tools (open, click, reply, profile view, meeting accept) 2. Normalize → Engagement_Event; classify intent (HIGH/MEDIUM/LOW) 3. Link triggering_message_id where traceable; persist discrete event 4. If reply: run sentiment; update Contact.interaction_lineage + Market_Interaction_History 5. Emit to M8 (exit/advance), M2 (rescore), M5 (window), M11 (feedback) 6. If positive intent → trigger M10 human handoff path 7. If no engagement after N touches → emit learn...

OPP Rules (4)

OPP-CONTAINER-001
Account retains account-level state shared across opportunities: Organizational_DNA, Market_Interaction_History, Connector access, relationship capital, and lifecycle. Opportunity inherits these from its parent Account but maintains its own committee, budget, window, incumbent, strategy, and score. Shared-vs-specific is the dividing line.

OPP-INITIATIVE-001
Opportunity links to one or more Initiatives but is not identical to one. An Initiative is account-side reality (intelligence); an Opportunity is Decimal's addressable journey (pursuit). The mapping may be one-to-one, one-initiative-to-many-opportunities, or many-initiatives-to-one-opportunity.
Rationale: *#4 — Why must opportunities within one account be scored and sequenced independently? Because they compete for different budgets and have different windows and incumbents. Decimal might pursue SNB's Open Banking opportunity aggressively (open window, weak incumbent) while holding its Digital Lending opportunity (entrenched incumbent, closed window). One account, opposite strategies. Independent scoring is what makes that possible.*

OPP-INDEPENDENT-001
Each Opportunity carries its own Three_Score_Record, Buying_Window, and Account_Strategy (now Opportunity_Strategy). Portfolio allocation (M6) ranks OPPORTUNITIES, not accounts, so two journeys in the same account can receive opposite postures and capital levels simultaneously.
Rationale: *#5 — Deepest reason the Opportunity re-anchoring matters? Because it aligns the system's fundamental unit with how enterprise buying actually works — many parallel journeys per organization. Without it, the system is structurally blind to most of the real pipeline inside a large account, and will repeatedly mis-read one journey's signal as another's. It is the correction that lets the system see the full opportunity surface, not a flattened account average.*

OPP-PRIMACY-001
Account-level aggregation (e.g. 'SNB is HOT') is a derived ROLL-UP of its opportunities, never a primary score. Any decision, score, window, or strategy that previously resolved to an Account MUST now resolve to a specific Opportunity, except for the explicitly account-level shared state in OPP-CONTAINER-001. OPPORTUNITY — Field Specification Field Type Business Rule Source opportunity_id UUID Immutable; the new decision anchor OPP-ANCHOR- 001 account_id UUID FK to parent Account (container) OPP-CONTAIN ER-001 opportunity_name String e.g. 'SNB — Open Banking origination' OPP-ANCHOR- 001 linked_initiative_ids Array[UUID] One or many Initiatives (not identical) OPP-INITIATIVE -001 opportunity_type Enum OPEN_BANKING / API_MGMT / DIGITAL_LENDIN...

ORCH Rules (5)

ORCH-ROUTER-001
n8n routes System_Events between modules, schedules time-based triggers (poll cadences, allocation cycles), and integrates external tools (email, LinkedIn, CRM, enrichment APIs). Modules communicate via events through the orchestrator, not direct coupling.
Rationale: *#2 — Why must n8n hold NO business logic? Because scoring, reasoning, timing, and allocation logic must be versioned, testable, and governed in services and the Rule_Registry — not buried in opaque workflow nodes. Mixing logic into orchestration makes it untestable and ungovernable.*

ORCH-NOLOGIC-001
n8n contains orchestration ONLY: routing, scheduling, integration. It must NOT contain scoring, reasoning, AI generation, conflict resolution, or any business logic — those live in services governed by the Rule_Registry (per the original n8n philosophy).
Rationale: *#3 — Why route everything through System_Events rather than direct calls? Because the event log is the system's audit spine and coordination mechanism at once. Routing via events means every cross-module action is traceable, causally linked, and reconstructable — and the denormalized caches can always be rebuilt from the log.*

ORCH-EVENT-001
All cross-module coordination flows through System_Events (immutable, causally linked per SYSEVENT-COORD-001). The orchestrator subscribes/publishes events; any cache must be reconstructable from the event log (SYSEVENT-PRIMACY-001).
Rationale: *#4 — Why does the orchestrator enforce the pipeline order (M0 first, etc.)? Because governance and reasoning gates only work if they cannot be bypassed. The orchestrator wires the mandatory sequence — M1 detect → M0 bless → M2 score → M3 enrich → M4 reason → M7 generate → M10 review (if gated) → M8 send → M9 engage → M11/M12 learn — so no stage can be skipped.*

ORCH-PIPELINE-001
The orchestrator enforces mandatory pipeline gates: no scoring before M0 blessing, no generation before M4 reasoning, no send before M8 (and M10 review where gated). These ordering constraints are wired in orchestration and cannot be bypassed by any module.
Rationale: *#5 — Deepest reason the orchestration layer exists? Because a system of independent intelligent modules needs a reliable, traceable nervous system to make them act as one coherent whole — without coupling them or hiding logic where it can't be governed. n8n is the connective tissue that keeps the system both modular and coordinated.*

ORCH-PRIMACY-001
The orchestrator is coordination infrastructure, not an actor. It may route, schedule, and integrate, but it never makes business decisions. Authority always rests with the modules and the Rule_Registry, never the orchestration layer. ① Workflow Logic (step-by-step) 1. Subscribe to System_Events from all modules 2. Schedule time-based triggers (signal poll cadences, M6 allocation cycle, baseline checks) 3. On event → route to the correct subscribing module(s) per pipeline rules 4. Integrate external tools: enrichment APIs, email (Instantly/Smartlead), LinkedIn (Heyreach), CRM (HubSpot) 5. Enforce mandatory ordering gates (M0/M4/M8/M10) — reject out-of-order requests 6. Log every routing action as part of the System_Event spine ② Entities Co...

Section 4: Integrations, APIs, Failure Handling (93 rules)

EPIS Rules (18)

EPIS-SRC-01
Every source carries TWO independent scores: Reliability (accuracy track record) and Incentive-Bias (motive to distort — success/sell/enforce/attention). They are never collapsed into one number.

EPIS-SRC-02
Incentive-Bias is applied when a source's framing (not just its facts) feeds a hypothesis; high-bias framing cannot raise a hypothesis's confidence on its own.

EPIS-COV-01
Every account carries an Intelligence Coverage Score = observable share of its buying system (executives/vendors/signals visible vs budget/politics/opposition/priorities invisible).

EPIS-COV-02
Coverage caps confidence: low coverage mechanically reduces the RCM confidence of any account-level recommendation; the engine cannot be more certain than its coverage allows.

EPIS-COV-03
Every recommendation declares Knowns, Unknowns and Critical Unknowns (the unknowns that would most change the decision), feeding Principle 3's uncertainty-reduction loop.

EPIS-VOI-01
Every intelligence action carries an Acquisition Cost (API + human + compute + time).

EPIS-VOI-02
Information is acquired only if Expected Decision Improvement > Acquisition Cost (value-of-information gate); below threshold, the engine deliberately stays uninformed and records why.

EPIS-HALF-01
Every intelligence category has an expected decay curve (semantic half-life), e.g. job posting ~30d, conference speech ~180d, vendor contract ~365d, organization identity ~5y — replacing generic freshness TTL.

EPIS-HALF-02
Decay state is part of the RCM stamp; an aged belief loses confidence on its own curve, not a global clock.

EPIS-TRUST-01
Trust is measured PER subsystem (intelligence, enrichment, scoring, AI, automation, data quality), each with Trust Score + Error Rate + Human-Override Rate; trust loss in one does not collapse the others.

EPIS-TRUST-02
A subsystem whose override rate or error rate breaches threshold is flagged low-trust; its outputs are de-weighted and surfaced for review, not silently trusted.

EPIS-RCM-01
Every fact, hypothesis and recommendation carries an RCM stamp: confidence + supporting evidence + contradicting evidence + source reliability/bias + coverage + decay state.

EPIS-RCM-02
RCM governs the INT-SIG reasoning streams (4.1): no SIG-HYP/POWER/TENSION/RELEVANCE/VSAT output is surfaced or fed to scoring without an RCM stamp.

EPIS-RCM-03
Confidence is calibrated and falsifiable: every confidence figure links to the evidence that produced it and the evidence that would lower it (‘what would change our mind’). Free-floating AI-generated percentages are rejected.

EPIS-RCM-04
Low-confidence beliefs are labelled, de-weighted or withheld — never presented to a human as established fact (anti-false-confidence, ties to Principle 3 and Module 0 hallucination governance).

EPIS-RCM-05
The engine never fabricates confidence: absence of evidence is recorded as a Critical Unknown, not filled with an inferred value.

EPIS-ETH-01
Reasoning about individuals (influence, belief, tension) is bounded to PUBLICLY observable signals with sources; the engine never infers private, protected or sensitive attributes of a named person, and operates within PDPL (Constraint Hierarchy top tier).

EPIS-ETH-02
Political/power/tension intelligence is competitive analysis of public signals only; any inference lacking a public-source basis is recorded as ~inferred at capped confidence and excluded from automated action. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

FAIL Rules (2)

FAIL-LEARN-01
Every failure generates a Failure Pattern Record: root cause, precursor signals, recovery effectiveness, and prevention strategy — failures are treated as lessons, not just events. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes Rule ID Specification

FAIL-LEARN-02
Failure Pattern Records form a searchable organizational memory; recurring precursor patterns (e.g. signals preceding a LinkedIn ban) become predictive so the engine can pre-empt failures before they occur. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

GOV Rules (1)

GOV-ICS-001
CONFLICT RESOLUTION (FIN-C02): An account may not be promoted to HOT/TIER_1 unless its Intelligence Confidence Score (ICS) is >= 40. Below 40 the account remains in WATCH and enrichment continues. Human force-promotion below ICS 40 is permitted but triggers a logged SOFT WARNING (per AIHUMAN-DISAGREE-002). This is the rule previously referenced as GOV-001 in EXPLAIN-003. Rationale: the ICS>=40 floor was used consistently but its defining rule was unnamed.

INT Rules (42)

INT-ENR-01
Provider fallback chain is ordered Apollo → Clearbit → Clay; resolution stops at the first verified result. Order is configurable but the verified-only gate is not.

INT-ENR-02
Only verified emails make a contact email-eligible; unverified → LinkedIn-only outreach path (per FSD ENR decision logic).

INT-ENR-03
Enrichment is gated by tier + reachability: HOT/WARM and path-reachable accounts enrich first; COLD/unreachable are deferred.

INT-ENR-04
Cross-provider conflict-merge: most-recent + highest-source-trust wins; unresolved title/role conflicts are flagged ~ and excluded from automated send.

INT-ENR-05
Every enriched record carries source + timestamp + freshness TTL; on expiry the contact is re-enriched before reuse.

INT-ENR-06
Buying-committee coverage is required: single-threaded enrichment (one contact per account) is flagged as insufficient coverage to scoring.

INT-ENR-07
Enrichment cost/volume is budgeted and logged; spend per account is capped and visible to the Scarcity (capital) layer. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-LIN-01
Per-account daily caps on connection requests + DMs + profile views are enforced below LinkedIn's human-plausible thresholds; caps are configurable downward only.

INT-LIN-02
Ban-risk circuit breaker: on early-warning signals (captcha challenge, view restriction, accept-rate collapse, withdrawal spike) the account is auto-paused and flagged for human review.

INT-LIN-03
Channels are coordinated, not concurrent: email and LinkedIn touches to the same contact are spaced per sequence design; simultaneous same-day multi-channel contact is blocked.

INT-LIN-04
Sending-account assignment matches the resolved SIG-PATH where a P1/P2 Decimal-person path exists; round-robin only for P4/P5 cold paths.

INT-LIN-05
Connection-accept and DM-reply update the contact state machine (→ OPENED/REPLIED) and trigger reply-detection + handoff logic.

INT-LIN-06
Account warmup: new sending accounts ramp action volume gradually before reaching standard caps.

INT-LIN-07
A banned/restricted account immediately emits SIG-PATH path-break events for every P1 edge it anchored.

INT-EML-01
SPF, DKIM and DMARC must be configured and validated per sending domain before that domain is permitted to send.

INT-EML-02
Domain warmup is mandatory: new domains ramp volume on a schedule and cannot send at full volume until warmed.

INT-EML-03
Per-domain volume throttles + sending-identity rotation enforce a low-volume, high-relevance posture; mass blasting is structurally blocked.

INT-EML-04
Bounce, complaint and spam-placement telemetry feed a per-domain reputation score; below a floor, the domain is auto-paused from sending.

INT-EML-05
Only INT-ENR-verified emails may be sent to; unverified addresses are never sent email (LinkedIn-only path).

INT-EML-06
Reply received pauses the sequence and updates contact state (→ REPLIED) per FSD EXEC-002.

INT-EML-07
Duplicate-outreach prevention: the same contact is never sent the same touch twice across workflows (FSD EXEC-003).

INT-CRM-01
Sync is idempotent on (account_id, event_hash); re-delivery of an already-synced event is a no-op.

INT-CRM-02
Authority split: ABM is authoritative for signals/scores/engagement; CRM is authoritative for deal stage/owner. No field is dual-authoritative.

INT-CRM-03
Failed syncs route to a dead-letter queue with full payload + error; never a silent drop.

INT-CRM-04
On state conflict, CRM deal-stage wins and automated outreach to that account is paused pending human review.

INT-CRM-05
Reply / positive engagement triggers: pause automation → assign owner → generate AI meeting brief → Slack notify → create CRM task (FSD handoff sequence).

INT-CRM-06
CRM credentials are encrypted secrets; sync runs under least-privilege scopes (FSD security architecture).

INT-SLK-01
Handoff notification fires in real time on the engagement trigger, carrying the AI meeting brief.

INT-SLK-02
Brief payload contract: account, triggering signal, resolved warm path/owner, and suggested angle — all grounded in INT-SIG facts.

INT-SLK-03
Notifications are deduplicated and priority-ranked; volume is rate-limited to prevent alert fatigue.

INT-SLK-04
Delivery is confirmed; an unconfirmed/failed notification is retried then escalated, never silently dropped. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-ANL-01
Metabase reads from a read replica only; no analytical query touches the primary operational database.

INT-ANL-02
Analytics has no write access to operational tables; it is strictly read-only.

INT-ANL-03
KPI definitions are contracted and versioned (signal, persona, messaging, conversion performance) so the optimization loop tunes on stable metrics.

INT-ANL-04
Optimization outputs (weight/prompt changes) flow through a governed, reviewed, versioned loop — not applied directly from the BI layer. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

INT-ORC-01
n8n contains orchestration only: no business logic, no scoring logic, no AI reasoning. These live in versioned APIs/services.

INT-ORC-02
Every workflow step is idempotent; retries and re-runs cannot double-send outreach or double-write records.

INT-ORC-03
Workflows are grouped into Signal / Intelligence / Execution / Analytics domains with isolated failure boundaries.

INT-ORC-04
Every workflow follows the standard pattern: Trigger → Validation → Decision → Execution → Logging → State update.

INT-ORC-05
All external integration calls route through n8n so retry, rate-limit and observability policy are applied in one place.

INT-ORC-06
Workflow execution, decisions and errors are logged to the observability layer (OBS) on every run.

INT-ORC-07
n8n calls services over the defined API contracts (4.9); it never reaches into service databases directly. ABM Business Logic Bible — Section 4 · Integrations, APIs & Failure Modes

SIG Rules (30)

SIG-HYP-01
Every material signal stores competing explanations (hypotheses), each with a calibrated, RCM-stamped confidence; the engine reasons causally, not descriptively.

SIG-HYP-02
Hypothesis confidences are evidence-linked and falsifiable (EPIS-RCM-03); they are revised as new evidence arrives and never presented as settled fact.

SIG-HYP-03
Hypotheses feed Strategy reasoning (Principle 5) as explicit causal premises, not hidden assumptions. ACCESS-01 ACCESS_STRENGTH is modelled separately from path existence (SIG-PATH), scored on relationship freshness, trust, reciprocity and introduction likelihood. ACCESS-02 Effective path rank = path tier × access strength; a P3 with strong access can outrank a P1 with weak/stale access. Sender assignment uses effective rank. ACCESS-03 Access strength is sourced from observable interaction signals only; it never asserts a private relationship

SIG-PATH-01
Path priority is strictly ordered P1>P2>P3>P4>P5 (LinkedIn 1st-degree > mutual connection > shared node > generalized connector > cold email). Highest available path is selected for sender/route assignment.

SIG-PATH-02
Decimal employee-network ingestion: the engine maintains an opt-in graph of Decimal personnel LinkedIn connections + email domains, refreshed continuously, used as the P1/P2 resolution source. Personal network data is access-controlled and never exported off-platform.

SIG-PATH-03
Every warm path carries a decay TTL. On expiry the path is re-resolved; an unresolvable previously-valid path emits a path-break event.

SIG-PATH-04
Executive departure from a target emits BOTH a target-vacancy intent signal (to SIG-EXEC/scoring) AND a follow-the-person event (re-anchor the relationship at the new employer).

SIG-PATH-05
Cross-group power-node detection: any individual resolved as holding roles across ≥2 target/priority entities is flagged as a high-value node and surfaced to every account they touch.

SIG-PATH-06
IT-subsidiary gatekeeper rule: where a target has a wholly-/majority-owned implementation arm (EJADA-equivalent), no C-suite outreach is sequenced until the gatekeeper relationship is established or explicitly waived by a human.

SIG-PATH-07
Mutual-connection confidence = f(shared-node count, shared-node seniority, recency of connection). Paths below a confidence floor are demoted to P4/P5.

SIG-PATH-08
Generalized connector registry (KPMG/Mastercard/Deloitte/Tarabut/Fintech-Saudi) is maintained as reusable P4 bridges with confirmed-reach metadata; a connector's reach is per-bank-verified, never assumed global.

SIG-PATH-09
No warm path above P5 → account is flagged 'reachable: cold only'; scoring applies an unreachability penalty so attention is not over-allocated to unreachable targets.

SIG-PATH-10
SIG-PATH never fabricates a path or a mutual connection. Unverifiable paths are recorded as ~inferred and excluded from automated sender assignment until human-confirmed. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.2 SIG-EXEC — C-SUITE DOSSIER & SPEECH CAPTURE SIG-EXEC — C-Suite Dossier & Speech Capture

SIG-EXEC-01
Every C-suite/decision-maker contact resolves to a real named person where publicly findable, with a verification badge (✓✓ 2+ sources, ✓ 1 source, ~ inferred, ■ sources conflict, ✗ not found). Role-level desk only as labelled fallback with LinkedIn search string + email pattern.

SIG-EXEC-02
Public speeches, keynote quotes, interview statements are captured with source + date and stored as grounded personalization material linked to the named exec.

SIG-EXEC-03
Conflicting role-holders are flagged ■ verify and excluded from automated outreach until resolved; the engine never asserts one contested name confidently.

SIG-EXEC-04
Executive-move detection triggers: (a) dossier re-resolution, (b) target-vacancy intent signal, (c) SIG-PATH re-anchor / follow-the-person.

SIG-EXEC-05
Only SIG-EXEC-sourced, badged facts may be used as grounding for message personalization; ungrounded executive claims are rejected by QC. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.3 SIG-VENDOR & SIG-SUBS — VENDOR REGISTRY, GAP ANALYSIS & SUBSIDIARY ACTIVITY

SIG-VENDOR-01
Vendor registry is strictly per-account; a vendor fact confirmed at one bank is never propagated to another without independent confirmation (exclusivity discipline).

SIG-VENDOR-02
Each vendor entry carries a Decimal displacement/complement gap classification (displace · complement · no-fit) with source + freshness.

SIG-VENDOR-03
Vendor activity events (incumbent hiring surge, contract renewal/loss, case study) update the displacement-window state and may emit scoring triggers.

SIG-VENDOR-04
Displacement-gap magnitude + freshness feeds the account's Effective Opportunity, not only message content.

SIG-SUBS-01
Subsidiaries are resolved from the bank's official annual-report PDF (authoritative); subsidiary execs resolved to named persons (Phase-7 discipline) with badges.

SIG-SUBS-02
Subsidiary activity (launches, hires, regulatory moves) is ingested as account-linked signals; an IT/implementation subsidiary is additionally tagged as a SIG-PATH gatekeeper node. ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.4 SIG-NEWS · SIG-EVENT · SIG-FIN · SIG-REG — DISCLOSURE & ACTIVITY STREAMS

SIG-NEWS-01
All four streams normalize + entity-resolve + deduplicate into one account intelligence graph; the same event from N sources collapses to one badged fact (badge rises with corroborating sources).

SIG-NEWS-02
Every ingested item carries source + date + a freshness TTL; expired items are demoted from active scoring but retained in Market Memory.

SIG-REG-01
Source-trust weighting: regulator (SAMA/CMA/Tadawul) > bank official > tier-1 press > aggregator > forum/rumour. Low-trust-only facts cannot cross the scoring-trigger threshold alone.

SIG-FIN-01
Financial disclosures (quarterly results, annual report, rating changes) map to budget-availability signals consumed by Effective Opportunity's Budget Modifier.

SIG-REG-02
Regulatory circulars map to compliance-urgency windows with explicit open/close timing fed to the Timing module (Window Multiplier).

SIG-EVENT-01
Event/speaking detections cross-link to SIG-EXEC (capture the speech as grounded material) and to the Timing module (event proximity = engagement window). ABM Business Logic Bible — Section 4 · Checkpoint 4.1 · INT-SIG Account Intelligence Acquisition Engine 4.1.5 INT-SIG CONSOLIDATED RULE INDEX (34 new rules) All rules surfaced in this checkpoint, grouped by sub-stream. These are additive to the Section 1–3 rule base

Section 5: Resource Allocation Operating System (25 rules)

TIER-PHIL-001
A tier represents Decimal's Resource Commitment Level, not merely account attractiveness. HOT commits enrichment budget, SDR attention, founder visibility, AI compute, and review effort. WARM commits less. COLD commits minimal. Every tier-driven action is a capital deployment.

TIER-PHIL-002
Separate Opportunity Tier from Relationship Tier. An account may be Opportunity=WARM and Relationship=STRATEGIC simultaneously; the two scores move independently. Tier demotion on opportunity decay does not erase relationship investment (see Layer 3).

TIER-PHIL-003
Every tier exists within portfolio capacity. Promotion eligibility (score ≥ 78) is not promotion guarantee. HOT admission is bounded by a configurable HOT capacity (Layer 2); accounts exceeding capacity enter HOT_WAITLIST. Tiers are portfolio-relative, not absolute.

TIER-PHIL-004
Tier and Timing are separate dimensions. HOT + WAIT must be expressible. A high-opportunity account may correctly require deliberate non-action (e.g. new CTO 60-day onboarding); the engine must support tier without forcing immediate execution (see Layer 8).

TIER-PHIL-005
Routing objective is maximize (Account Potential × Human Fit), not round-robin fairness. Humans are not interchangeable resources; the engine treats them as strategic assets whose expertise materially changes conversion probability (see Layer 4). Resolves OPEN-Q4 architecturally. ABM Business Logic Bible — §5 v3 (Architectural) · Resource Allocation Operating System

TIER-PORT-001
The engine maintains a configurable HOT capacity derived from SDR count, reviewer count, enrichment budget, and founder availability. Accounts qualifying for HOT that exceed capacity enter HOT_WAITLIST rather than HOT immediately. WAITLIST is a real state, not a queue label.

TIER-PORT-002
HOT admission ranking is Expected Opportunity Value ÷ Expected Resource Cost, not score alone. An 82-score account with low resource cost can outrank a 91-score account with extreme cost. The waitlist re-ranks every recalc; admission is highest-ratio first.

TIER-PORT-003
Every HOT promotion records its Displaced Opportunity — which waitlisted account would have received resources if this one had not. Displacement is a first-class field for portfolio learning, not just a debug log.

TIER-PORT-004
The engine tracks Opportunity Cost Score per HOT account = Expected Value of the best alternative opportunity it displaced. Failed accounts are learned-from at the portfolio level: not 'account failed' but 'account failed AND displaced a better opportunity'. This requires a multi-period outcome record. PARTIAL RESOLUTION OF OPEN-Q13 OPEN-Q13 (oscillation prevention in score→tier→enrichment→signal loop) is partially resolved by Layer 2. A capacity-gated HOT can't oscillate freely because a promotion must displace an existing HOT account; displacement is a heavyweight decision the engine cannot make casually. The 24h tier-change cooldown remains the Phase 1 safe

TIER-REL-001
Maintain a Relationship Equity Score per account, independent of the composite score, computed from: meetings held, introductions received, referrals made, executive familiarity, trust history (positive CRM notes), and the durability of warm-path connectors. Equity accumulates; opportunity decays. They move independently.

TIER-REL-002
Accounts with Relationship Equity above threshold cannot fall below RELATIONSHIP_MONITOR tier regardless of opportunity decay. RELATIONSHIP_MONITOR is a fifth-state alongside HOT / WARM / COLD / WAITLIST: opportunity engagement is paused, but relationship maintenance (occasional value sends, executive check-ins, conference outreach) continues. The relationship asset is preserved against opportunity-axis demotion.

TIER-ROUTE-001
The engine maintains a Human Expertise Graph across dimensions: geography (KSA / GCC / global), industry (Islamic banking, digital banking, BNPL, embedded finance), product (lending, origination, open banking, KYC), relationship network (who the human knows at target accounts), and language (EN, AR, fluency level).

TIER-ROUTE-002
Owner Assignment Score = Opportunity Score × Human Fit Score. The Human Fit Score is derived from the Expertise Graph against the account's profile. Highest combined score wins assignment. Resolves OPEN-Q4.

TIER-ROUTE-003
Round-robin is a tie-breaker only, never the primary assignment logic. When two humans score equally on Opportunity × Human Fit (or when the Human Fit Score is not yet computable in Phase 1), round-robin selects between them. The Phase 1 safe default (pure round-robin everywhere) is upgraded to TIER-ROUTE-002 when the Expertise Graph is populated. RESOLUTION OF OPEN-Q4 The architectural answer to OPEN-Q4 is expertise-aware routing (TIER-ROUTE-001/002). The Phase 1

TIER-ESC-001
The engine maintains an Escalation Eligibility Score per account derived from: account value (deal-size potential), strategic importance (flagship reference value), relationship path quality (P1–P5 from SIG-PATH), urgency (timing window from M5), and target executive level. Score determines which escalation tiers are available.

TIER-ESC-002
Founder intervention requires Expected Incremental Value > Founder Attention Cost. If the same outcome is achievable with an AE or Domain Expert, the founder is not invoked. Founder time is allocated like venture capital, not like SDR rotation.

TIER-ESC-003
Escalation ladder is explicit and five-level: L1 SDR (standard outreach), L2 Account Executive (qualified discovery), L3 Domain Expert (technical / Islamic-banking / open-banking depth), L4 Founder (strategic accounts, reference banks, high-value handoffs), L5 Board / Advisor (board-level intros, governance-grade relationships). Each level consumes increasing strategic capital and requires the prior level's gate to pass. ABM Business Logic Bible — §5 v3 (Architectural) · Resource Allocation Operating System

TIER-ATTN-001
Every account receives an Attention Budget measured in concrete units: SDR-hours, reviewer-minutes, enrichment credits, and executive-attention units. The budget scales with tier (HOT receives more, COLD receives minimal) and is bounded by team-level totals.

TIER-ATTN-002
Every action consumes attention budget at a defined rate. Examples: manual dossier review = 30 attention units; founder outreach = 100 units; enrichment cycle = 20 units; AI-assisted message review = 5 units. Spend is logged per action; budget overruns require an explicit reallocation.

TIER-ATTN-003
The stated purpose of tiering is allocation of scarce attention under uncertainty, not account classification. Every §5 rule must be interpretable as an attention-allocation rule; rules that fail this test are not §5 rules. This is the architectural test, not a Phase 2 implementation — it binds spec design now.

TIER-HUMAN-001
Every account receives a Cognitive Load Score derived from: stakeholder count (size of buying committee), buying-center count (decentralized vs centralized), political complexity (SIG-TENSION from §3), signal volatility (rate of new signals), relationship-path complexity (P1–P5 mix), and escalation level (L1–L5).

TIER-HUMAN-002
Owner capacity is measured in Complexity Units, not account count. A typical SDR capacity is ~100 Complexity Units; a simple account may consume 5 units, a complex KSA bank account may consume 40. Assignment respects unit-budget, not account-count caps.

TIER-HUMAN-003
The routing engine optimizes load balance by complexity, not by account volume. Two SDRs with equal account counts can have very different effective workloads; the engine prefers the better-balanced complexity allocation. Burnout prevention is a first-class consideration. ABM Business Logic Bible — §5 v3 (Architectural) · Resource Allocation Operating System

TIER-PATIENCE-0
The routing engine may intentionally allocate Monitoring Capital instead of 03 Engagement Capital. Monitoring is a valid attention deployment, not a workflow stall. The engine asks not 'should we engage?' but 'is engagement currently the highest-value use of attention?' — and may answer no while still committing the account to HOT. ABM Business Logic Bible — §5 v3 (Architectural) · Resource Allocation Operating System

TIER-PATIENCE-001
..003 Mixed M5 Timing & Window,

Section 6: Knowledge-Acquisition Engine (33 rules)

ENR-COMP-01
Every potential enrichment action carries an Expected Information Gain estimate computed BEFORE execution. Finding a missing committee contact when the dossier has 2 known contacts has high estimated gain; finding the 8th has low. The estimate is the gate.

ENR-COMP-02
Enrichment STOPS when Expected Information Gain < Acquisition Cost — not when every dossier field is filled. Perfect dossiers are economically irrational. The 7-step workflow runs to completion only when each remaining step still clears the marginal-value gate.

ENR-BLOCK-01
Every dossier field is classified at design time as Blocking or Non-Blocking. Blocking: buying committee unknown, primary decision-maker unknown, no outreach path, no verified contact. Non-Blocking: secondary phone, office location, biography details.

ENR-BLOCK-02
Enrichment prioritization is Blocking-Information-First, not completeness percentage. The engine fills blocking gaps before any non-blocking, regardless of how easy a non-blocking field would be (closes the availability-bias trap).

ENR-BLOCK-03
Blocking-field gaps trigger LOW_COMMITTEE_COVERAGE and equivalent per-field flags; the flag binds §4.3 Reachability scoring and gates EXEC stage 6. Non-Blocking gaps do not gate downstream stages. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-STOP-01
Every enrichment workflow carries an explicit Stopping Condition. Default for HOT: buying-committee coverage ≥ 80% AND ≥2 reachable contacts (Response Reachability per §6 v2) AND ≥1 outreach path resolved. When all three pass, enrichment stops.

ENR-STOP-02
Additional enrichment beyond the stopping threshold requires explicit justification: a named manual trigger or an event-driven trigger (§6 v2 ENR-REF-DEEP-01). Continuing past stopping by default is forbidden.

ENR-STOP-03
Stopping is observable: every enrichment cycle logs whether it stopped on the threshold (clean), on marginal-value gating (Layer 1), on cost ceiling (Layer 6), or on irreducibility (Layer 10). Stop-reason is a first-class telemetry field.

ENR-QUAL-01
Every enriched fact carries an Information Quality Score: decision relevance, uniqueness, actionability, confidence (RCM stamp), recency. The Quality Score is stored alongside the fact and consumed by the Layer 6 ROI engine.

ENR-QUAL-02
Enrichment success is measured by Decision Utility — how much the acquired information changed downstream decisions — not by Record Volume. KPI 'records enriched per account' is replaced by 'decision-impact-weighted information acquired'.

ENR-QUAL-03
A smaller, higher-quality dossier is preferred to a larger, lower-quality one. The 7-step workflow is configurable: a step that consistently produces low-Quality output for a segment may be skipped without completeness penalty. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-DECAY-01
Every fact is tagged with one of four decay layers: Operational (7–30d), Tactical (30–90d), Strategic (6–12m), Structural (3–5y). Tag is set at acquisition; categories are not negotiable per fact.

ENR-DECAY-02
Decay state is read at fact-consumption time (EPIS-HALF-02), per category. Strategic and Structural facts remain authoritative far longer than the V2 30-day cadence implies — §6 v2

ENR-DECAY-03
§6 v2 Iteration 3 Ramadan/Eid 1.5× decay-rate adjustment applies to Operational and Tactical, NOT Structural — organizational identity does not decay faster during Ramadan. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-ECON-01
Every account receives a Maximum Intelligence Budget = f(Expected Opportunity Value × Stage Probability × Strategic Importance). A $5M strategic bank receives a larger budget; the budget is dynamic and updates with the underlying inputs.

ENR-ECON-02
Enrichment spend is tracked as Intelligence ROI = Decision Improvement ÷ Information Cost. This KPI replaces 'API cost per account' as the engine's optimization target.

ENR-ECON-03
Research budget dynamically INCREASES for: strategic accounts, high-uncertainty accounts (low EPIS coverage), and high-value opportunities. Uniform per-tier caps are Phase 1 only; Phase 2 routes by per-account economics. OPEN-DESIGN-QUESTION OPEN-Q6.5 — BUDGET UNIT OF ACCOUNT Question: Is the Maximum Intelligence Budget (ENR-ECON-01) denominated in money (real currency) or in attention units (the §5 v3 TIER-ATTN-001 currency)? Candidate options: (a) Money — each account has a real-dollar budget; aligns with finance reporting. (b) Attention

ENR-SEARCH-01
Every enrichment target carries a Dependency Value — the count of downstream unknowns this fact could resolve. A fact with Dependency Value = 4 is more valuable than a fact with Dependency Value = 0.

ENR-SEARCH-02
Search order is optimized by Expected Information Gain × Dependency Value ÷ Acquisition Cost. The engine selects the next search that maximizes this ratio, not the next step in the workflow's literal order.

ENR-SEARCH-03
The engine identifies and tracks Keystone Unknowns — missing facts whose resolution unlocks multiple downstream decisions. The Primary Initiative Sponsor is typically a Keystone Unknown for a HOT bank account.

ENR-PORT-01
Every intelligence artifact carries a Reusability Score. Ecosystem facts (connector intelligence, regulatory landscape, advisory networks) score high; per-account vendor relationships score zero (exclusivity-bound).

ENR-PORT-02
Knowledge acquisition priority increases when Expected Cross-Account Reuse is high. Researching Mastercard KSA leadership (high reuse) is prioritized over a single local manager at one account (low reuse) at equivalent immediate value.

ENR-PORT-03
Shared Intelligence Assets are first-class objects, separate from per-account dossiers: KSA banking ecosystem map, Open Banking vendor map, Regulatory intelligence map, Advisory council graph. Queried as supplementary context, not duplicated into each dossier.

ENR-PORT-04
Vendor exclusivity discipline binds: per-account vendor facts (Backbase=SNB only, Efigence=Riyad only) are NEVER promoted into Shared Intelligence Assets. The Reusability classifier hard-rejects any fact tagged as bank-specific-vendor. ABM Business Logic Bible — §6 v3 (Information Economics) · Knowledge-Acquisition Engine

ENR-HUMAN-01
Research tasks (human and AI) must explicitly ask: what would disprove the current working hypothesis? The disconfirmation question is a required field in any structured research output for HOT accounts. Counters confirmation bias.

ENR-HUMAN-02
Research completion is measured by uncertainty reduction, not by field completion. A dossier with every field filled but no critical-unknown resolved is incomplete; a dossier with three fields filled that resolved all blocking unknowns is complete.

ENR-HUMAN-03
For major accounts (strategic or flagship tier), the dossier is not marked complete until a Contradictory Evidence Check has been performed and logged: 'we looked for evidence against the working narrative; here is what we found / didn't find.'

ENR-HUMAN-04
Availability bias counter: workflow order is set by Layer 7 (Dependency Value), not by what's easiest to find. The engine MAY override this order only with a logged justification.

ENR-BUDGET-01
Every identified unknown is classified at design time: Reducible (more research will likely resolve), Difficult (resolvable but expensive), or Currently Unknowable (no public source path; do not chase). Classification is a first-class field, not a comment.

ENR-BUDGET-02
Research stops on an unknown when Expected Cost of Further Reduction > Expected Value of the reduced uncertainty. Unknowable items stop immediately; Difficult items stop at their cost ceiling; Reducible items continue until the Layer 1 marginal-value gate binds.

ENR-BUDGET-03
Every account dossier maintains a Residual Uncertainty Register: the named list of things the engine has decided it cannot currently know (e.g. 'budget status unknown', 'executive private preference unknown', 'internal resistance unknown'). The Register is the operational counterpart to EPIS-COV-03 and binds downstream: §7 personalization cannot ground a message in a fact present in the Register.

ENR-REF-DEEP-01
). Continuing past stopping by default is forbidden.

ENR-REF-DEEP-04
Refresh cadence floor: ENR-REF-DEEP-04 turns the 30-day refresh interval into a floor (minimum), working with the EPIS half-life decay model and the event-driven refresh trigger (ENR-REF-DEEP-01). Enrichment data is refreshed no less frequently than every 30 days, and sooner when an event-driven trigger fires.

Section 7: Trust and Influence Construction (42 rules)

AI Rules (39)

AI-NARR-01
Every account is assigned an Active Narrative (e.g. API Modernization, Digital Origination, AI Banking, Open Banking) derived from the dossier's Public Strategic Initiatives (§6 v2 ENR-INIT-DEEP-01). The narrative is the account-level story Decimal is telling, not a per-message choice.

AI-NARR-02
Agent 5 may vary the specific angle within a message but must remain inside the account's Active Narrative. A pivot to a different narrative requires an explicit account-level narrative change, not a per-message improvisation.

AI-NARR-03
Narrative drift — measured as the semantic distance between consecutive message angles for the same account — above a configured threshold triggers a review flag. Consistency is what creates trust; drift is observable and gated, not silently tolerated.

AI-NARR-04
Every strategy angle is tagged with a Narrative Stage: Emerging, Growing, Established, or Inevitable, derived from the strength and recency of the account's and the market's signal activity around that narrative (SIG-NEWS/EVENT volume + SIG-REG urgency).

AI-NARR-05
Message framing changes by Narrative Stage: Emerging → education; Growing → differentiation; Established → execution; Inevitable → urgency. Agent 5 reads the stage and selects the matching frame; a mismatch (urgency framing on an Emerging narrative) is rejected by QC.

AI-NARR-06
Agent 5's objective prioritizes Narrative Adoption progress before Feature Awareness — the strategy brief states which stage-transition this message is intended to advance, not merely which feature it highlights. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System

AI-SEQ-01
Agent 5 reads the contact's Interaction Stage: First Contact, Familiar, Engaged, or Active Discussion — derived from §10 engagement history (touch count, opens, replies) and contact_state (V2 §2.2).

AI-SEQ-02
Every message is assigned a Portfolio Role: Awareness, Credibility, Curiosity, Proof, or Commitment. Not every message should ask for a meeting — an Awareness-role message ends in a low-friction curiosity hook; a Commitment-role message ends in a direct CTA.

AI-SEQ-03
Portfolio Role is determined jointly by Interaction Stage and the sequence touch number (V2 §9.2 Touch 1–4): Touch 1 defaults to Awareness/Credibility; Touch 4 defaults to Proof/Commitment. Agent 5 may override the default with explicit justification.

AI-SEQ-04
Message optimization occurs at the sequence level: Agent 5's strategy brief states not just this message's angle but its role in the planned arc of touches, so a later message doesn't repeat an earlier message's role.

AI-TRUST-01
Every contact carries a Trust Accumulation Score built from: personalization accuracy (did prior claims hold up?), response quality, relevance ratings, and engagement depth over time. The score is append-only and visible in the dossier.

AI-TRUST-02
Aggressive CTAs (direct meeting asks, pricing discussions) are forbidden until the Trust Accumulation Score crosses a configured threshold OR the path carries high Trust Transfer Potential (AI-TRUST-04). Cold trust gets low-friction asks ('worth comparing notes?'); high trust gets direct asks ('15-minute call?').

AI-TRUST-03
Agent 5 optimizes Trust Growth before Meeting Acquisition when Trust Accumulation Score is below threshold: the strategy brief's stated objective for early-stage contacts is explicitly 'build trust', not 'book meeting', even though the message still carries a CTA.

AI-TRUST-04
Every resolved SIG-PATH carries a Trust Transfer Potential score derived from: relationship strength of the connector, introducer credibility, and the connector's ecosystem position (per §3 SIG-PATH connector registry and §5 v3 TIER-ROUTE-001 Expertise Graph).

AI-TRUST-05
High Trust Transfer Potential paths (warm introductions through credible connectors) permit more direct CTAs and more ambitious asks earlier in the sequence than the Trust Accumulation Score alone would allow — borrowed trust substitutes partially for earned trust.

AI-TRUST-06
Agent 5's strategy brief must explicitly name the trust source being leveraged when one exists (e.g. 'warm path via [connector]') — not merely cite personalization facts. Trust sources are a distinct input from content facts. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System

AI-POL-01
Every personalization claim derived from an Inferred Buying Driver (§6 v2 ENR-INIT-DEEP-01) receives a Political Sensitivity Rating: Low, Medium, or High. The rating is computed strictly from SIG-TENSION's sourced friction signals (public statements, contradictory initiatives, budget conflicts) — NEVER from open-ended inference about internal politics. A claim with no SIG-TENSION basis defaults to Low sensitivity, not High; the engine does not invent risk.

AI-POL-02
High-sensitivity claims require human review regardless of the account's tier or the §8

AI-POL-03
When political risk is elevated (Medium or High), Agent 5 prefers Public Strategic Initiatives over Inferred Buying Drivers as the message's grounding — the safer, sourced, less politically-loaded angle is chosen, consistent with §6 v2 ENR-INIT-DEEP-03's anti-false-confidence default.

AI-POL-04
Political Sensitivity Rating is recomputed whenever the underlying SIG-TENSION signal changes; a rating is never assigned once and treated as permanent.

AI-REP-01
Every outreach action updates a Reputation Impact Score: Positive, Neutral, or Negative, derived from engagement quality (reply sentiment, silence after high-personalization effort, explicit negative feedback).

AI-REP-02
Contact-level Reputation Impact rolls up into Account Reputation; Account Reputation rolls up into Market Reputation (an ecosystem-wide rolling score). This uses the same roll-up pattern as §4's score architecture (append-only, never overwritten).

AI-REP-03
Agent 5's formal optimization objective is Expected Opportunity Value combined with Expected Reputation Value — not conversion probability alone. This is the engine's deepest objective-function change in §7 v2; every other layer in this document feeds into this combined objective.

AI-REP-04
Every interaction is assigned a Reputation Yield estimate — expected future value generated through ecosystem memory, independent of this interaction's immediate conversion outcome.

AI-REP-05
Account-level success metrics (§13 Analytics) report BOTH Immediate Outcome (reply, meeting, opportunity) and Reputation Impact, never immediate outcome alone.

AI-REP-06
The engine may deprioritize a higher-conversion action in favor of a higher-reputation action ONLY when the Reputation Yield delta exceeds a configured floor (guardrail, not a vibe). Absent that explicit threshold being met, Agent 5 defaults to pursuing the highest-conversion action available — reputation is a constraint that can override conversion, but only when the trade-off is measurable and material, not as a default disposition. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System

AI-ATTN-01
Every drafted message receives an Attention Cost estimate — the mental effort required to process it, derived from length, structural complexity, and number of distinct ideas presented.

AI-ATTN-02
Agent 5 optimizes Attention-to-Value Ratio, not message length minimization alone. A 50-word high-relevance message with low cognitive effort beats a 300-word high-relevance message with high effort — this formalizes V2's existing 150/80-word caps as a ratio objective, not just a ceiling.

AI-ATTN-03
Different personas carry different Attention Budgets: C-suite (very low — shortest, most distilled framing); Director (moderate); Manager (higher — more detail tolerated). Agent 6's message complexity adapts to the target's seniority-derived attention budget, extending AI-AG6-07's seniority-matched framing rule with an explicit cognitive-cost dimension.

AI-ATTN-04
Attention Cost is tracked against actual engagement outcomes (open-to-reply conversion by Attention Cost bracket) to calibrate the cost model over time — requires the Tier 6 outcome loop.

AI-INF-02
Stakeholder-specific objectives mean reducing legitimate friction through clarity and reassurance — never engineering around a stakeholder's legitimate concerns. 'Procurement Neutralization' means addressing real cost/process concerns with concrete, honest answers, not minimizing or routing around procurement's role. A message strategy that asks Agent 6 to obscure a real constraint is rejected by QC-007 (compliance check).

AI-INF-03
Messages to different stakeholders at the same account may pursue different objectives: CTO receives enthusiasm-building content; Compliance receives risk-reduction content (sourced, specific, never dismissive); Procurement receives friction-reduction content (clear process, pricing transparency). Same opportunity, different persuasion objective per role — this is the operational form of multi-threaded outreach already required by §4 (Reachability multi-thread bonus) and §6 (3–7 committee contacts).

AI-INF-04
Account-level success tracking includes a Committee Influence Progress metric — how many buying-committee members have moved from UNCONTACTED/SEQUENCED toward QUALIFIED — reported alongside (not instead of) reply rate.

AI-INF-05
A Stakeholder Influence Strategy assignment must be consistent with the Political Sensitivity Rating (Layer 4): a Procurement Neutralization message cannot rely on a High-sensitivity Inferred Buying Driver claim; it defaults to Public Strategic Initiative grounding per AI-POL-03. ABM Business Logic Bible — §7 v2 · Trust & Influence Construction System §7 v2 RULE ROLL-UP & PLACEMENT STATUS

AI-POL-001
binds strictly to SIG-TENSION's public, sourced friction signals — the engine never free-reasons about ‘is this politically dangerous’ from inference alone. Stakeholder-specific objectives are reassurance, not manipulation.

AI-AG6-07
's seniority-matched framing rule with an explicit cognitive-cost dimension.

AI-INF-002
defines 'Procurement Neutralization' and similar objectives as reducing legitimate friction through clarity, never as engineering around a stakeholder's legitimate concerns. The reputation-over-conversion trade-off is threshold-bound, not vibes-based.

AI-REP-006
requires a measurable Reputation Yield delta to exceed a configured floor before the engine may deprioritize conversion — otherwise Agent 5 defaults to pursuing the close.

AI-REP-003
makes this the formal optimization target for Agent 5. THREE GUARDRAILS MADE EXPLICIT IN THIS PASS Political sensitivity stays sourced-only.

ENR Rules (2)

ENR-INIT-DEEP-03
's anti-false-confidence default.

ENR-INIT-DEEP-01
). The narrative is the account-level story Decimal is telling, not a per-message choice.

QC Rules (1)

QC-HUMAN-004
post-MVP relaxation status. A High-sensitivity message never auto-sends, even after the 90-day relaxation criterion (§7 OPEN-Q7.1) is met for that segment.

Section 8: Trust Preservation and Irreversibility Management (20 rules)

QC-PHIL-01
Every QC rule (QC-RULE-001 through 010) maps to one or more Irreversible Risk Categories: Reputation, Relationship, Compliance, Market Memory, Political. A rule with no mapped category is incompletely specified.

QC-PHIL-02
QC severity = Probability × Magnitude × Reversibility Factor (0 for irreversible, approaching 1 for recoverable) — NOT raw frequency. A rare catastrophic failure (e.g. QC-RULE-009) gets highest protection despite low trigger rate.

QC-PHIL-03
Mapping: QC-001/002→Reputation; QC-004/005→Relationship; QC-006→Reputation; QC-007→Compliance; QC-RULE-008→Relationship+Market Memory; QC-RULE-009→Political+Compliance; QC-RULE-010→Compliance+Relationship.

QC-RISK-01
Each of the seven mechanical checks carries an explicit False Positive Cost (good message blocked, measured against §4 timing-window decay) and False Negative Cost (bad message sent, measured against Layer 1). Both are visible in QC configuration.

QC-RISK-02
QC optimization target for QC-RULE-001..007 is Expected Total Risk = (FP Rate × FP Cost) + (FN Rate × FN Cost), NOT maximum strictness. Thresholds may loosen where FP Cost is empirically shown to dominate.

QC-RISK-03
GUARDRAIL: applies ONLY to QC-RULE-001..007. Does NOT apply to QC-RULE-008/009/010, which remain strictness-maximizing regardless of false-negative cost. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management

QC-HUMAN-08
Every QC reviewer carries a Reviewer Calibration Score: agreement rate with eventual outcomes, consistency across similar cases, and override-of-AI accuracy.

QC-HUMAN-09
Approval accuracy, override accuracy, and rejection accuracy tracked as THREE SEPARATE metrics per reviewer — a reviewer may excel at compliance issues but miss narrative-drift nuance.

QC-HUMAN-10
Periodic blind review audits: the same message routed to 2+ reviewers independently. Disagreement rate tracked; persistent disagreement on a check type triggers a calibration session, not blame.

QC-HUMAN-11
Reviewer Calibration Score feeds the same per-subsystem trust model as AI components (EPIS-TRUST-01): a degrading reviewer is flagged exactly as a degrading AI subsystem is flagged.

QC-SYS-01
§8 tracks Failure Clusters: a statistically significant spike in any check type within a rolling window, segmented by account, segment, persona, or agent-version.

QC-SYS-02
A Failure Cluster breaching threshold triggers a System Health Investigation routed to engineering/prompt-ops, NOT per-message regeneration. Regenerating 200 failing messages treats symptoms; investigation finds the shared root cause.

QC-SYS-03
Failure-cluster telemetry is the engine's early-warning system: §8 sees every output from every upstream agent, so a rising hallucination rate is detectable here first. Binds to FAIL-LEARN-01 as a QC-specific instance of that broader pattern. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management

QC-TRUST-01
Every QC failure is categorized by Trust Impact: Low, Medium, High, Critical — distinct from Irreversible Risk Category. Trust Impact measures damage to confidence IN THE SYSTEM, not the specific relationship.

QC-TRUST-02
§8 tracks a Trust Preservation Score per upstream module (Agent 5, Agent 6, §6 Enrichment, §4 Scoring, Narrative layer): rate of QC intervention weighted by Trust Impact. A declining score is the system's early signal that humans are about to start distrusting that module.

QC-TRUST-03
§8's success metric explicitly includes Trust Preserved, not only Messages Approved. §13 Analytics KPIs are extended to report Trust Preservation Score per module alongside QC pass rate — a high pass rate with declining trust score is a warning, not a success. ABM Business Logic Bible — §8 v2 · Trust Preservation & Irreversibility Management §8 v2 RULE ROLL-UP & PLACEMENT STATUS

QC-RULE-009
) gets highest protection despite low trigger rate.

QC-RULE-010
→Compliance+Relationship. LAYER 2 — FALSE POSITIVE / NEGATIVE BALANCE (RISK) Rule ID Specification

QC-RULE-008
→Relationship+Market Memory;

QC-RULE-001
through 010) maps to one or more Irreversible Risk Categories: Reputation, Relationship, Compliance, Market Memory, Political. A rule with no mapped category is incompletely specified.

Section 9: Dynamic Relationship Management (31 rules)

OUT-ATTN-01
Every planned send receives an Attention Availability Score derived from: board-cycle proximity, reporting/budget-cycle timing, regulatory-period activity (SIG-REG), major-initiative launches, and executive-travel signals where publicly sourced (conference attendance, public schedule).

OUT-ATTN-02
Send timing optimizes against Attention Availability in addition to KSA business-hours windows (V2 §17-001/002, §9.1 send windows). A touch otherwise due in a low-availability window is held and rescheduled to the next high-availability slot within the sequence's timing tolerance.

OUT-ATTN-03
Attention Availability inputs must be sourced from public signals only (SIG-REG deadlines, SIG-EVENT conference dates, SIG-NEWS earnings dates) — the engine never infers private calendar information about a named individual.

OUT-INTER-01
Every touch reads Sequence Memory — the full record of what this contact has already seen, across BOTH channels — before Agent 5/6 generate the next touch. This extends §7 v2 AI-NARR-01's Active Narrative to the per-contact touch history specifically.

OUT-INTER-02
Every touch receives a Reinforcement Score: does it strengthen the Active Narrative the contact has already been shown, or dilute it? A touch scoring as dilution is rejected and re-queued to Agent 5, using the same hard-gate mechanism as §8 QC-RULE-008 (narrative drift) — message interference is a specific, cross-channel instance of narrative drift.

OUT-INTER-03
Reinforcement scoring applies ACROSS channels: an email's narrative and a LinkedIn touch's narrative for the same contact must score as reinforcing, not merely each channel internally consistent with itself. ABM Business Logic Bible — §9 v2 · Dynamic Relationship Deployment System

OUT-RESP-01
Engagement Signal (opens, clicks, replies) is tracked separately from Influence Signal (profile views, website visits, content consumption, colleague forwarding/engagement — where observable via §10 tracking infrastructure).

OUT-RESP-02
Sequence success is measured by Influence Progress, not reply rate alone. A sequence with zero replies but rising Influence Signal is NOT classified as failed; it remains active per its normal cadence rather than being prematurely abandoned.

OUT-RESP-03
Influence Signal feeds the Trust Accumulation Score (§7 v2 AI-TRUST-01) as an additional input category, NOT as a separate competing trust number — one trust ledger, richer inputs (Integration Decision 4).

OUT-RECIP-01
Every touch receives a Value Contribution Score: did this touch give the recipient something useful (an insight, a relevant benchmark, a specific observation) independent of any ask?

OUT-RECIP-02
Before a high-friction ask (direct meeting, pricing discussion), Cumulative Value Contribution across the sequence must exceed a configured threshold. This is the SAME gate as §7 v2

OUT-MEM-01
Every outreach interaction updates a Relationship Memory Record on the contact: Positive, Neutral, or Negative, derived from engagement quality — the same evidence basis as §7 v2 AI-REP-01 (Reputation Impact Score) applied at the individual-contact level rather than account/market level.

OUT-MEM-02
Relationship Memory propagates when SIG-MOBILITY (§3) detects the contact moving to a new account: the Memory Record transfers with the person, becoming a Trust Transfer Potential input (§7 v2 AI-TRUST-04) at the new account rather than starting cold.

OUT-MEM-03
The engine's account-level success metric explicitly includes Lifetime Relationship Value alongside Sequence Conversion Rate (extends §7 v2 AI-REP-05's dual-metric reporting to the per-sequence level).

OUT-BEH-01
Each stakeholder receives an Engagement Archetype (Explorer / Skeptic / Delegator / Protector / Builder) inferred from observable response patterns over the relationship history — NOT from personality inference (EPIS-ETH-01 binds: this is behavioral-pattern classification from actual interaction data, never psychological profiling).

OUT-BEH-02
Sequence interpretation adapts by archetype: a non-response from a Delegator is NOT treated as a negative signal the way a non-response from an Explorer is. §10 sentiment/engagement classification reads the archetype before assigning a positive/neutral/negative interpretation to silence.

OUT-BEH-03
Optimal cadence varies by archetype: Skeptics receive more proof-heavy touches spaced further apart; Explorers receive earlier, more direct touches. Agent 5's sequence-level strategy brief reads the archetype as an input. ABM Business Logic Bible — §9 v2 · Dynamic Relationship Deployment System

OUT-MTHREAD-01
A Stakeholder State Map is maintained per active committee member: unaware / aware / interested / supportive / resistant, derived from each contact's individual engagement history.

OUT-MTHREAD-02
Sequence-level strategy decisions (Agent 5) read the Stakeholder State Map for the whole committee, not just the individual contact's state, when generating any single contact's next touch.

OUT-COMP-01
Every opportunity tracks a Competitive Awareness Probability derived STRICTLY from sourced signals: SIG-VENDOR contract/pricing-activity changes, SIG-NEWS coverage of the incumbent, and observable account behavior (e.g. a sudden incumbent renewal announcement). The engine NEVER infers awareness from unsourced speculation about the incumbent's internal knowledge or strategy (mirrors EPIS-ETH-01/02 + §7 v2 AI-POL-01's sourced-only discipline).

OUT-COMP-02
When incumbent entrenchment is high (per §6 v3 SIG-VSAT displacement-probability inputs) AND Competitive Awareness Probability is elevated, Agent 5's strategy shifts toward more proof points and differentiation, and away from direct incumbent-attack framing.

OUT-COMP-03
Major sourced competitor moves (SIG-VENDOR contract events, pricing news) become direct inputs to the active sequence's strategy brief, not merely account-dossier updates consumed at the next enrichment cycle — a live sequence can react within its current cadence.

OUT-PORT-01
Every active sequence inherits a Portfolio Priority computed via the SAME formula as §5 v3 TIER-PORT-002 (Expected Opportunity Value ÷ Expected Resource Cost) applied to the sequence's specific remaining-touch cost, NOT a new parallel scoring mechanism (Integration Decision 3).

OUT-PORT-02
Execution resources (the shared daily send-budget per OUT-SEQ-07, SDR follow-up time, founder attention) are allocated portfolio-wide using Portfolio Priority — resolving §9 v1's OPEN-Q9.2 directly via this rule rather than a separate allocation scheme.

OUT-PORT-03
When execution capacity is constrained, the highest Portfolio Priority sequence is served first, NOT the oldest or the one closest to its next scheduled touch — consistent with §5 v3's displacement discipline (TIER-PORT-003): a lower-priority sequence delayed is logged as a displaced opportunity. ABM Business Logic Bible — §9 v2 · Dynamic Relationship Deployment System

OUT-COMPREL-01
Every contact maintains a Relationship Momentum value — the rate of change of Trust Accumulation Score over the recent interaction window — stored alongside (not instead of) the Trust Accumulation Score itself (Integration Decision 4: one ledger, an added rate-of-change field).

OUT-COMPREL-02
Sequence continuation decisions read BOTH level and momentum: a moderate trust level with rising momentum favors continued investment; a higher trust level with declining momentum triggers a pause-and-reassess (this generalizes §9 v1's OUT-SEQ-08, which already pauses on a Trust Accumulation drop — momentum makes the trend explicit rather than reactive only to an already-realized drop).

OUT-COMPREL-03
The engine's optimization objective for sequence-level decisions is Lifetime Relationship Value, consistent with §7 v2 AI-REP-03's account-level Expected Reputation Value objective — extended down to the individual contact/sequence level so the two objectives are the same value function at different granularities, not two different goals. ABM Business Logic Bible — §9 v2 · Dynamic Relationship Deployment System

OUT-SEQ-07
, SDR follow-up time, founder attention) are allocated portfolio-wide using Portfolio Priority — resolving §9 v1's OPEN-Q9.2 directly via this rule rather than a separate allocation scheme.

OUT-SEQ-08
, which already pauses on a Trust Accumulation drop — momentum makes the trend explicit rather than reactive only to an already-realized drop).

OUT-SEQ-03
(§9 v1) requires that a reply on ANY contact immediately pauses ALL sequences for the ENTIRE account.

Section 10: Reality Calibration Engine (20 rules)

ENG-EPIST-01
Every engagement event carries an RCM stamp (EPIS-RCM-01) with an Intent Confidence component distinct from Event Confidence (did this event technically occur vs. what does it imply about motive). Meeting Booked carries very high Intent Confidence (externally verified, unambiguous); a single Email Open carries very low Intent Confidence (multiple non-intent explanations exist).

ENG-EPIST-02
Observed Behavior and Inferred Intent are stored as separate fields on every engagement record, never collapsed into one. Downstream consumers (§4 scoring, §7 v2 Trust Accumulation) read Inferred Intent, never raw Observed Behavior directly — preventing the over-learning that comes from treating every click as equally meaningful.

ENG-NULL-01
Every contact with sufficient history (per §10 v1 ENG-ARCH archetype minimum-evidence floor) gets an Expected Engagement Pattern: typical reply latency, typical engagement cadence, derived from their own history.

ENG-NULL-02
A material deviation from the Expected Engagement Pattern (e.g. a contact who normally replies within 3 days going silent for 30) generates a Negative Signal Event — silence becomes a first-class logged event, not merely an absence of logged events.

ENG-NULL-03
Negative Signal Events are interpreted through Engagement Archetype (§10 v1 ENG-ARCH, §9 v2 OUT-BEH) exactly as ENG-SENT-06/ENG-SILENCE already require — this layer generates the event; archetype interpretation (already specified) consumes it. No new interpretation mechanism. ABM Business Logic Bible — §10 v2 · Reality Calibration & Institutional Learning Engine

ENG-SENSOR-01
Every event SOURCE (not just each individual event) carries a Reliability tier, reusing EPIS-SRC-01's reliability/incentive-bias separation: Meeting Booked → Very High (externally verified by calendar/CRM); Reply → High (requires actual human action, though sentiment can still be misread); Click → Medium (link-prefetch and security-scan false positives are common); Open → Low (pixel-blocking and auto-fire are common).

ENG-SENSOR-02
§13 Analytics and any learning loop (§4 SCORE-RECAL, §7 v2 prompt optimization) weight observations by source reliability — a Click-sourced signal contributes less to a learned conclusion than a Reply-sourced signal of equal apparent sentiment. This rule is what makes Layer 1's Intent Confidence and this layer's source reliability operate as one coherent weighting system, not two.

ENG-CAUSE-01
Every significant engagement outcome (positive reply, meeting booked) receives a Cause Attribution Distribution over named candidates: outreach, pre-existing relationship, narrative/market timing, competitor action, unknown. This is a distribution (e.g. outreach 40% / relationship 35% / market timing 25%), not a single confident label — consistent with EPIS-RCM-03's anti-false-confidence discipline.

ENG-TRUTH-01
Every significant engagement event is tagged with the specific prior belief(s) it bears on — named explicitly: a timing hypothesis (§5 v3 Patience layer), a persona hypothesis (§7 Agent 3), a narrative hypothesis (§7 v2 AI-NARR), or a trust hypothesis (§7 v2 AI-TRUST). An event with no identifiable prior belief it bears on is logged but does not feed any update.

ENG-TRUTH-02
§10's defined purpose is Model Correction, not Event Recording — this is the explicit architectural test for §10 rules going forward (mirrors §5 v3 TIER-ATTN-003 and §6 v3's information-economics test as the same kind of binding self-test). A proposed §10 rule that does not update some specific upstream belief is not a §10 rule.

ENG-REAL-01
Engagement Reality (the observed response) is explicitly separated from Opportunity Reality (the inferred underlying buying state, per §4's Effective Opportunity / SIG-VSAT displacement / SIG-HYP). Engagement score alone NEVER directly sets or moves Opportunity score — it is one weighted input among the §4 dimensions, gated through Layer 4's cause attribution first.

ENG-REAL-02
Positive engagement with no supporting independent signal (no SIG-NEWS/SIG-REG/SIG-VENDOR corroboration) creates Curiosity — a flagged hypothesis for further verification — not Certainty. This binds via EPIS-COV-02 (coverage caps confidence): an account whose only positive evidence is a single enthusiastic reply has, by definition, low Intelligence Coverage, and the engine's confidence is capped accordingly regardless of how positive the reply sounds. ABM Business Logic Bible — §10 v2 · Reality Calibration & Institutional Learning Engine

ENG-TRUTH-03
A Stakeholder Truth Map extends the §9 v2/§10 v1 Stakeholder State Map with: support level, resistance level, influence level (per §3 SIG-POWER), and confidence level per committee member — a richer per-contact record, still observational only pending OPEN-Q9.3.

ENG-TRUTH-04
Account-level truth is derived from WEIGHTED stakeholder aggregation (weighted by each member's SIG-POWER influence level), not from any single contact's engagement — this is read-only synthesis for human/analytics consumption (§11, §13), not a sequence-execution input, consistent with OPEN-Q9.3's unresolved status.

ENG-TRUTH-05
Conflicting stakeholder signals (e.g. CTO supportive AND Compliance resistant, both at high confidence) produce an Organizational Ambiguity State on the account. DATA-MODEL IMPLICATION — ORGANIZATIONAL AMBIGUITY STATE BELONGS IN §2 Iteration 8 proposes Organizational Ambiguity State as a 'first-class entity.' That is a §2 (Master Entity Model) decision, not something a §10 document can quietly create. §10 v2 specifies the STATE and its derivation rule (ENG-TRUTH-05) but flags the entity itself as a pending §2 addendum — to be formally added to the entity model (likely Tier 3 Decision, alongside Hypothesis and InfluenceMap) in a future §2 revision, rather than introduced informally here.

ENG-LONG-01
Every contact maintains a Relationship Trajectory — the time series of Trust Accumulation Score, Engagement Score, and Intent Confidence values, not just current snapshots. This is the data record; §9 v2 OUT-COMPREL-01 (Relationship Momentum) is the rate-of-change computation derived FROM this trajectory — one record, one derived metric, not two parallel histories.

ENG-LONG-02
Trend direction is read alongside current value in any downstream decision that uses Trust Accumulation or Engagement Score: a rising-40 and a falling-80 are NOT treated as equivalent inputs even when comparing two contacts at similar current values. This generalizes §9 v2

ENG-MEM-02
Patterns extracted from Learning Artifacts (e.g. 'this narrative angle underperforms with Compliance personas at Islamic banks') are stored as reusable organizational knowledge in Tier 6, NOT attributed to or siloed by the original account_owner — the knowledge survives personnel turnover by construction, because it lives in Tier 6, not in any individual's notes.

ENG-MEM-03
§10's defined output, per ENG-TRUTH-02's architectural test, includes its Tier 6 contribution as a first-class deliverable — a §10 implementation that logs events but never produces Learning Artifacts is incomplete, regardless of how complete its event-capture coverage is. ABM Business Logic Bible — §10 v2 · Reality Calibration & Institutional Learning Engine §10 v2 RULE ROLL-UP & PLACEMENT STATUS

ENG-SENT-06
/ENG-SILENCE already require — this layer generates the event; archetype interpretation (already specified) consumes it. No new interpretation mechanism.

Section 11: Human-AI Decision Partnership (26 rules)

HAND-COMP-01
Every handoff package leads with an Executive Summary Layer: a maximum 60-second read (~150 words) that sits ABOVE the full §11 v1 nine-component package, not instead of it. The full package remains available for the rep who wants depth; the summary is what most reps will actually read first.

HAND-COMP-02
Every insight in the package is ranked by Decision Importance (does this change what the rep should say or do in the next 10 minutes?), not by informational richness or recency. The richest, most recent enrichment fact does not automatically rank above an older, less detailed fact if the older fact has higher decision relevance — reuses §6 v3 Layer 4's Information Quality vs Quantity discipline (ENR-QUAL-01), applied to package ordering rather than enrichment acquisition.

HAND-COMP-03
The Executive Summary explicitly answers: 'if the rep remembers only three things, what are they?' — capped at exactly three items, forcing real prioritization rather than a longer 'top 5' that defeats the compression purpose.

HAND-JUDGE-01
Every suggested discussion angle (§11 v1 HAND-PKG component, sourced from §7 Agent 5) includes its Reasoning Chain in the package: not 'lead with regulatory urgency' alone, but 'lead with regulatory urgency because SIG-REG contributed the largest share of this account's §10 v2 ENG-CAUSE attribution for the triggering engagement.' The reasoning, not just the conclusion, transfers.

HAND-JUDGE-02
Every strategy recommendation in the package includes its Rejected Alternatives — the other angles Agent 5 considered and why each scored lower — so the human understands the tradeoff space, not just the winning choice. This is the package-facing view of the same comparison Agent 5 already performs internally; no new reasoning step, only added transparency. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-DEC-01
Every §11 handoff that reaches a material next-action decision (which stakeholder to approach, which angle to use, whether to escalate) generates a Decision Record: options considered, selected option, rationale, assumptions, and — if the human's choice differs from the package recommendation — an explicit Override Reason selected from a defined taxonomy (local knowledge, relationship context, political consideration, timing concern, other-with-required-free-text).

HAND-DEC-03
Override Reasons aggregated across many Decision Records become a feedback signal for §7 v2 Agent 5 strategy calibration: a recommendation pattern that gets overridden for 'relationship context' reasons repeatedly at a specific persona/segment combination indicates Agent 5's strategy model is under-weighting a factor the human reps consistently see.

HAND-DEC-04
Future reps and analysts can query Decision Records for a given account or persona type to see
Rationale: *prior decisions were made, not only what happened — institutional memory stores reasoning, retrievable independent of whether the original rep is still at the company. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine*

HAND-ACC-01
Every handoff records an explicit Decision Ownership Boundary with three separated stages: AI Recommendation (what §7 Agent 5 suggested and why), Human Decision (what the human actually decided, per the Layer 3 Decision Record), and Human Execution (how the decision was carried out in the actual conversation/email/call). Each stage is independently attributable.

HAND-ACC-02
Post-outcome reviews (§13 Analytics) classify failures into one of four categories: reasoning failure (the AI's underlying analysis was wrong — e.g. a hallucinated or miscalibrated fact), recommendation failure (the analysis was right but the strategic recommendation built on it was poor), execution failure (the recommendation was sound but human execution diverged or fell short), or external reality failure (everything upstream was reasonable; the outcome was driven by something genuinely unforeseeable — the §6 v3 ENR-BUDGET-01 'Currently Unknowable' category surfacing downstream).

HAND-ACC-03
Failure classification feeds the correct calibration target: a reasoning failure updates §7/§6 model calibration; a recommendation failure updates §7 v2 Agent 5 strategy weighting; an execution failure feeds §8 v2 QC-HUMAN reviewer/owner Calibration Score (the same mechanism §11 v1 HAND-WF-07 already extended to account owners); an external reality failure updates nothing — it is logged and explicitly NOT used to penalize any component, consistent with EPIS-RCM-05's anti-fabrication discipline (don't manufacture a lesson from pure noise). ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-AUTH-01
Every handoff package states explicitly, in plain language,
Rationale: *automation stopped — rendered from the specific GOV Autonomy Tier and trigger that fired (GOV-ESC-01/02), not a new taxonomy: 'Escalated because: Political Sensitivity Rating is High on this account's current narrative (§7 v2 AI-POL-02 permanent gate)' or 'Escalated because: Trust Accumulation Score crossed the threshold for direct pricing discussion, which is always Human-led (§11 v2 Layer 10).'*

HAND-AUTH-02
The handoff package includes Remaining Unknowns — sourced directly from the account's §6 v3 Residual Uncertainty Register (ENR-BUDGET-03) — not just Knowns. A human walking in confident about everything the engine knows, blind to what it has explicitly flagged as unknowable, is exactly the overconfidence EPIS-RCM-04 exists to prevent, now extended to the human-facing side of the boundary.

HAND-BIAS-01
Every recommendation displays its EPIS-RCM Confidence Level and Key Assumptions directly alongside the recommendation text — not buried in an appendix. 'System is usually right' should not be separable from seeing exactly how confident the system actually is on THIS recommendation.

HAND-BIAS-02
High-confidence recommendations (above a configured RCM threshold) REQUIRE a Contrarian View in the package: 'what might make this recommendation wrong?' — this is EPIS-RCM-03's falsifiability requirement ('what would change our mind') made mandatory and human-facing specifically because high confidence is precisely when automation bias is most dangerous.

HAND-BIAS-03
The package's design objective is explicitly to STIMULATE judgment, not replace it — this is an architectural test applied to every §11 v2 rule going forward, parallel to §5 v3's attention-allocation test and §10 v2's belief-correction test: a proposed §11 rule that makes the human's review faster but less critical fails this test. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-KNOW-01
After any handoff interaction, the human can submit Missing Context Feedback (hidden sponsor identified, budget freeze mentioned, political concern surfaced, relationship detail) through a structured input — not a CRM free-text field that nothing downstream reads.

HAND-KNOW-02
Missing Context Feedback enters the engine through the EXISTING signal pipeline: it is created as a MANUAL-source signal (V2 §3.1's existing Manual Input source type) with the human as the named source, NOT a new ingestion mechanism. §3 SIG-DET normalization, dedup, and freshness rules apply to it exactly as they apply to any other signal.

HAND-KNOW-03
Human-sourced signals carry a full RCM stamp (EPIS-RCM-01) like any other signal: source reliability (calibrated per-reporter over time, similar in spirit to §8 v2's Reviewer Calibration Score), confidence, and a decay profile appropriate to the content (a 'budget freeze mentioned' fact decays faster than a 'longtime personal relationship with the CTO' fact, per §6 v3 Layer 5's multi-layer decay categories). ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-EXP-01
Every escalated engagement receives an Expertise Requirement Profile: tags drawn from the same dimension set as §5 v3's Human Expertise Graph (geography, industry, product, relationship network, language) plus discussion-specific tags (technical, regulatory, procurement, executive, integration) inferred from the triggering reply content and §7 v2 AI-POL Political Sensitivity context.

HAND-EXP-02
Where the Expertise Requirement Profile indicates a need beyond the assigned owner's profile, the system recommends Support Roles — named individuals or role types to loop in — using the same Opportunity × Human Fit scoring as §5 v3 TIER-ROUTE-002, applied per-requirement rather than producing a single primary-owner match.

HAND-EXP-03
Complex opportunities are explicitly modeled as multi-human collaboration objects: the handoff package and CRM task (§12) can carry more than one assigned human, each tagged with their specific role (primary owner, technical support, regulatory support), rather than forcing a single-owner field to awkwardly represent a team.

HAND-CDL-01
Every decision TYPE (not individual action — that's GOV's Autonomy Ladder) in the ABM pipeline is classified as AI-led, Human-led, or Joint. Examples: signal aggregation → AI-led; pricing negotiation → Human-led; opportunity strategy → Joint. CDL sits one level above GOV: it classifies decision CATEGORIES, which then resolve to specific GOV Autonomy Tiers for the individual actions within them.

HAND-CDL-02
The handoff package explicitly identifies Human Judgment Areas for this specific engagement — the points where §7 v2's EPIS confidence is insufficient for the engine to have a strong view, flagged so the human knows where their judgment is the binding constraint, not a formality.

HAND-CDL-03
§13 Analytics' success metric for handoff quality is explicitly Combined Decision Quality — the joint outcome of AI recommendation quality and human decision quality together — never AI quality or human quality scored in isolation. This is the §11-level instance of §8 v2's Trust Preservation principle: the system succeeds or fails as a partnership, not as two separately-graded components. ABM Business Logic Bible — §11 v2 · Human-AI Decision Partnership Engine

HAND-WF-07
already extended to account owners); an external reality failure updates nothing — it is logged and explicitly NOT used to penalize any component, consistent with

Section 12: Distributed Reality Coordination (27 rules)

CRM Rules (26)

CRM-TIME-01
Every sync conflict is classified as Temporal (both values were true at their respective timestamps; the disagreement resolves itself once the later event is applied) or Logical (the values are fundamentally contradictory regardless of timing — e.g. ABM=HOT and CRM=CLOSED_LOST cannot both be simply 'true at different times' in a way that self-resolves).

CRM-TIME-02
Temporal conflicts resolve automatically by applying the later Event Time's value (Layer 7) without escalation. Only Logical conflicts trigger INT-CRM-04's conflict-resolution escalation — this distinction is what prevents the unnecessary escalation volume that would otherwise drown the genuinely meaningful conflicts in noise.

CRM-REAL-01
Authority (per the Layer 6 Registry) means final decision rights on a field's VALUE, not superior knowledge about the underlying reality. A field's Authority Owner can be wrong; the Registry resolves WHO decides, not WHAT is objectively true.

CRM-REAL-02
Every conflict resolution explicitly logs the Lost Information — the losing side's value and its supporting context — even though it does not win. CRM wins deal-stage conflicts (INT-CRM-04), but ABM's discarded signal-based context is retained as a Tier 6 input, not deleted; it may still be useful even when it didn't carry the field. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-LAT-01
Every CRM-sourced field carries a Freshness Confidence derived from: the field owner's historical update lag, the field's typical update frequency, and time since last change. This is distinct from the field's Authority (Layer 6) — CRM remains authoritative even when Freshness Confidence is low; low confidence flags the value as POSSIBLY stale, not WRONG.

CRM-LAT-02
Repeated delayed updates from the same owner or team accumulate into a CRM Staleness Risk score, feeding the same per-subsystem trust model as §8 v2 QC-HUMAN-08 and §11 v2 HAND-WF-07 — a rep whose CRM updates are chronically late is a degrading subsystem exactly like a degrading AI module, tracked the same way.

CRM-LAT-03
The engine distinguishes Unknown (no value has ever been recorded) from Outdated (a value exists but Freshness Confidence is low) as genuinely different states — an Outdated deal stage still gates automation per INT-CRM-04, but the handoff package (§11 v2 HAND-AUTH-02) surfaces 'last confirmed N days ago' rather than presenting it with the same confidence as a value updated an hour ago. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-FUSION-01
Every synchronized account maintains a Unified Reality Timeline: signals (§3), outreach touches (§9), engagement events (§10), handoffs and Decision Records (§11 v2), and CRM-pulled deal events (§12.2), merged chronologically by Event Time (Layer 7), not by which system happened to record them.

CRM-FUSION-02
§13 Analytics consumes the Unified Reality Timeline as its primary input for opportunity-history analysis, NOT isolated per-system event logs — a KPI computed from ABM events alone or CRM events alone is, by construction, working from a partial reality (Layer 2) and is flagged as such in any §13 report.

CRM-EPIST-01
The success metric for §12 is Reality Convergence, operationalized concretely as: a DECLINING Logical-conflict rate (Layer 1) on the same field over time, not merely 'sync completed' or 'fields match.' Two systems that agree because one blindly overwrote the other have NOT converged; two systems that increasingly agree because the underlying ambiguity was resolved HAVE.

CRM-EPIST-02
A field with a persistently elevated Logical-conflict rate (not declining) becomes a Reality Investigation Opportunity flagged to §8 v2 QC-SYS-01 as a Failure Cluster — persistent disagreement usually indicates a missing signal source, a miscalibrated authority assignment, or a process gap, not random noise. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-OWN-01
Every synchronized field has exactly ONE declared Authority Owner: ABM, CRM, Governance, Human, or Derived (computed from other authoritative fields, itself carrying its own authority). No field has shared or co-equal ownership, ever.

CRM-OWN-02
A field with no declared Authority Owner CANNOT enter synchronization — this is enforced at the point a new field is added to the sync configuration, not discovered later when a conflict occurs.

CRM-OWN-03
The Authority Registry is a runtime-enforced artifact (not documentation): INT-CRM-02's existing ABM/CRM split and §11 v2 HAND-ACC-01's Decision Ownership Boundary (AI Recommendation / Human Decision / Human Execution) both register into it as existing entries, alongside every field added since. The Registry, not separate prose rules, is what §12's conflict-resolution logic actually queries at runtime. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-ORDER-01
Every synchronized event stores BOTH an Event Timestamp (when it actually happened, per the source system) and an Arrival Timestamp (when the ABM engine received it) as separate fields, never collapsed into one.

CRM-ORDER-02
State transitions are computed from Event Timestamp, never Arrival Timestamp. An event that arrives late but has an earlier Event Timestamp than the current state's last-applied event does NOT blindly overwrite — it triggers Layer 1's conflict classification against what's already been applied.

CRM-ORDER-03
A late-arriving event with an earlier Event Timestamp than already-applied events triggers State Reconciliation: the timeline (Layer 4) is recomputed with the correctly-ordered event inserted, and any state derived from the now-corrected sequence is re-evaluated — never a blind overwrite that could silently revert a correct later state to an incorrect earlier one. OPEN-DESIGN-QUESTION OPEN-Q12.3 — EVENT TIMESTAMP RELIABILITY ACROSS SOURCES Question: CRM-ORDER-01/02 assume every source can supply a trustworthy Event Timestamp. Not all sources are equal: HubSpot webhooks typically carry reliable event-time metadata; some signal sources (§3 INT-SIG) may only have a detection time, with the true underlying event time inferred or approximate. Wh...

CRM-AUDIT-01
Every field change stores Change Provenance: previous value, new value, actor (which system/person), timestamp, and reason where available — using the SAME append-only discipline as V2 §2.3 (signals) and SCORE-CALC-002 (scores): never overwritten, always a new historical record.

CRM-AUDIT-02
Every synchronized object supports Reality Replay: reconstructing the complete history of any field's values over time from its Change Provenance records, consumed by the Unified Reality Timeline (Layer 4) and available for compliance/audit requests.

CRM-AUDIT-03
No destructive updates anywhere in the CRM sync layer — this generalizes the append-only principle the Bible has used since §2 (SIGNAL, ACCOUNT_SCORE) to its full scope: every synchronized field, not only the two V2 originally called out. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-LINEAGE-01
Every synchronized fact carries Origin Metadata using EPIS-SRC-01's existing reliability/incentive-bias axes (Human / AI / CRM / External Source / Partner each get their own reliability and bias profile) — not a new, separate provenance taxonomy.

CRM-LINEAGE-02
Every fact maintains a Transformation Chain: origin → any derived calculations performed on it → current usage. A score derived from a CRM-sourced fact carries that fact's lineage forward, not just its current numeric value.

CRM-LINEAGE-03
Any model or scoring function (§4, §7 v2, §10 v2) that consumes a CRM-synchronized fact reads Fact + Lineage together, exactly as EPIS-RCM-01 already requires for every other fact in the architecture — CRM-sourced facts are not exempt from the RCM stamp just because they crossed a system boundary. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

CRM-ECON-01
Every synchronized field has a Materiality Threshold below which a change does not trigger a push: e.g. a score change under 5 points does not push to HubSpot. The threshold is field-specific, not a single global cutoff (a tier CHANGE is always material; a score WOBBLE within a tier is not).

CRM-ECON-02
Sync is optimized for Decision Impact, not data freshness — reuses §6 v3 Layer 1's marginal-information-value discipline (ENR-COMP-02), applied to outbound propagation instead of inbound acquisition: push when the change would plausibly alter what a human does next, not whenever the underlying number ticks.

CRM-ECON-03
The materiality test is explicitly: 'will this change alter human behavior?' A tier change, a deal-stage-relevant state transition, or a new Decision Record clears this test readily; a sub-threshold score fluctuation does not. Sub-threshold changes still accumulate in Change Provenance (Layer 8) — they are not lost, only not pushed. ABM Business Logic Bible — §12 v2 · Distributed Reality Coordination System

SCORE Rules (1)

SCORE-CALC-002
) generalized to every CRM-synchronized field — not a new logging mechanism. Origin Metadata IS

Section 14: Opportunity State Machine (31 rules)

ENR Rules (1)

ENR-REACH-04
) AND an Agent-5-approved strategy brief exists AND QC is not in a blocked state.

OPP Rules (12)

OPP-LIFE-01
An Opportunity is created only when ALL THREE conditions hold simultaneously: (1) an initiative is identified (a named, sourced strategic priority — reuses §6 v2 ENR-INIT-DEEP-01's Public Strategic Initiatives layer as the evidence source); (2) a plausible Decimal solution is mapped to it (a named capability, not a generic 'we might be relevant'); (3) economic value is believed possible (a non-zero Effective Opportunity estimate per §5 v3's existing scoring, not merely a hunch). A signal alone never satisfies this test.

OPP-LIFE-02
Signals (§3 INT-SIG) create HYPOTHESES, not Opportunities. A SIG-HYP hypothesis (§3 reasoning streams) may mature into an Opportunity once OPP-LIFE-01's three conditions are met, but the hypothesis and the Opportunity are explicitly different entities at different confidence levels — this preserves EPIS-RCM-03's anti-false-confidence discipline at the Opportunity-creation boundary specifically.

OPP-LIFE-03
Opportunity creation is an EXPLICIT, logged event (creator, timestamp, the three satisfied conditions cited) — never silently inferred by a background process. This is itself a new Authoritative-Writer transition (Layer 1): Opportunity creation's Authoritative Writer is §5/§6's Opportunity Graph Engine (M19, §2/§3), not any single upstream signal-detection module. ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

OPP-ID-01
Every Opportunity has an explicit Opportunity Thesis: {Account, Initiative, Business Outcome} — e.g. 'SAB + Open Banking + API Monetization.' The thesis is the Opportunity's identity, not its account_id alone (which would conflate every initiative at one bank into one Opportunity, defeating the entire re-anchoring).

OPP-ID-02
A new signal attaches to an EXISTING Opportunity when its Initiative-overlap similarity exceeds a configured threshold against an existing open Opportunity's thesis — reusing the SAME dedup-similarity discipline already established for signals (§3 SIG-DEDUP-001's matching logic), applied at the Opportunity-thesis level rather than the individual-signal level. Below threshold, a new Opportunity is created (subject to OPP-LIFE-01).

OPP-ID-03
Opportunity creation triggers a duplication check against all open Opportunities at the same account BEFORE creation completes — a likely-duplicate match (above the OPP-ID-02 threshold but below auto-merge confidence) routes to human review rather than auto-creating or auto-merging silently.

OPP-END-01
Opportunity closure requires a mandatory, typed Closure Reason — never a bare WON/LOST binary: WON, LOST_COMPETITOR, LOST_BUDGET, LOST_PRIORITY, LOST_TIMING, MERGED (absorbed into another Opportunity per OPP-ID-02), ABANDONED (no clear reason, sequence simply went cold), PAUSED (not a closure — see Part B state table).

OPP-END-02
§13 v1/v2 learning loops consume the typed Closure Reason, not a binary outcome — LOST_BUDGET and LOST_COMPETITOR imply completely different corrective actions (timing/patience tuning vs. competitive-positioning tuning, §9 v2 Layer 8) and must never be collapsed into one 'LOST' bucket for learning purposes.

OPP-END-03
Closed (WON/LOST_*) Opportunities remain permanently searchable (append-only, per the Bible's standing discipline) — a LOST_TIMING or LOST_BUDGET Opportunity can resurrect into a NEW Opportunity later (new thesis, new creation event per OPP-LIFE-03) with the prior closed record retained as Tier 6 historical context, not deleted or hidden. ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

OPP-COMP-01
Every account maintains an Opportunity Portfolio — the full set of its Opportunities (open and closed), each independently in its own Opportunity State. This REPLACES §14 v1's

OPP-COMP-02
Account-level reporting uses a Portfolio Summary (e.g. '3 Active, 1 Proposal, 1 Lost') instead of a single most-advanced-Opportunity proxy — §13 v2 dashboards and §11 v2 handoff packages read the Portfolio Summary, not a collapsed single value.

OPP-COMP-03
Account State (§14 v1, unchanged in its state list) is formally reframed in MEANING, not states: it represents Relationship Health — the container's standing with Decimal — never sales progression. account_state values (WATCHING, HOT, ENGAGED, etc.) describe how actively Decimal is engaging the RELATIONSHIP; they do NOT describe how any individual Opportunity is progressing. This is the conceptual core of the whole resolution. ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

SIG Rules (1)

SIG-DEDUP-001
's matching logic), applied at the Opportunity-thesis level rather than the individual-signal level. Below threshold, a new Opportunity is created (subject to

STATE Rules (17)

STATE-GOV-01
§14 v1's STATE-ACCT-07 (EXEC Stage 5 sole writer of HOT-bound transitions) is GENERALIZED: every state transition in EITHER machine has exactly one declared Authoritative Writer. Examples: WATCHING→ENRICHING = EXEC Stage 5; HOT→OUTREACH_ACTIVE = §9 Outreach Orchestrator; ENGAGED→MEETING_BOOKED = Human + CRM (joint, per §11 v2 HAND-ACC-01's Decision Ownership Boundary); OPPORTUNITY→CUSTOMER = §12 v2's CRM Authority (INT-CRM-02). No transition exists without a named writer; this registers into the §12 v2 Authority Registry (CRM-OWN-03) as the state-transition instance of that general mechanism.

STATE-GOV-02
A state has MANY readers, exactly ONE writer — never multiple writers for the same transition. Where a transition appears jointly owned (e.g. ENGAGED→MEETING_BOOKED), it is specified as a single joint-writer pairing with an explicit protocol (human confirms, system records), not as two independent writers that could race.

STATE-GUARD-01
Every transition has explicit Entry Conditions beyond 'is this destination reachable from here.' Example: HOT→OUTREACH_ACTIVE requires enrichment complete (§6 ENR-STOP-01 stopping condition met) AND ≥1 Response-Reachable contact (§6 v2 ENR-REACH-04) AND an Agent-5-approved strategy brief exists AND QC is not in a blocked state.

STATE-GUARD-02
A transition request that fails its Entry Conditions creates a Transition Rejection Event (logged, not silently dropped) — retained for §13 v2 analytics (a recurring rejection pattern on the same transition is itself a Failure Cluster candidate, §8 v2 QC-SYS-01).

STATE-GUARD-03
State machine validation checks BOTH destination validity (is this an allowed edge) AND readiness (do entry conditions hold) — §14 v1 only specified the former; this closes the gap. ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

STATE-TIME-01
Every state in both machines has an Expected Residence Time: ENRICHING ≤ 48 hours; OUTREACH_ACTIVE ≤ 30 days (matches the 4-touch/14-day sequence plus buffer); HOT_UNENRICHED ≤ 14 days; (Opportunity states get their own residence times, specified in Part B).

STATE-TIME-02
Breaching Expected Residence Time triggers a State Health Alert — NOT an automatic transition. An account stuck in HOT_UNENRICHED for 20 days is flagged for human review, never auto-demoted or auto-retried indefinitely without visibility.

STATE-TIME-03
State Aging (current residence time vs Expected Residence Time, per entity) becomes a standing §13 v2 operational KPI — a system with many entities aging past expectation in the same state is itself a signal worth the same Failure Cluster treatment as a QC anomaly.

STATE-DER-01
Every state field is classified Authoritative (stored, per §12 v2's Authority Registry) or Derived (computed on read from other authoritative fields, never independently stored). Example: 'does this account have any live Opportunity' is DERIVED from the Opportunity Portfolio (Part B), never a separately-stored boolean that could drift out of sync with the portfolio it should reflect.

STATE-DER-02
Values reliably derivable from already-authoritative fields are NEVER additionally stored — this directly reduces §12 v2's sync-conflict surface (CRM-OWN-01's Authority Registry already requires exactly one owner per field; a derived field's 'owner' is simply the computation itself, with no write path to conflict).

STATE-DER-03
Every entity's field-by-field Authoritative/Derived classification is documented explicitly at the §2 entity-model level, not left implicit — critical for §12 CRM sync, since only Authoritative fields are sync candidates at all (a Derived field is recomputed locally by whichever system needs it, never pushed/pulled). ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

STATE-OPP-06
The Opportunity State Machine (9 states above) exists INDEPENDENTLY of the Account State Machine (§14 v1, 13 states). They are never merged into one field, and no Opportunity state maps 1:1 onto an Account state.

STATE-OPP-07
Multiple Opportunities may exist simultaneously at one account, each independently positioned in the Opportunity State Machine — this is not an edge case to be handled, it is the NORMAL case the machine is designed for, consistent with §2's original re-anchoring rationale.

STATE-OPP-08
Account State (§14 v1) NEVER derives directly from any single Opportunity's state. Where a summary view is needed (§11 v2 handoff, §13 v2 dashboards), it reads the Portfolio Summary (OPP-COMP-02) — an aggregate across ALL the account's Opportunities — never one Opportunity's state treated as a proxy for the account.

STATE-OPP-09
PAUSED is explicitly NOT a closure state — it preserves the Opportunity's PRIOR active state for resumption (OPP-LIFE pattern: a budget freeze doesn't erase the qualifying work already done) and is reviewed on a fixed 30-day cadence rather than left indefinitely open or forced into a premature LOST closure. ABM Business Logic Bible — §14 v2 · Opportunity State Machine & Portfolio Architecture

STATE-ACCT-07
(EXEC Stage 5 sole writer of HOT-bound transitions) is GENERALIZED: every state transition in EITHER machine has exactly one declared Authoritative Writer. Examples: WATCHING→ENRICHING = EXEC Stage 5; HOT→OUTREACH_ACTIVE = §9 Outreach Orchestrator; ENGAGED→MEETING_BOOKED = Human + CRM (joint, per §11 v2

STATE-OPP-02
) was honest about being temporary. Rule ID Phas Specification e

Section 15: Reliability, Resilience, Antifragility (21 rules)

FAIL-SEV-01
Every failure receives a Business Severity (P0 Critical / P1 High / P2 Medium / P3 Low), assigned INDEPENDENTLY of its §15 v1 failure type. The same failure type (e.g. CRM sync failure) can carry different severities depending on the specific object affected — a sync failure on a PROPOSAL-stage Opportunity is P0; the same failure type on a WATCHING account is P3.

FAIL-SEV-02
Retry urgency and recovery-queue priority are driven by Business Severity, not failure type alone — a P0 transient failure and a P3 transient failure both get exponential backoff per FAIL-02, but the P0 case escalates to human attention far sooner if backoff doesn't resolve it.

FAIL-CASC-01
Every failure stores its Upstream Dependency — what system or service it required to succeed. An enrichment-API outage that subsequently prevents contact discovery, which prevents outreach, which prevents engagement, which degrades scoring, is recorded as ONE root failure with a dependency chain, not four-plus unrelated incidents.

FAIL-CASC-02
Failure dashboards display a Root Failure Tree (per OBS, §4 v2), not merely aggregate error counts — a spike in four different downstream error types that share one upstream dependency renders as one root-cause branch, not four alarms.

FAIL-CAP-01
A Recovery Backlog is tracked per failure category, consumed from the SAME attention-unit currency as §5 v3 TIER-ATTN-001 (reviewer-minutes) — recovery review is not a free, infinitely-available action; it draws from the engine's overall attention budget like any other human-effort expenditure in the Bible.

FAIL-CAP-02
The recovery queue is prioritized by Business Severity (Layer 1) × Opportunity Impact, NOT arrival order — the same ranking discipline as §5 v3 TIER-PORT-002 (Expected Opportunity Value ÷ Resource Cost), applied to recovery-item ordering rather than HOT-admission ordering.

FAIL-CAP-03
DLQ health (backlog size, age distribution, recovery-capacity utilization) is a first-class §13 v2 operational KPI — unrecovered failures are hidden technical debt, and a growing backlog with stable daily review capacity is itself a Layer 9 resilience-investment signal (more recovery capacity is needed, or the inflow rate itself needs reducing at the root).

FAIL-COST-01
Every P0/P1 failure receives a Business Impact estimate (opportunity delay, trust erosion, reputation risk, revenue risk) computed via the SAME Expected-Value family already used throughout the Bible (§6 v3 marginal-value estimation, §5 v3 TIER-PORT-002) — not a separate, third valuation method invented for failures specifically.

FAIL-COST-02
Operational reporting (§13 v2) includes Failure Cost (estimated Business Impact, aggregated) alongside Failure Volume — a high-volume, low-cost failure category and a low-volume, high-cost one are never collapsed into one 'failure rate' number.

FAIL-TRUST-01
Every failure is additionally classified by Reliability Impact (Low/Medium/High/Critical) — distinct from Business Severity (Layer 1), measuring damage to confidence IN THE PLATFORM specifically (will humans keep trusting the engine's outputs), mirroring §8 v2 QC-TRUST-01's Trust Impact taxonomy applied to infrastructure failures rather than QC failures.

FAIL-TRUST-02
Operational reviews evaluate Reliability Preservation (is institutional trust in the platform holding or eroding) alongside uptime metrics — high uptime with declining Reliability Impact scores (frequent small breakages that erode confidence even without major outages) is reported as a warning, the §15 instance of the same pattern §13 v2 ANL-KPI-04 already established for Trust Preservation Score.

FAIL-LAT-01
Every major failure tracks MTTD (Mean Time To Detect — failure occurrence to detection) and MTTR (Mean Time To Recover — detection to verified resolution, per Layer 7) as SEPARATE metrics, never blended into one 'incident duration' figure.

FAIL-LAT-02
Failure reviews prioritize closing detection bottlenecks BEFORE recovery bottlenecks when MTTD exceeds MTTR for a given failure category — a fast recovery process is wasted value if the failure isn't noticed for hours; FAIL-08's no-silent-failure principle (§15 v1) is necessary but not sufficient — 'logged' is not the same as 'detected promptly.'

FAIL-LAT-03
Detection Lag is recorded as a first-class metric on every major (P0/P1) failure record, feeding the same §13 v2 operational dashboard as DLQ health (Layer 3).

FAIL-VERIFY-01
Every recovery action produces a Recovery Confidence Score — not merely a success/failure flag on the retry attempt itself. A CRM replay that executes without throwing an error has not necessarily verified that all records synced correctly, ordering was preserved (§12 v2 Layer 7), or no duplicates were introduced.

FAIL-VERIFY-02
Critical (P0/P1) recoveries require explicit Post-Recovery Validation — a check that the recovered state actually matches expected reality, not merely that the recovery action completed without throwing an exception.

FAIL-REPEAT-01
Every failure recurrence (matched against prior Failure Pattern Records, FAIL-LEARN-01) increments a Failure Recurrence Count for that specific failure signature, tracked across ALL §15 failure types — GENERALIZING §8 v2 QC-SYS-01 / §13 v2's Failure Cluster mechanism from QC-specific clusters to every category in §15. QC-SYS-01 is recognized as the earlier, narrower instance of this more general mechanism.

FAIL-REPEAT-02
Recurrence automatically escalates Business Severity (Layer 1): a 3rd occurrence of the same failure signature triggers a mandatory Root Cause Review regardless of its original severity rating — a P3 failure that recurs repeatedly is no longer treated as P3, because the recurrence itself is evidence of an unaddressed systemic issue.

FAIL-INVEST-01
A Resilience Opportunity Register tracks candidate reliability improvements (better detection, better recovery automation, root-cause elimination) as a standing backlog, parallel in structure to §6 v3's Shared Intelligence Assets pattern — a durable, queryable register, not an ad-hoc list.

FAIL-INVEST-02
Resilience fixes are prioritized by Expected Loss Prevented (reusing Layer 4's Business Impact estimation method), NOT by raw incident count — a rare P0 failure with high Business Impact can outrank a frequent P3 annoyance on the Register, exactly mirroring §5 v3 TIER-PORT-002's ratio-based ranking rather than a simple frequency sort.

FAIL-ANTI-02
Failure reviews explicitly answer: 'what new capability exists because this happened?' — a review that produces only a root-cause explanation, with no proposed capability improvement, is incomplete by this standard.

Section 16: Temporal Intelligence Layer (27 rules)

SCORE Rules (1)

SCORE-SIG-01
) — a step-function signal's value should not be modeled as gradually fading when its real behavior is a cliff.

TIME Rules (26)

TIME-SCHED-01
A System Clock Map visualizes every recurring timer in §16.0–16.8 on one timeline, making simultaneous-fire clusters visible at a glance — built from the existing §16 v1 matrix, not a new timer inventory.

TIME-SCHED-02
New recurring timers are deliberately offset from existing ones at the SAME cadence to avoid clustering (e.g. if Loop 1 and Loop 4 both run 'every 30 days,' they are scheduled on different days of that cycle, not the same hour of the same day).

TIME-SCHED-03
Timer design considers system load (API rate limits, compute cost, concurrent-job ceiling) alongside business need — a timer change proposal (per §16 v1 TIME-MATRIX-05's maintenance obligation) must show it doesn't create a new collision before approval.

TIME-LAT-01
Every critical workflow (signal→enrichment→scoring→outreach) receives an explicit End-to-End Response SLA: e.g. Executive-move signal → first outreach ≤ 48 hours, summing across every component timer in the chain (§16.0–16.3), not just each stage's local SLA.

TIME-LAT-02
§13 v2 tracks Reality-to-Action Latency as its own KPI — distinct from any single component's performance — because a chain of five individually-fast stages can still produce an unacceptably slow end-to-end response if their handoffs aren't coordinated. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-CLASS-01
Every timer in §16.0–16.8 is classified Real-Time (engagement webhooks, reply detection), Operational (CRM pulls, signal polling), Strategic (enrichment refresh, opportunity-window tracking), or Governance (Learning Review Board, Resilience Investment review).

TIME-CLASS-02
A breached timer's response priority is set by its Class, not by raw count of breaches — ten missed Governance-class reviews in a quarter is less urgent than one missed Real-Time reply-detection window, even though the latter is a single incident.

TIME-HUM-01
Human-facing SLAs (§16.5: review SLA, handoff SLA, escalation SLA) carry a Capacity Context — Ramadan, conference weeks, quarter-end, individual leave calendars — sourced from the same KSA-calendar constraints §17 will formally specify, plus any team-level capacity data already available.

TIME-HUM-02
Human SLA calculations adjust for Available Capacity, not clock time alone: a 4-hour review SLA during a known low-capacity period (Ramadan reduced hours, a public holiday) is interpreted against capacity-adjusted business hours, not literal wall-clock hours — this does NOT relax the SLA's intent, it correctly accounts for when 'business hours' actually exist. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-ECON-01
Every timer's specification (going forward; not retroactively re-justifying all 60+ existing ones) requires an Economic Justification — why this frequency, why not faster, why not slower — recorded alongside the timer in §16.

TIME-ECON-02
Timer reviews (part of §16 v1 TIME-MATRIX-05's maintenance obligation) evaluate Value Generated vs Cost Consumed using the SAME Expected-Value family as §6 v3/§15 v2 (marginal-value gating, Business Impact estimation) — not a fourth independent value-estimation method for timing specifically.

TIME-OPP-01
Every signal category receives an Opportunity Window Profile — the typical duration the underlying BUYING OPPORTUNITY (not the signal's freshness) stays open: Leadership Hire 90–180 days; Regulatory Change 30–90 days; Funding Event 60–120 days; RFP 15–45 days. This is a NEW clock, separate from §3/§6 v3's signal-freshness decay.

TIME-OPP-02
§5's Effective Opportunity scoring (§1/§5 v3) takes Opportunity Window Remaining as an ADDITIONAL multiplier input alongside Signal Strength — a strong signal deep into its window's close scores differently than the same strength signal early in its window. This modifies the EXISTING formula; it does not create a parallel scoring path.

TIME-OPP-03
An account approaching its Opportunity Window's expiry receives a priority boost in §5 v3's portfolio-ranking (TIER-PORT-002) specifically because of urgency — a closing window is itself a reason to act sooner, distinct from and additive to the account's raw Effective Opportunity value. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-DEP-01
Named Temporal Escalation Events (CEO/CTO hire, strategic acquisition, major RFP, SAMA mandate, Vision 2030 initiative announcement) are recognized as a SUBSET of §3's existing P1_CRITICAL signal classification, cross-referenced against GOV's Autonomy Ladder (which tier of action they unlock) and the EDGE catalogue (which scenario, if any, they also trigger) — not a new classification system.

TIME-DEP-02
A recognized Temporal Escalation Event uses the FAST-TRACK processing path already implied by P1_CRITICAL's existing 1-hour score-recalculation / 2-hour enrichment-trigger SLA (§3 §3.4) — §16 v2 makes explicit that this IS the fast-track path, rather than describing a separate queue that doesn't exist.

TIME-DEP-03
Variable process speed (some workflows compress, most don't) is bounded by GOV: a Temporal Escalation Event can accelerate TIMING but never bypass an AUTHORITY gate (e.g. it cannot skip QC-RULE-009's permanent political-sensitivity human review just because it's time-urgent) — speed and authority are independent dimensions; this rule keeps them that way. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-DECAY-01
Every signal category receives a Decay Function Type — Linear, Exponential, Step-function, or Plateau — layered ONTO §6 v3 ENR-DECAY-01's existing Operational/Tactical/Strategic/Structural duration categories, not a replacement for them. Example: LEADERSHIP_HIRE (executive departure sub-type) = Step-function (high value until a successor is named, then near-zero); DIGITAL_INITIATIVE = gradual Exponential decay.

TIME-DECAY-02
§4/§5 scoring applies the signal-specific Decay Function Type, not a universal smooth curve, when computing freshness-weighted Signal Strength (§4 v1 SCORE-SIG-01) — a step-function signal's value should not be modeled as gradually fading when its real behavior is a cliff.

TIME-DECAY-03
§13 v2's Model Drift mechanism (ANL-DRIFT-01/02) periodically validates decay-shape assumptions against actual outcome data — a signal category whose assumed decay shape no longer matches observed value-over-time is itself a drift signal, using the SAME Predictive Stability Score mechanism, not a new validation method specific to decay curves. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-STRAT-01
Initiative Progression Models (e.g. Cloud → API → Partner → Marketplace) are specified and maintained as named patterns, derived from observed account histories (Tier 6 historical data) — not invented from intuition.

TIME-STRAT-02
INTEGRATION DECISION: a predicted future opportunity from an Initiative Progression Model is created as a SIG-HYP hypothesis (§3 reasoning streams) with a full EPIS-RCM stamp — NOT an unstamped prediction. Its confidence is explicitly falsifiable (EPIS-RCM-03) and it is never presented to a human as a confirmed future event.

TIME-STRAT-03
FLAGGED TENSION, RESOLVED: 'outreach may begin before the opportunity fully materializes' is permitted ONLY as explicitly-labeled Hypothesis-driven outreach — it does NOT satisfy or bypass §14 v2 OPP-LIFE-01's three-condition Opportunity birth test. An Opportunity entity is not created merely because a Progression Model predicts one is likely; outreach grounded in a SIG-HYP hypothesis is clearly distinguished (in the dossier, in QC, in the handoff package) from outreach grounded in a confirmed Opportunity, so a human never mistakes a good guess for a qualified deal. ABM Business Logic Bible — §16 v2 · The Temporal Intelligence Layer

TIME-ADV-01
Signal-to-Action Time is tracked for every opportunity — Decimal's OWN elapsed time from signal detection to first qualified outreach — as a standing §13 v2 KPI, feeding the Layer 2 End-to-End Latency metric.

TIME-ADV-02
INTEGRATION DECISION: any claim about COMPETITOR response speed is bound to §9 v2 OUT-COMP-01's sourced-only discipline (SIG-VENDOR/SIG-NEWS evidence only) — §16 v2 never assumes or estimates a competitor's speed without a sourced signal. Decimal's own Signal-to-Action Time, measured directly, is the primary and always-available metric; competitive comparison is a sometimes-available secondary one.

TIME-MATRIX-05
's maintenance obligation) must show it doesn't create a new collision before approval. LAYER 2 — LATENCY CHAIN (LAT) Rule ID Phas Specification e

Section 17: Market Intelligence (KSA) (34 rules)

ENR Rules (2)

ENR-COMM-02
) remains the floor, but a thinner contact list with a strong connector graph is not penalized relative to a fuller list with no connector path.

ENR-COMM-01
) prioritizes resolving Sponsor status alongside the standard persona/seniority classification, not as an optional enrichment nice-to-have.

KSA Rules (29)

KSA-REL-01
Every KSA opportunity tracks Formal Influence (§5/§7 v2's existing scoring/strategy signals) and Relationship Influence (warm-path strength per §3 SIG-PATH, Trust Transfer Potential per §7 v2 AI-TRUST-04) as SEPARATE tracked dimensions — not blended into one opportunity score.

KSA-REL-02
A warm introduction (high SIG-PATH P1/P2 tier) registers as Relationship Influence movement BEFORE any change in Opportunity-level scoring — the two move on different clocks, and Relationship Influence is permitted to lead Opportunity Influence for KSA accounts specifically.

KSA-REL-03
§14 v2's Opportunity State Machine recognizes, for KSA accounts, that meaningful relationship-development activity may precede OPP_IDENTIFIED (§14 v2 OPP-LIFE-01's birth conditions) — this is the KSA-specific instance of §16 v2 TIME-STRAT-03's Hypothesis-driven outreach: relationship-building activity is tracked and valued even before an Opportunity formally exists, explicitly as pre-Opportunity activity, never mislabeled as an open deal. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-SPON-02
Sponsor identification is a HIGH-PRIORITY enrichment objective for KSA Tier-1 accounts specifically — §6's committee-discovery workflow (ENR-COMM-01) prioritizes resolving Sponsor status alongside the standard persona/seniority classification, not as an optional enrichment nice-to-have.

KSA-SPON-03
Opportunity health (§5 v3/§14 v2) includes Sponsor Strength as a distinct factor from raw engagement — an opportunity with high contact engagement but NO identified Sponsor is flagged as structurally weaker than the engagement numbers alone would suggest, for KSA accounts specifically.

KSA-NET-01
INTEGRATION DECISION: an account's Ecosystem Connectivity is computed from the SAME SIG-PATH connector registry (§3 SIG-PATH-08: KPMG/Mastercard/Deloitte/Tarabut/Fintech-Saudi) and §6 v3 Shared Intelligence Assets (ENR-PORT-03) already built — not a new graph structure. §17 v2 confirms these mechanisms are FIRST-ORDER priorities for KSA accounts, not background context.

KSA-NET-02
For KSA accounts, §5 v3's path-priority ladder (P1–P5) is consulted BEFORE cold outreach is even considered as the default — network entry-point discovery is the first action on a newly-HOT KSA account, not a parallel option weighed equally against direct outreach.

KSA-NET-03
Account-level relationship graphs (the SIG-PATH connector network for a given account) are treated as more decision-relevant than raw contact-list completeness for KSA Tier-1 accounts — §6's committee-discovery minimum (3–7 contacts, ENR-COMM-02) remains the floor, but a thinner contact list with a strong connector graph is not penalized relative to a fuller list with no connector path. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-TIME-01
§9 v2's Engagement Archetype interpretation (§10 v1 ENG-SILENCE, §9 v2 OUT-BEH-02) and §5 v3's Strategic Patience layer (HOT_WAIT substate, Action Readiness Score) use KSA-CALIBRATED decay/silence benchmarks for KSA accounts, not the Bible's general-purpose defaults — the underlying mechanisms are unchanged; the calibration inputs are KSA-specific.

KSA-TIME-02
Silence thresholds before a Negative Signal Event fires (§10 v2 ENG-NULL-02) are LONGER for KSA Tier-1 banks specifically, reflecting genuinely slower typical cycles rather than disengagement — bound to §17 v1 KSA-030's existing Tier-1 procurement-complexity default as the same population this calibration applies to.

KSA-TIME-03
Opportunity abandonment/closure logic (§14 v2 OPP_* residence-time expectations, §14 v2 state table) uses KSA-LOCAL cycle benchmarks for KSA accounts — e.g. OPP_EVALUATING's general ≤90-day expected residence may be insufficient for a KSA Tier-1 institution and should not trigger a State Health Alert (§14 v2 STATE-TIME-02) at the same threshold as a faster-cycle market. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-TRUST-01
For KSA accounts, Trust Accumulation Score (§7 v2 AI-TRUST-01) is elevated from a CTA-friction MODIFIER to an explicit GATE on specific actions: direct meeting requests, pricing discussion, and any Proposal/Negotiation-stage engagement (§14 v2 OPP_PROPOSAL/OPP_NEGOTIATION) require Trust Accumulation above a KSA-specific (higher) threshold, regardless of how high the underlying Opportunity score is.

KSA-TRUST-02
INTEGRATION DECISION: this is an ELEVATION of §7 v2 AI-TRUST-02's existing mechanism (same Trust Accumulation Score, same gating logic), not a new trust model — KSA-020 (§17 v1) already set a cultural floor on Touch-1 CTA aggressiveness; KSA-TRUST-01 extends that floor explicitly through the Proposal/Negotiation stages, where the Bible's general rules would otherwise allow trust-independent progression once an Opportunity is sufficiently advanced.

KSA-TRUST-03
The PRIMARY objective of first-touch KSA outreach (Touch 1, §9 v1 OUT-SEQ-EMAIL-01) is explicitly RELATIONSHIP CREATION, not meeting creation, reply, or demo — §7 v2 Agent 5's Portfolio Role assignment (§9 v2 OUT-SEQ-EMAIL-02) defaults Touch 1 to Awareness/Credibility for ALL accounts already; KSA-TRUST-03 makes this the explicit, non-negotiable PRIMARY success criterion for KSA Touch 1 specifically, overriding any temptation to optimize Touch 1 copy for reply-rate at the expense of relationship framing. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-SEG-01
Every KSA account receives a Governance Archetype (STATE_INFLUENCED / COMMERCIAL / DIGITAL_CHALLENGER / SPECIALIZED), classified from PUBLICLY AVAILABLE information (ownership structure, listing status, stated market positioning — not private speculation about influence or control).

KSA-SEG-02
§7 v2 Agent 5's outreach strategy adapts to Governance Archetype: STATE_INFLUENCED accounts default toward Executive Sponsorship/Consensus Building strategies (§7 v2 AI-INF-01) with longer patience calibration (Layer 4); DIGITAL_CHALLENGER accounts may tolerate more direct, faster-paced strategies.

KSA-SEG-03
INTEGRATION DECISION: Governance Archetype REFINES §17 v1 KSA-030's existing Tier-1 procurement-complexity default rather than replacing it — KSA-030's HIGH/EXTREME default remains the floor for STATE_INFLUENCED and COMMERCIAL archetypes; DIGITAL_CHALLENGER is the one archetype permitted to default lower, per the table above. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-REG-01
SAMA-sourced regulatory signals (§3 SIG-REG, top-tier source-trust) receive an ELEVATED Opportunity Multiplier relative to generic market signals (funding, hiring, competitive moves) when computing Effective Opportunity (§1/§5 v3) for KSA accounts — this is an explicit weighting refinement within §4's existing 30% Regulatory Pressure dimension, not a new scoring dimension.

KSA-REG-02
Every KSA Opportunity is checked for a mappable Regulatory Driver (a specific SAMA/CMA mandate it responds to) where one exists — surfaced explicitly in the dossier and the §11 v2 handoff package, since a regulatory-driven opportunity's urgency and budget-availability story differs fundamentally from a discretionary one.

KSA-REG-03
The engine distinguishes Regulatory Initiative (driven by a mappable SAMA/CMA mandate, KSA-REG-02) from Discretionary Initiative (driven by competitive or internally-generated strategic priority) as a stored Opportunity attribute — §7 v2 Agent 5 reads this distinction: a Regulatory Initiative's messaging can lead with compliance urgency; a Discretionary Initiative cannot borrow that framing without becoming an unsourced, inflated claim (API-09 grounding still applies). ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-MOVE-01
INTEGRATION DECISION: Executive Mobility tracking in KSA banking is §3's existing SIG-MOBILITY mechanism (executive/board/consultant/vendor movement → Opportunity Transfer Events), NOT a new tracking system. §17 v2 confirms SIG-MOBILITY's priority is elevated for KSA accounts specifically, given how concentrated and interconnected the GCC banking executive population is relative to larger, less interconnected markets.

KSA-MOVE-02
When SIG-MOBILITY detects a tracked executive moving between two KSA accounts, the Relationship Influence (Layer 1) at the new account is seeded from the departing relationship's history, NOT reset to zero — this is §3 SIG-PATH-04's existing 'follow-the-person' re-anchoring rule, confirmed as the binding behavior for KSA executive moves specifically.

KSA-MOVE-03
A KSA executive move triggers an Opportunity Re-Evaluation at EVERY connected account where that executive previously held influence (not only the new and former employer) — reusing §3 SIG-PATH-05's cross-group power-node detection to identify which other accounts the moving executive touches. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-POL-01
INTEGRATION DECISION: Political Alignment Risk for an Opportunity is computed EXCLUSIVELY from §9 v2's existing SIG-TENSION mechanism (competing PUBLIC priorities, conflicting PUBLIC statements, contradictory PUBLIC initiatives, observable budget conflicts) — the same sourced-only discipline already binding SIG-TENSION, with zero new inference pathways.

KSA-POL-02
A sustained or escalating SIG-TENSION signal at a KSA account generates a Consensus Risk Alert on the relevant Opportunity, surfaced in §11 v2's handoff package — this is SIG-TENSION's existing output, given a KSA-specific name and a guaranteed surfacing point in the handoff flow rather than left as background dossier context.

KSA-POL-03
HARD BOUNDARY: relationship mapping for KSA accounts prioritizes discovering PUBLICLY VISIBLE decision-relevant connections (committee membership, public transformation-program leadership, public board composition) — ‘decision coalition discovery’ means resolving WHO PUBLICLY HOLDS WHAT ROLE more completely, never inferring private alliances, rivalries, or negotiating positions among named individuals. Any output that would require speculating about a specific person's private stance is rejected at §8 QC-007 (compliance) as a sourced-fact violation, not merely discouraged. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

KSA-ECO-01
INTEGRATION DECISION: the ‘Ecosystem Trust Graph’ is §3 SIG-PATH's connector registry plus §6 v3's Shared Intelligence Assets, READ AT THE FULL-ECOSYSTEM SCALE for KSA specifically (across SNB, Riyad, Al Rajhi, SAB, Alinma, and the broader KSA banking population), not a new graph structure layered on top of them.

KSA-ECO-02
Opportunity prioritization (§5 v3 TIER-PORT-002) for KSA accounts includes Ecosystem Influence Potential as an additional ranking input alongside Expected Opportunity Value: an account whose win would strengthen Decimal's connector position across MULTIPLE other KSA accounts (per the SIG-PATH/Shared Intelligence Assets graph) ranks higher than its own standalone Effective Opportunity value alone would suggest.

KSA-ECO-03
§13 v2's Reputation Yield metric (§7 v2 AI-REP-04, §13 v1 ANL-KPI-07) is read at KSA-ecosystem scale: a reputational event (positive or negative) at one KSA account is modeled as propagating measurable Reputation Impact to ecosystem-CONNECTED accounts specifically (via the same SIG-PATH/SIG-MOBILITY graph), not only to the account where the event occurred — this is §7 v2 AI-REP-02's existing account→market roll-up, made explicit at the KSA-ecosystem granularity rather than left implicit at the abstract 'market' level. ABM Business Logic Bible — §17 v2 · The Saudi Banking Market Intelligence Layer

OUT Rules (2)

OUT-SEQ-EMAIL-02
) defaults Touch 1 to Awareness/Credibility for ALL accounts already;

OUT-SEQ-EMAIL-01
) is explicitly RELATIONSHIP CREATION, not meeting creation, reply, or demo — §7 v2 Agent 5's Portfolio Role assignment (§9 v2

SIG Rules (1)

SIG-POWER-01
) crosses a KSA-specific threshold.

Section 19: Glossary and Reference (2 rules)

QC Rules (1)

QC-RULE-00
re-queues to STRATEGY (Agent 5), not wording (Agent 6) — because 8/009/010 the failure is in the plan, not the prose. Failure Cluster A statistically significant spike in one failure type, signaling an upstream §8 v2

TIER Rules (1)

TIER-DEEP-12
/ §6 v1 RELATIONSHIP_MONITOR A fifth account state preserving a high-equity relationship against generic §5 v3 opportunity-axis demotion.

WHAT DOCUMENT 3 (PERFECT) ADDS

Document 3 completes the engine. It adds: the 25 Meta-Principles as enforced constitutional law, Autonomy Ladder A3-A5, the advanced cognition modules (Consensus Formation, Champion Strength, Anti-Champion Detection, Political Power Mapping, Opportunity Death Prediction, Counterfactual Analysis, Market Narrative Tracking), Organizational Learning Governance (§13), the Antifragility Engine (§15), Temporal Intelligence (§16), and Edge Epistemics (§18). Total: 1065 rules (866 Production + 199 Perfect).